PEARL
Power and Energy Analysis and Research Laboratory

Escuela de Ingeniería Industrial y Aeroespacial Toledo
Universidad de Castilla-La Mancha

UCLM Universidad Castilla-La Mancha

Tools for the optimal operation and planning of power systems

Lecture 2: Operation and planning problems

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Solutions to the proposed exercises

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%202



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2. Operation
    1. Unit commitment
    2. Self-scheduling
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3. Planning
    1. Centralized generation expansion planning
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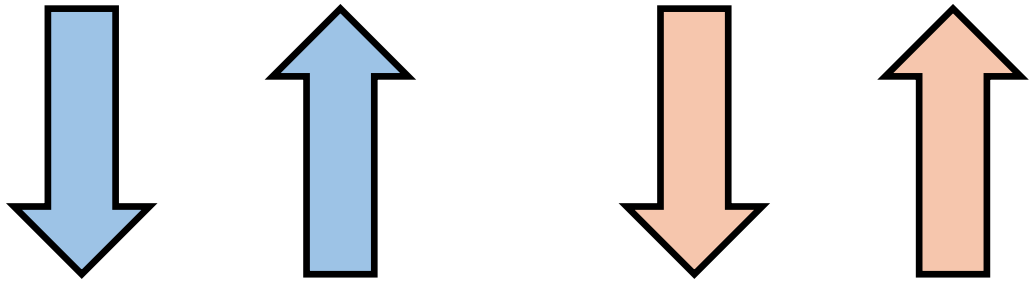
1. Introduction

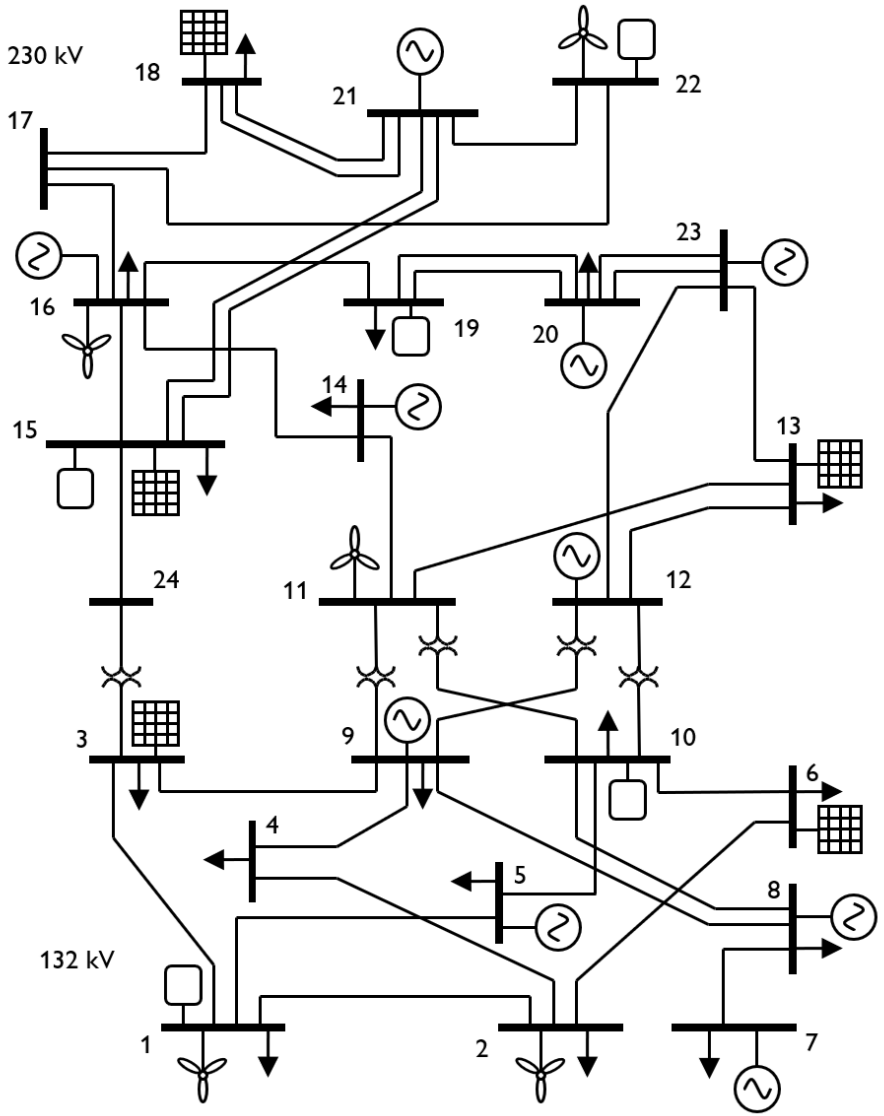
Applications to Power Systems

- Participation in electricity markets.
- Security of power systems.
- Participation in demand response programs.
- Investment in transmission lines, generating units, and storage facilities.

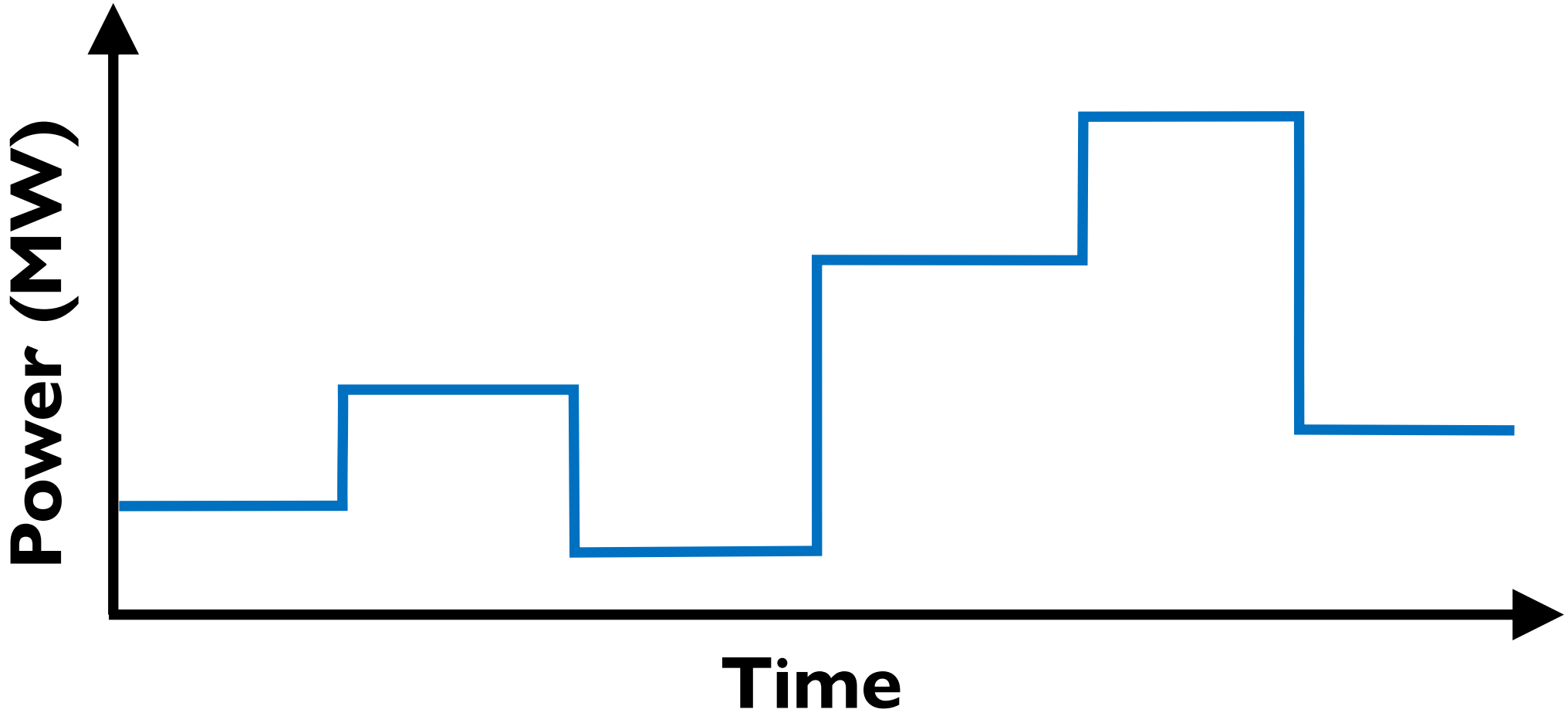
Balance of energy

Generation = Consumption

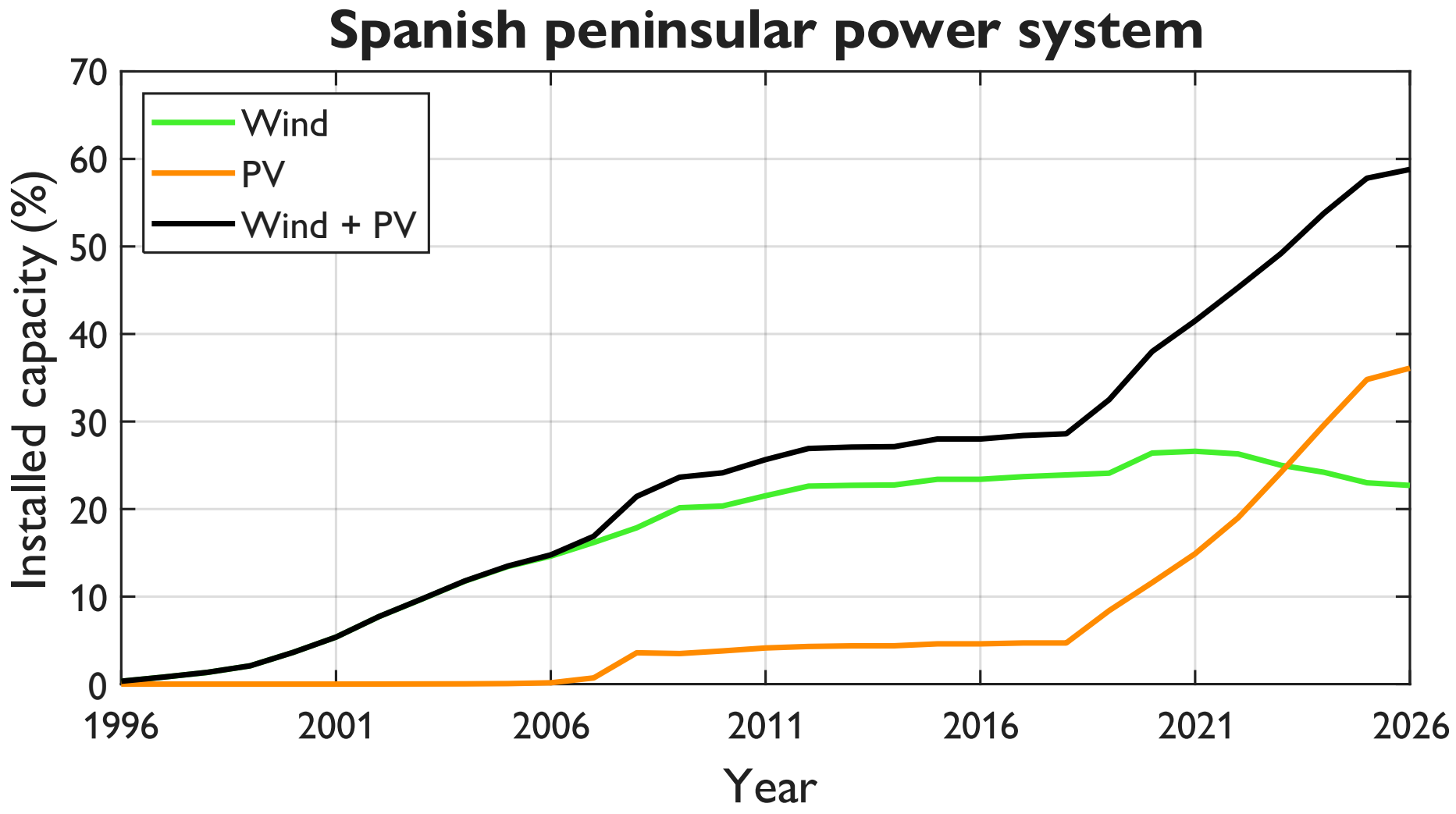




Scheduled production

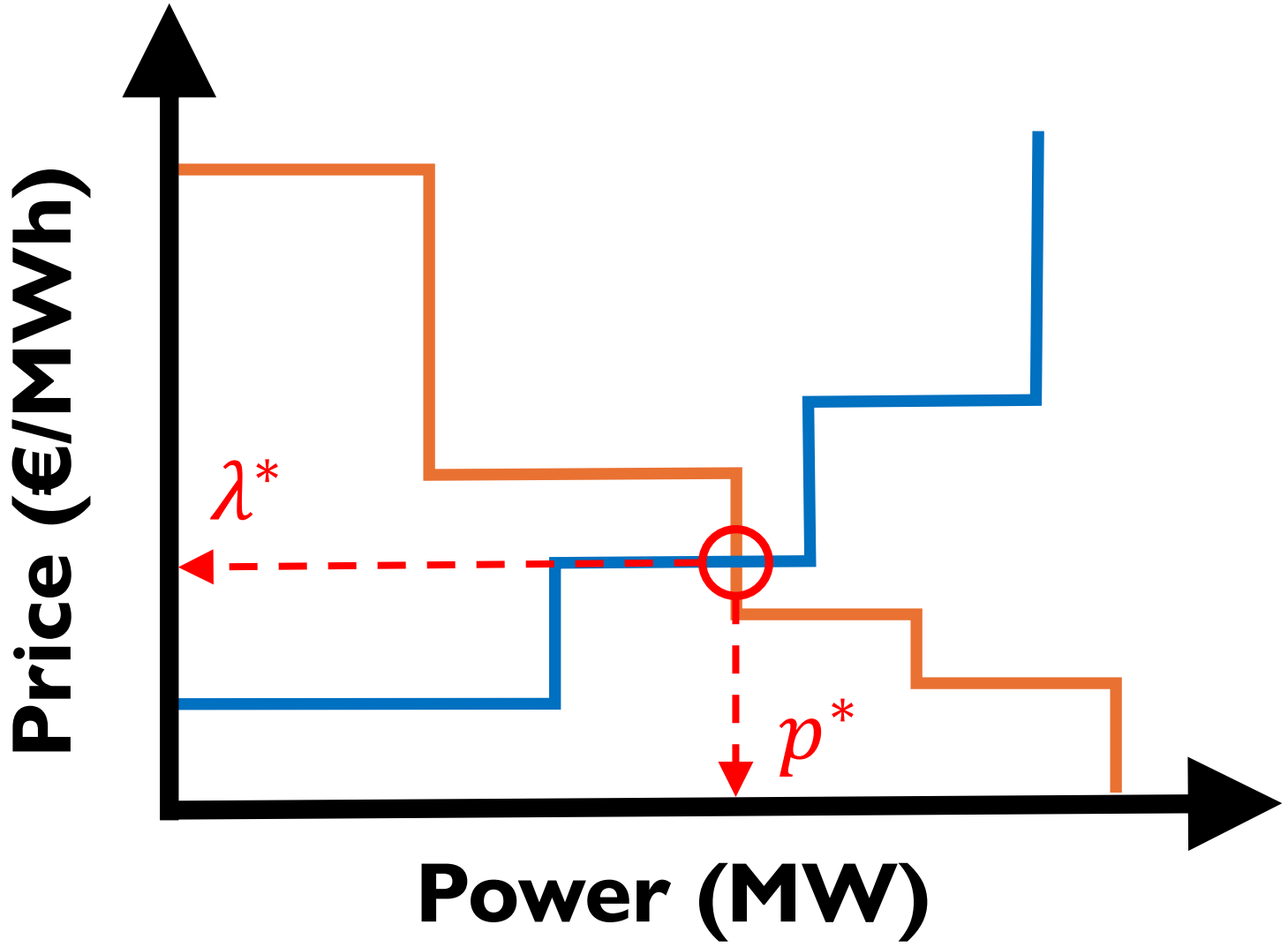


Modern power systems

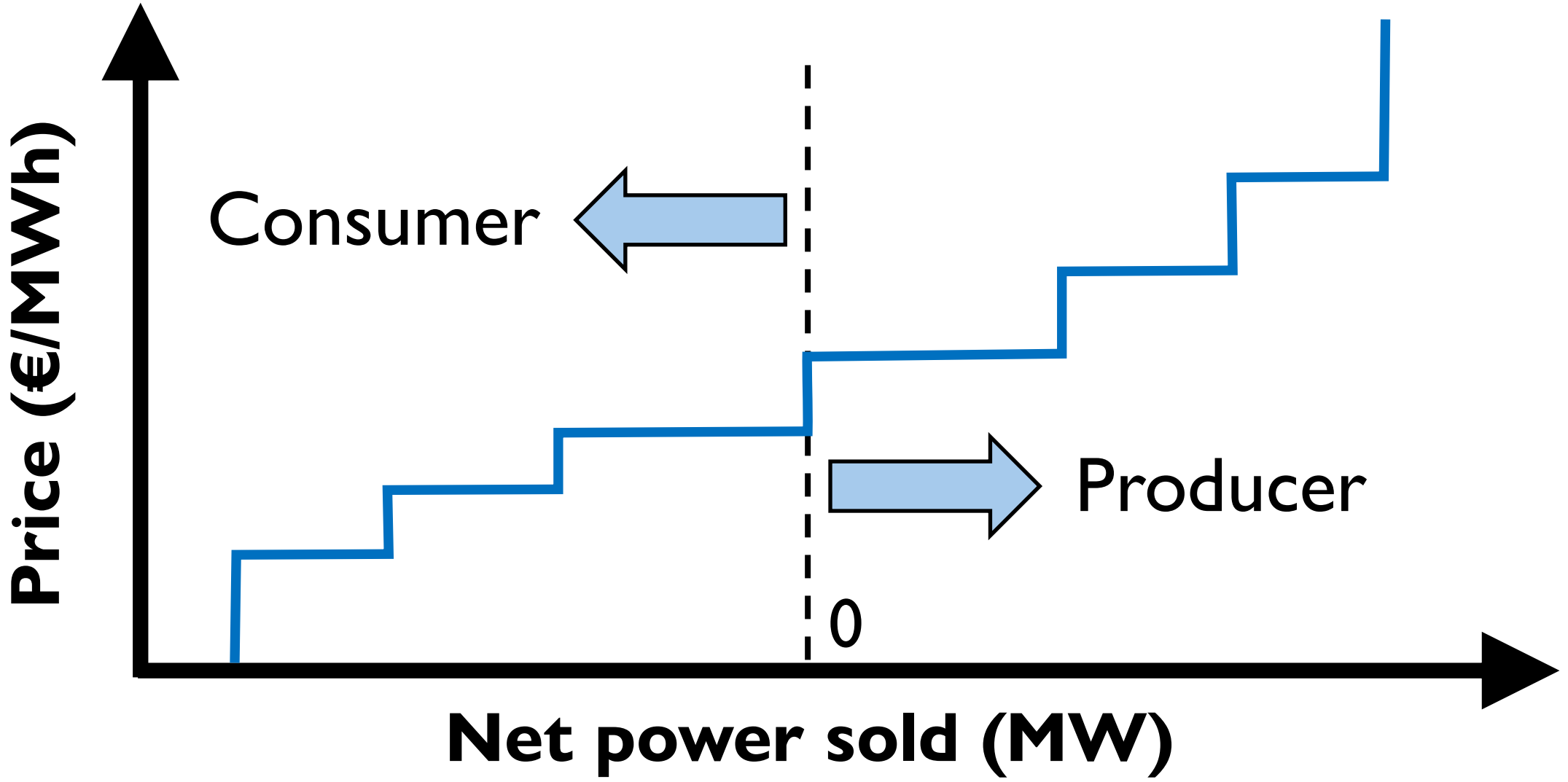


Source: https://www.ree.es/en/datos/generation/installed-capacity-generation

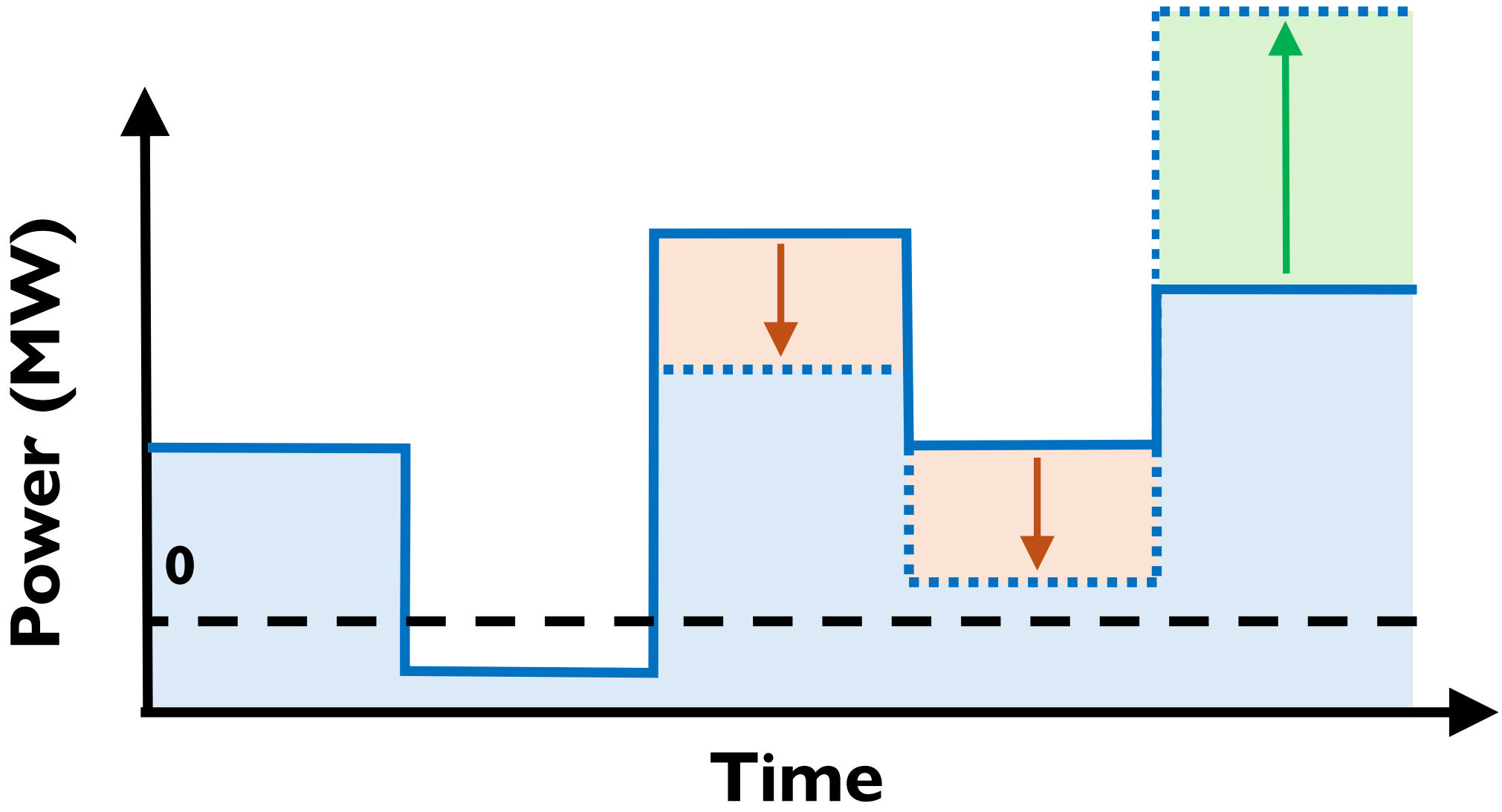
Day-ahead electricity market



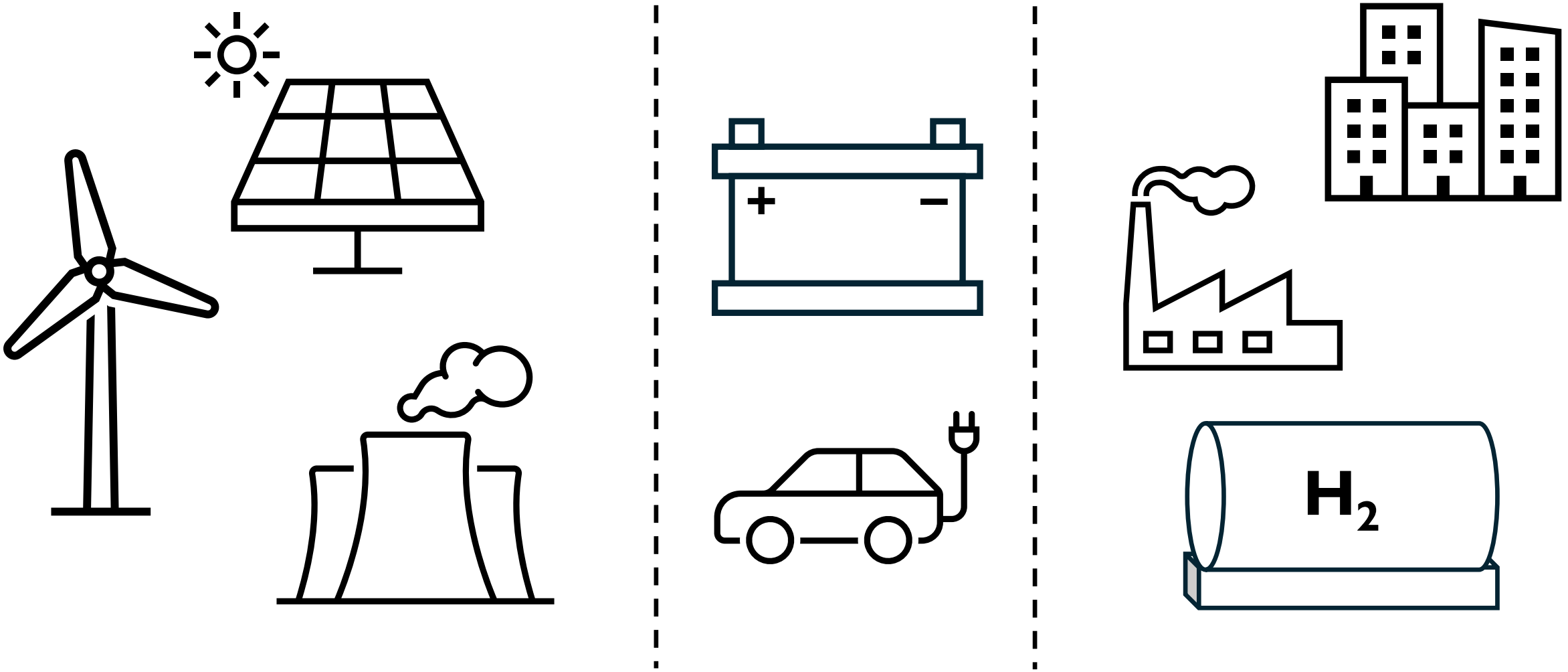
Bidding and offering curves



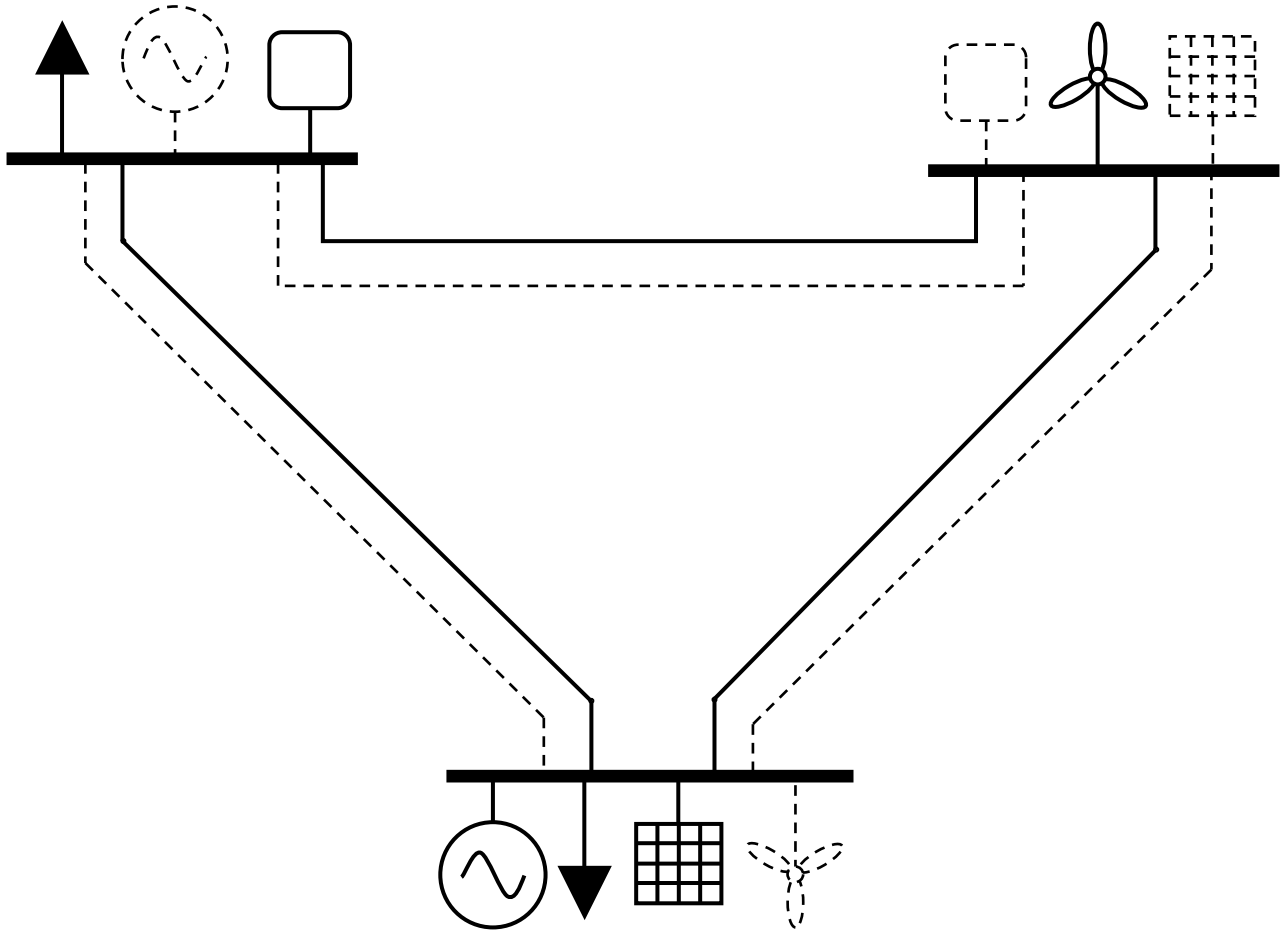
Demand Response Programs



Virtual Power Plant



Generation and transmission network expansion



What will you learn?

- To solve operation optimization problems from the viewpoint of:
  - A centralized entity (min cost).
  - Private entities (max profit / min cost).

- To solve planning optimization problems from the viewpoint of:
  - A centralized entity (min cost).
  - Private entities (max profit / min cost).

Notation

I will use the following notation in the course:

- Uppercase letters → Parameters (input data).
- Lowercase letters → Variables (output data).

Formulation

I will consider hourly time periods for the sake of simplicity.

Hence, power (MW) and energy (MWh) units are equivalent.

If this is not the case, the duration of the time periods should be included in the formulation.

2. Operation

Operation

Operation optimization problems solved by:

- A centralized entity → Unit commitment.
- Private entities → Self-scheduling.
- Cost function with intervals.

Operation

Operation optimization problems solved by:

- A centralized entity → **Unit commitment**.
- Private entities → Self-scheduling.
- Cost function with intervals.

Unit commitment

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \sum_{g \in \Omega^{\mathrm{G}}}$$

Unit commitment

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \sum_{g \in \Omega^{\mathrm{G}}} \left(\boldsymbol{C}_{g}^{\mathbf{G,F}} \boldsymbol{u}_{gt}^{\mathbf{G}} \right)$$

Unit commitment

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \sum_{g \in \Omega^{\mathrm{G}}} \left(C_g^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + \boldsymbol{C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}}} \right)$$

Unit commitment

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \sum_{g \in \Omega^{\mathrm{G}}} \left(C_g^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} + \boldsymbol{C_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}} \right)$$

Unit commitment

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \sum_{g \in \Omega^{\mathrm{G}}} \left(C_g^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}} + \boldsymbol{C_g^{\mathrm{G,SD}} z_{gt}^{\mathrm{G}}} \right)$$

Unit commitment – Status

$$u_{gt}^{\mathrm{G}} \in \{0,1\}, \forall g, \forall t$$

$$u_{gt}^{\mathrm{G}} = 1 \rightarrow \text{ON}$$

$$u_{gt}^{\mathrm{G}} = 0 \rightarrow \text{OFF}$$

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} \in \{0,1\}, \forall g, \forall t$$

$$y_{gt}^{\mathrm{G}} = 1 \rightarrow \text{Start-up}$$

$$z_{gt}^{\mathrm{G}} \in \{0,1\}, \forall g, \forall t$$

$$z_{gt}^{\mathrm{G}} = 1 \rightarrow \text{Shutdown}$$

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

u_{gt}^{G}	$u_{g,t-1}^{\mathrm{G}}$	y_{gt}^{G}	z_{gt}^{G}
0	0		
0	1		
1	0		
1	1		

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

u_{gt}^{G}	$u_{g,t-1}^{\mathrm{G}}$	y_{gt}^{G}	z_{gt}^{G}
0	**0**	**0/1?**	**0/1?**
0	1		
1	0		
1	1		

Unit commitment – Status

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \sum_{g \in \Omega^{\mathrm{G}}} \left(C_g^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} + \boldsymbol{C_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}} + \boldsymbol{C_g^{\mathrm{G,SD}} z_{gt}^{\mathrm{G}}} \right)$$

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

u_{gt}^{G}	$u_{g,t-1}^{\mathrm{G}}$	y_{gt}^{G}	z_{gt}^{G}
0	**0**	**0**	**0**
0	1		
1	0		
1	1		

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

u_{gt}^{G}	$u_{g,t-1}^{\mathrm{G}}$	y_{gt}^{G}	z_{gt}^{G}
0	0	0	0
0	**1**	**0**	**1**
1	0		
1	1		

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

u_{gt}^{G}	$u_{g,t-1}^{\mathrm{G}}$	y_{gt}^{G}	z_{gt}^{G}
0	0	0	0
0	1	0	1
1	**0**	**1**	**0**
1	1		

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

u_{gt}^{G}	$u_{g,t-1}^{\mathrm{G}}$	y_{gt}^{G}	z_{gt}^{G}
0	0	0	0
0	1	0	1
1	0	1	0
1	**1**	**0/1?**	**0/1?**

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

u_{gt}^{G}	$u_{g,t-1}^{\mathrm{G}}$	y_{gt}^{G}	z_{gt}^{G}
0	0	0	0
0	1	0	1
1	0	1	0
1	**1**	**0**	**0**

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

$$\boldsymbol{y_{gt}^{\mathrm{G}} + z_{gt}^{\mathrm{G}} \leq 1, \forall g, \forall t}$$

Unit commitment – Status

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - U_{g}^{\mathrm{G},0}, \forall g, t = 1$$

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t \in \Omega^{\mathrm{T}} \backslash \{1\}$$

Unit commitment – Power limits

$$\underline{P}_g^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}}$$

Unit commitment – Power limits

~~$\underline{P}_g^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}}$~~

$$\underline{P}_g^{\mathrm{G}} u_{gt}^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}} u_{gt}^{\mathrm{G}}, \forall g, \forall t$$

Unit commitment – Power limits

$$\underline{P}_g^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}}$$

$$\underline{P}_g^{\mathrm{G}} u_{gt}^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}} u_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$\boldsymbol{u_{gt}^{\mathrm{G}} = 0 \rightarrow 0 \leq p_{gt}^{\mathrm{G}} \leq 0 \rightarrow p_{gt}^{\mathrm{G}} = 0}$$

Unit commitment – Power limits

$$\underline{P}_g^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}}$$

$$\underline{P}_g^{\mathrm{G}} u_{gt}^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}} u_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$u_{gt}^{\mathrm{G}} = 0 \rightarrow 0 \leq p_{gt}^{\mathrm{G}} \leq 0 \rightarrow p_{gt}^{\mathrm{G}} = 0$$

$$\boldsymbol{u_{gt}^{\mathrm{G}} = 1 \rightarrow \underline{P}_g^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}}}$$

Unit commitment – Ramping limits

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} u_{g,t-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, \forall t$$

Unit commitment – Ramping limits

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} u_{g,t-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$u_{gt}^{\mathrm{G}} = 0 \text{ and } u_{g,t-1}^{\mathrm{G}} = 0 \ (y_{gt}^{\mathrm{G}} = 0):$$

$$0 - 0 \leq 0 + 0$$

Unit commitment – Ramping limits

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} u_{g,t-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$u_{gt}^{\mathrm{G}} = 0 \text{ and } u_{g,t-1}^{\mathrm{G}} = 1 \; (y_{gt}^{\mathrm{G}} = 0):$$

$$0 - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} + 0 \rightarrow -R_g^{\mathrm{G,U}} \leq p_{g,t-1}^{\mathrm{G}}$$

Note that $p_{g,t-1}^{\mathrm{G}} \geq -R_g^{\mathrm{G,U}}$ is an inactive constraint since it is less restrictive than $p_{g,t-1}^{\mathrm{G}} \geq \underline{P}_g^{\mathrm{G}}$.

Unit commitment – Ramping limits

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} u_{g,t-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$u_{gt}^{\mathrm{G}} = 1 \text{ and } u_{g,t-1}^{\mathrm{G}} = 0 \ (y_{gt}^{\mathrm{G}} = 1):$$

$$p_{gt}^{\mathrm{G}} - 0 \leq 0 + R_g^{\mathrm{G,SU}} \rightarrow p_{gt}^{\mathrm{G}} \leq R_g^{\mathrm{G,SU}}$$

Unit commitment – Ramping limits

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} u_{g,t-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$u_{gt}^{\mathrm{G}} = 1 \text{ and } u_{g,t-1}^{\mathrm{G}} = 1 \; (y_{gt}^{\mathrm{G}} = 0):$$

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} + 0 \rightarrow p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}}$$

Unit commitment – Ramping limits

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} u_{g,t-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$p_{gt}^{\mathrm{G}} - P_g^{\mathrm{G,0}} \leq R_g^{\mathrm{G,U}} U_g^{\mathrm{G,0}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, t = 1$$

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} u_{g,t-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, \forall t \in \Omega^{\mathrm{T}} \backslash \{1\}$$

Unit commitment – Ramping limits

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, \forall t$$

Unit commitment – Ramping limits

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, \forall t$$

$u^{\mathrm{G}}_{gt} = 0$ and $u^{\mathrm{G}}_{g,t-1} = 0$ ($z^{\mathrm{G}}_{gt} = 0$):

$$0 - 0 \leq 0 + 0$$

Unit commitment – Ramping limits

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, \forall t$$

$$u^{\mathrm{G}}_{gt} = 0 \text{ and } u^{\mathrm{G}}_{g,t-1} = 1 \ (z^{\mathrm{G}}_{gt} = 1):$$

$$p^{\mathrm{G}}_{g,t-1} - 0 \leq 0 + R^{\mathrm{G,SD}}_{g} \rightarrow p^{\mathrm{G}}_{g,t-1} \leq R^{\mathrm{G,SD}}_{g}$$

Unit commitment – Ramping limits

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, \forall t$$

$$u^{\mathrm{G}}_{gt} = 1 \text{ and } u^{\mathrm{G}}_{g,t-1} = 0 \ (z^{\mathrm{G}}_{gt} = 0):$$

$$0 - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} + 0 \rightarrow -R^{\mathrm{G,D}}_{g} \leq p^{\mathrm{G}}_{gt}$$

Note that $p^{\mathrm{G}}_{gt} \geq -R^{\mathrm{G,D}}_{g}$ is an inactive constraint since it is less restrictive than $p^{\mathrm{G}}_{gt} \geq \underline{P}^{\mathrm{G}}_{g}$.

Unit commitment – Ramping limits

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, \forall t$$

$$u^{\mathrm{G}}_{gt} = 1 \text{ and } u^{\mathrm{G}}_{g,t-1} = 1 \; (z^{\mathrm{G}}_{gt} = 0):$$

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} + 0 \rightarrow p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g}$$

Unit commitment – Ramping limits

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, \forall t$$

$$P^{\mathrm{G,0}}_{g} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, t = 1$$

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, \forall t \in \Omega^{\mathrm{T}} \backslash \{1\}$$

Unit commitment – Power balance

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}}, \forall t$$

Unit commitment – Security constraints

Some possible security constraints:

- Total power capacity $\geq$ Maximum demand + reserve.
- Inertia constraints.

Unit commitment – Variables

$$\Phi = \left\{ p_{gt}^{\mathrm{G}}, u_{gt}^{\mathrm{G}}, y_{gt}^{\mathrm{G}}, z_{gt}^{\mathrm{G}}, \forall g, \forall t \right\}$$

Unit commitment – Formulation

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \sum_{g \in \Omega^{\mathrm{G}}} \left(C_g^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,SD}} z_{gt}^{\mathrm{G}} \right)$$

s.t.:

$$u_{gt}^{\mathrm{G}}, y_{gt}^{\mathrm{G}}, z_{gt}^{\mathrm{G}} \in \{0,1\}, \forall g, \forall t$$

$$y_{gt}^{\mathrm{G}} - z_{gt}^{\mathrm{G}} = u_{gt}^{\mathrm{G}} - u_{g,t-1}^{\mathrm{G}}, \forall g, \forall t$$

$$y_{gt}^{\mathrm{G}} + z_{gt}^{\mathrm{G}} \leq 1, \forall g, \forall t$$

$$\underline{P}_g^{\mathrm{G}} u_{gt}^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}} u_{gt}^{\mathrm{G}}, \forall g, \forall t$$

Unit commitment – Formulation

$$p_{gt}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{G,U}} u_{g,t-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} y_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$p_{g,t-1}^{\mathrm{G}} - p_{gt}^{\mathrm{G}} \leq R_g^{\mathrm{G,D}} u_{gt}^{\mathrm{G}} + R_g^{\mathrm{G,SD}} z_{gt}^{\mathrm{G}}, \forall g, \forall t$$

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}}, \forall t$$

$$\Phi = \left\{ p_{gt}^{\mathrm{G}}, u_{gt}^{\mathrm{G}}, y_{gt}^{\mathrm{G}}, z_{gt}^{\mathrm{G}}, \forall g, \forall t \right\}$$

Exercise 1 – Unit commitment

Solve the unit commitment problem using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_01.gms

Solution:

Objective function value = €162.

Network-constrained unit commitment

Network Constraints + Unit Commitment (UC)

=

Network-Constrained Unit Commitment (NCUC)

NCUC – Power flows

AC model → Not addressed in this course.

Lossless DC model → **Approximation**.

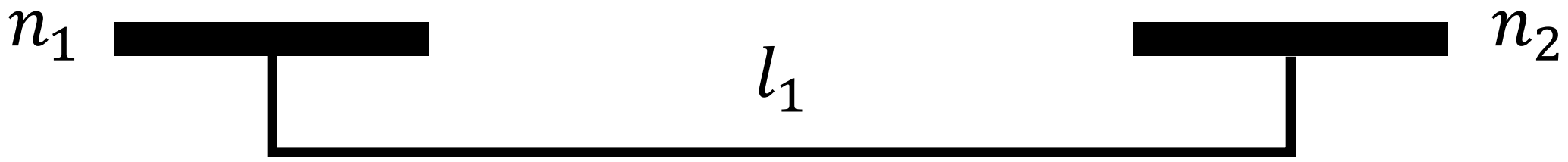
NCUC – Power flows (lossless DC model)

Assumptions of the lossless DC model (approximation):

1. Voltage magnitudes are equal to 1 in per unit.
2. Resistances are neglected.
3. Voltage angle differences across lines are small.

We must consider the per-unit system (modify the input data of the optimization problem accordingly)!

NCUC – Power flows (lossless DC model)



$$FR(l_1) = n_1$$

$$TO(l_1) = n_2$$

NCUC – Power flows (lossless DC model)

$$p_{lt}^{\mathrm{L}} = \frac{1}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right), \forall l, \forall t$$

NCUC – Power flows (lossless DC model)

$$p_{lt}^{\mathrm{L}} = \frac{1}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right), \forall l, \forall t$$

Direction of the power flow defined: $FR(l) \Rightarrow TO(l)$.

- $p_l^{\mathrm{L}} > 0 \rightarrow FR(l) \Rightarrow TO(l)$.
- $p_l^{\mathrm{L}} < 0 \rightarrow TO(l) \Rightarrow FR(l)$.

NCUC – Power flow limits and reference

$$-\overline{P}_l^{\mathrm{L}} \leq p_{lt}^{\mathrm{L}} \leq \overline{P}_l^{\mathrm{L}}, \forall l, \forall t$$

$$\theta_{nt}^{\mathrm{N}} = 0, n{:}\, ref., \forall t$$

NCUC – Nodal power balance

$$\sum_{g \in \boldsymbol{\Omega}_n^{\mathrm{G}}} p_{gt}^{\mathrm{G}} - \sum_{\boldsymbol{l} \in \boldsymbol{\Omega}^{\mathrm{L}} | \boldsymbol{FR}(\boldsymbol{l}) = \boldsymbol{n}} \boldsymbol{p}_{\boldsymbol{lt}}^{\mathbf{L}} + \sum_{\boldsymbol{l} \in \boldsymbol{\Omega}^{\mathrm{L}} | \boldsymbol{TO}(\boldsymbol{l}) = \boldsymbol{n}} \boldsymbol{p}_{\boldsymbol{lt}}^{\mathbf{L}} = P_{\boldsymbol{nt}}^{\mathrm{D}}, \forall n, \forall t$$

NCUC – Variables

$$\Phi = \left\{ p_{gt}^{\mathrm{G}}, u_{gt}^{\mathrm{G}}, y_{gt}^{\mathrm{G}}, z_{gt}^{\mathrm{G}}, \forall g, \forall t; \boldsymbol{p_{lt}^{\mathrm{L}}, \forall l, \forall t; \theta_{nt}^{\mathrm{N}}, \forall n, \forall t} \right\}$$

NCUC – Formulation

$$\min_{\Phi} \sum_{t \in \Omega^{T}} \sum_{g \in \Omega^{G}} \left(C_g^{G,F} u_{gt}^{G} + C_g^{G,V} p_{gt}^{G} + C_g^{G,SU} y_{gt}^{G} + C_g^{G,SD} z_{gt}^{G} \right)$$

s.t.:

$$u_{gt}^{G}, y_{gt}^{G}, z_{gt}^{G} \in \{0,1\}, \forall g, \forall t$$

$$y_{gt}^{G} - z_{gt}^{G} = u_{gt}^{G} - u_{g,t-1}^{G}, \forall g, \forall t$$

$$y_{gt}^{G} + z_{gt}^{G} \leq 1, \forall g, \forall t$$

$$\underline{P}_g^{G} u_{gt}^{G} \leq p_{gt}^{G} \leq \overline{P}_g^{G} u_{gt}^{G}, \forall g, \forall t$$

$$p_{gt}^{G} - p_{g,t-1}^{G} \leq R_g^{G,U} u_{g,t-1}^{G} + R_g^{G,SU} y_{gt}^{G}, \forall g, \forall t$$

NCUC – Formulation

$$p^{\mathrm{G}}_{g,t-1} - p^{\mathrm{G}}_{gt} \leq R^{\mathrm{G,D}}_{g} u^{\mathrm{G}}_{gt} + R^{\mathrm{G,SD}}_{g} z^{\mathrm{G}}_{gt}, \forall g, \forall t$$

$$p^{\mathrm{L}}_{lt} = \frac{1}{X^{\mathrm{L}}_{l}} \left(\theta^{\mathrm{N}}_{FR(l),t} - \theta^{\mathrm{N}}_{TO(l),t} \right), \forall l, \forall t$$

$$-\overline{P}^{\mathrm{L}}_{l} \leq p^{\mathrm{L}}_{lt} \leq \overline{P}^{\mathrm{L}}_{l}, \forall l, \forall t$$

$$\theta^{\mathrm{N}}_{nt} = 0, n\text{: } ref., \forall t$$

$$\sum_{g \in \Omega^{\mathrm{G}}_{n}} p^{\mathrm{G}}_{gt} - \sum_{l \in \Omega^{\mathrm{L}} | FR(l)=n} p^{\mathrm{L}}_{lt} + \sum_{l \in \Omega^{\mathrm{L}} | TO(l)=n} p^{\mathrm{L}}_{lt} = P^{\mathrm{D}}_{nt}, \forall n, \forall t$$

$$\Phi = \left\{ p^{\mathrm{G}}_{gt}, u^{\mathrm{G}}_{gt}, y^{\mathrm{G}}_{gt}, z^{\mathrm{G}}_{gt}, \forall g, \forall t; p^{\mathrm{L}}_{lt}, \forall l, \forall t; \theta^{\mathrm{N}}_{nt}, \forall n, \forall t \right\}$$

Operation

Operation optimization problems solved by:

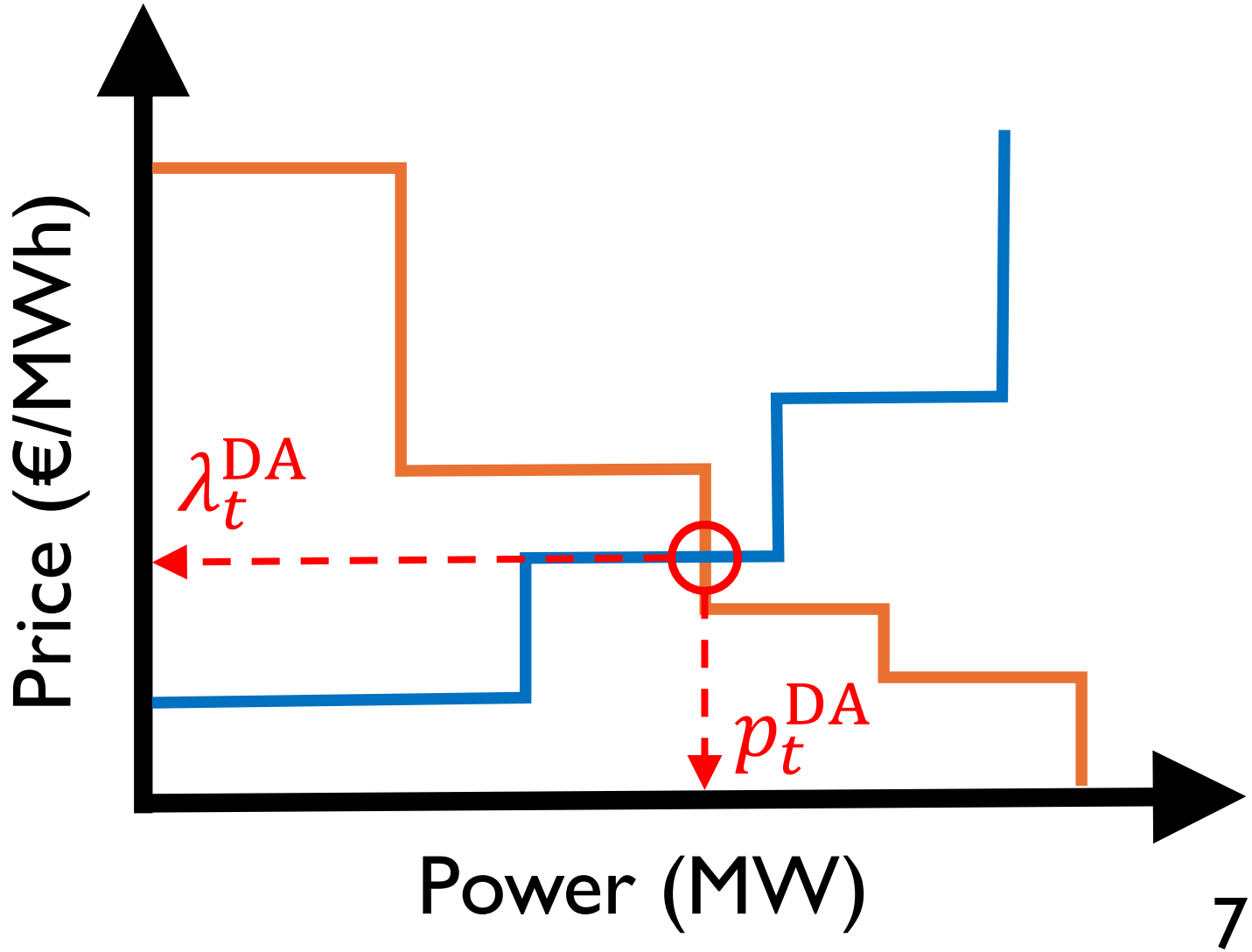
- A centralized entity → Unit commitment.
- Private entities → **Self-scheduling**.
- Cost function with intervals.

Self-scheduling

Viewpoint:

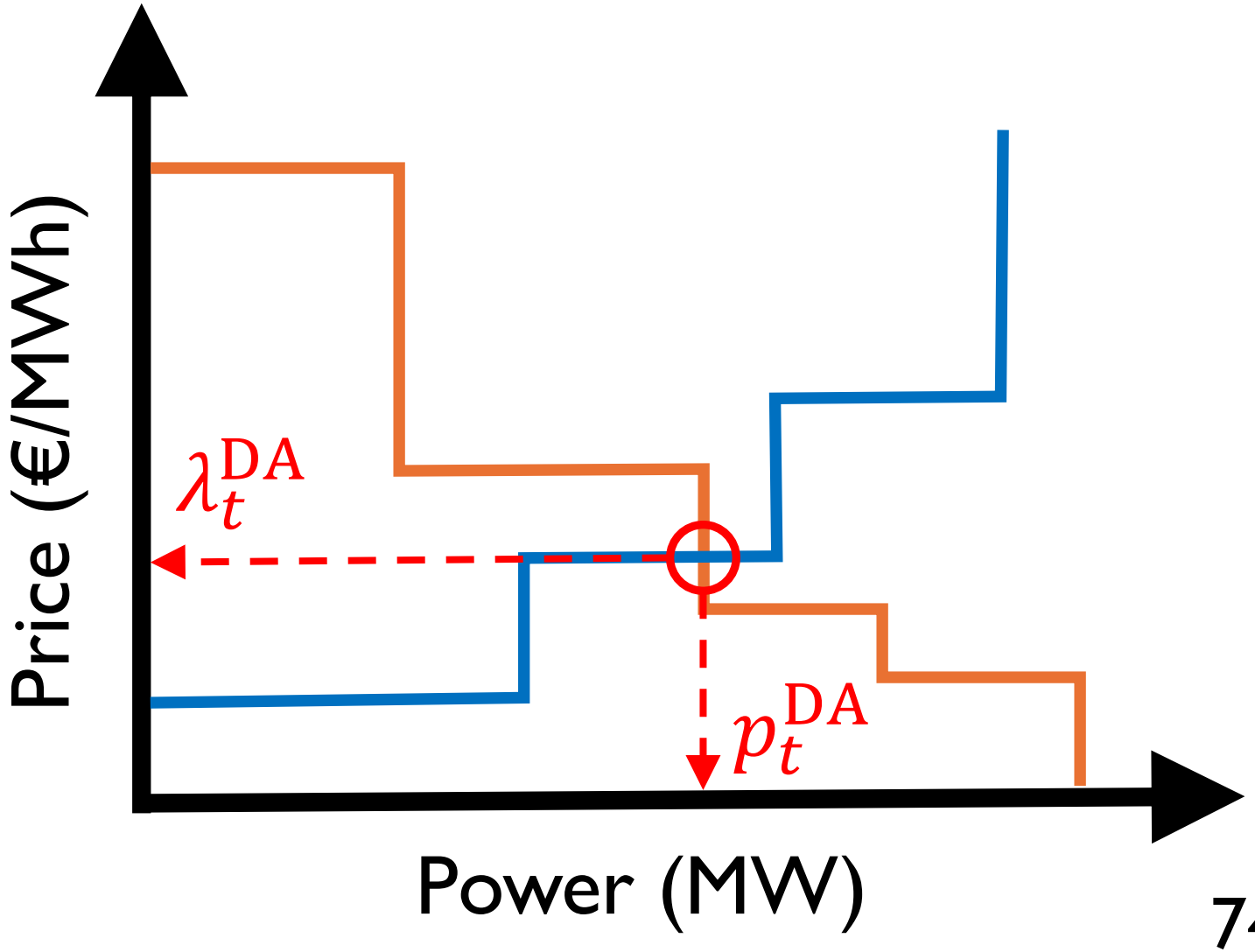
- Conventional producer.
- Inelastic and flexible demands.
- Renewable energy source & battery.
- Virtual power plant.

Price-taker vs price-maker



Price-taker vs price-maker

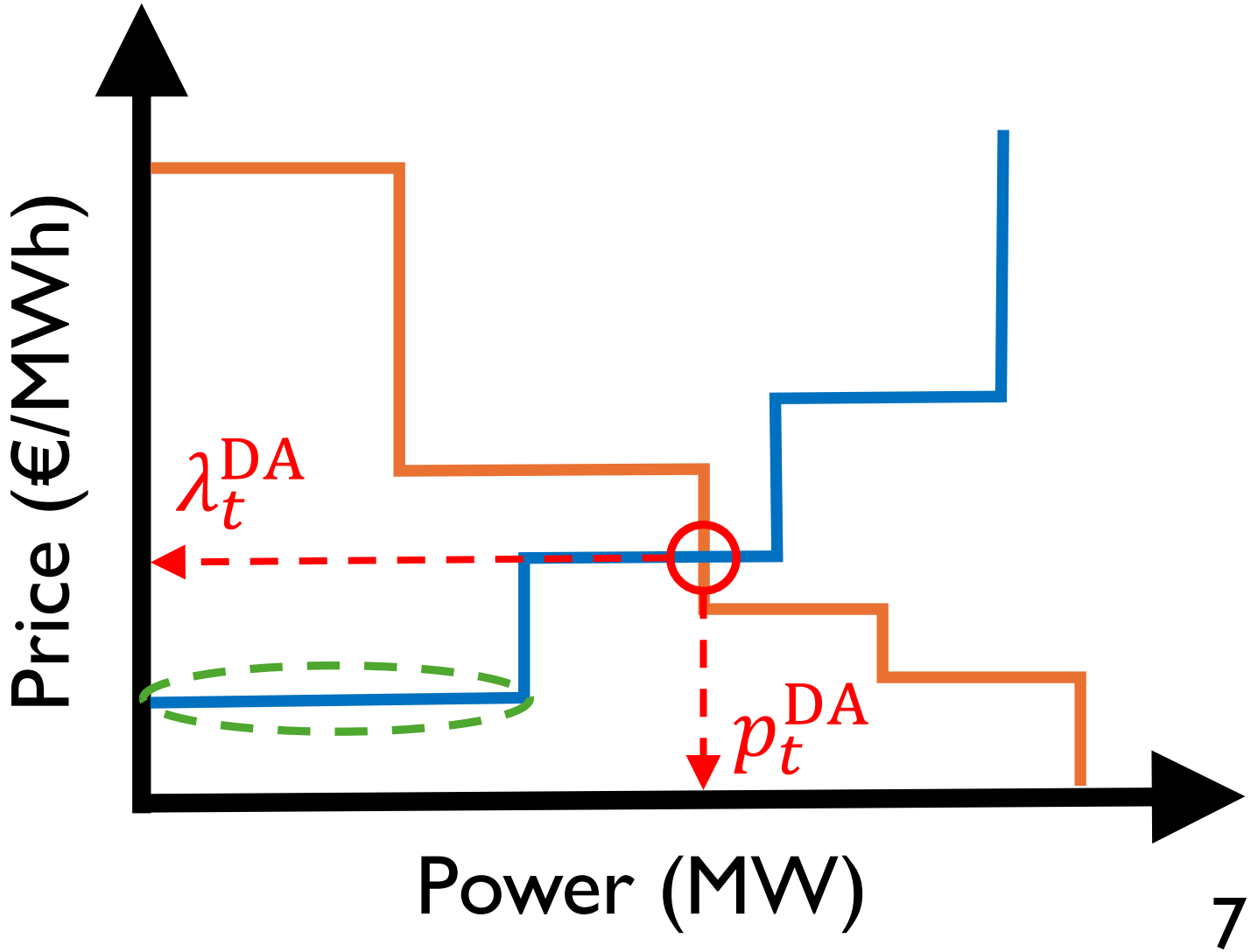
Price-taker: the participant cannot influence λ_t^{DA}.



Price-taker vs price-maker

Price-taker: the participant cannot influence λ_t^{DA}.

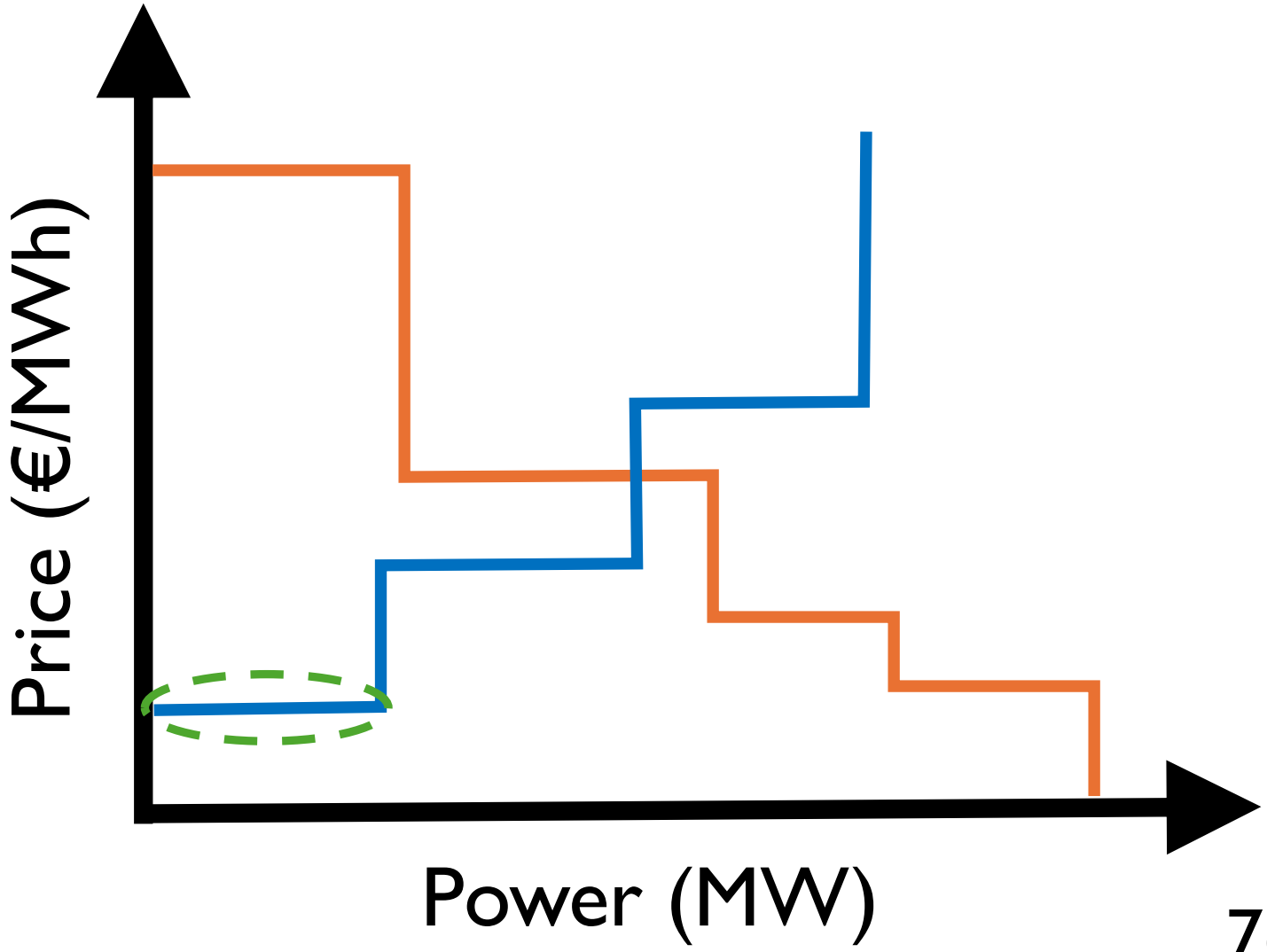
Price-maker: the participant can influence λ_t^{DA}.



Price-taker vs price-maker

Price-taker: the participant cannot influence λ_t^{DA}.

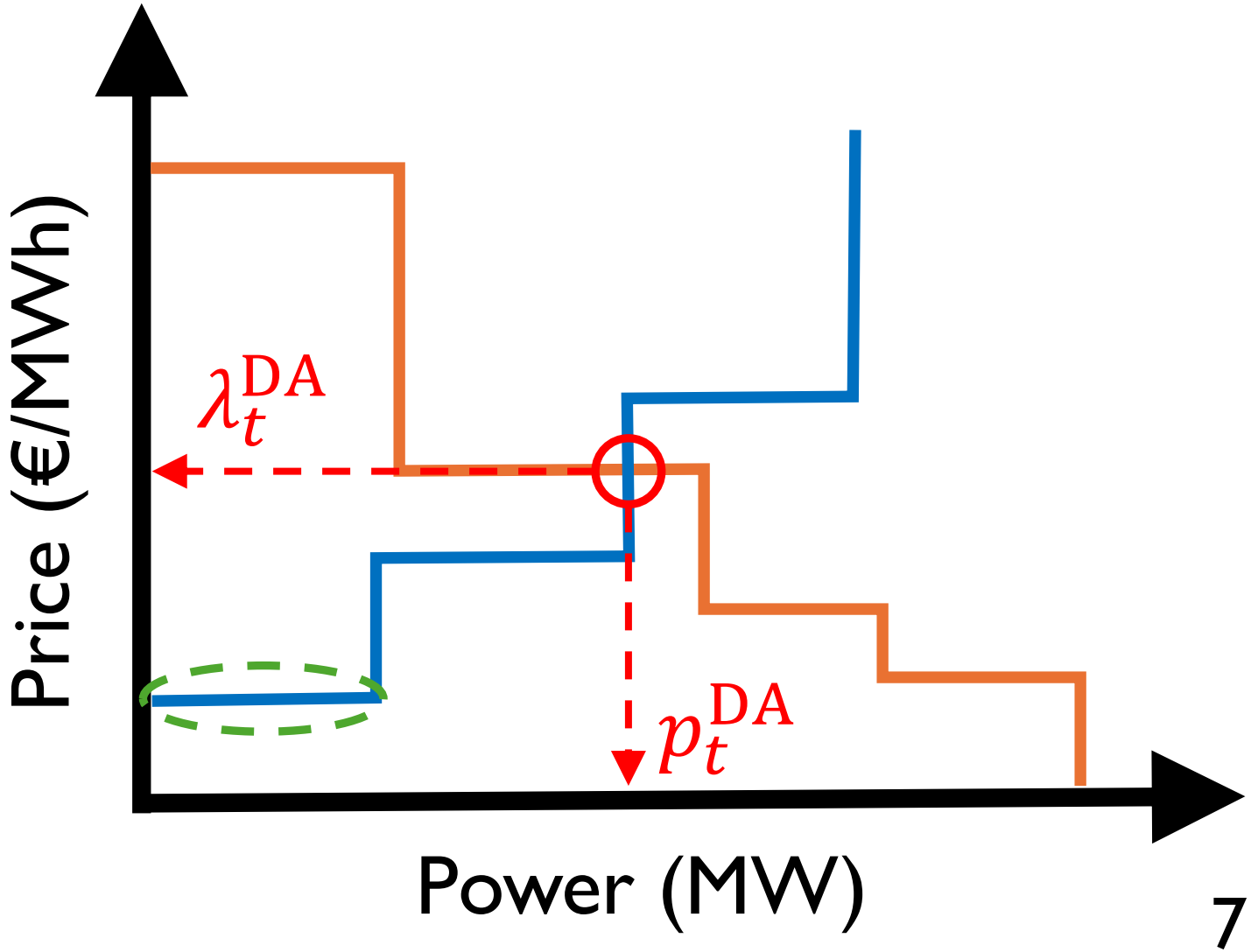
Price-maker: the participant can influence λ_t^{DA}.



Price-taker vs price-maker

Price-taker: the participant cannot influence λ_t^{DA}.

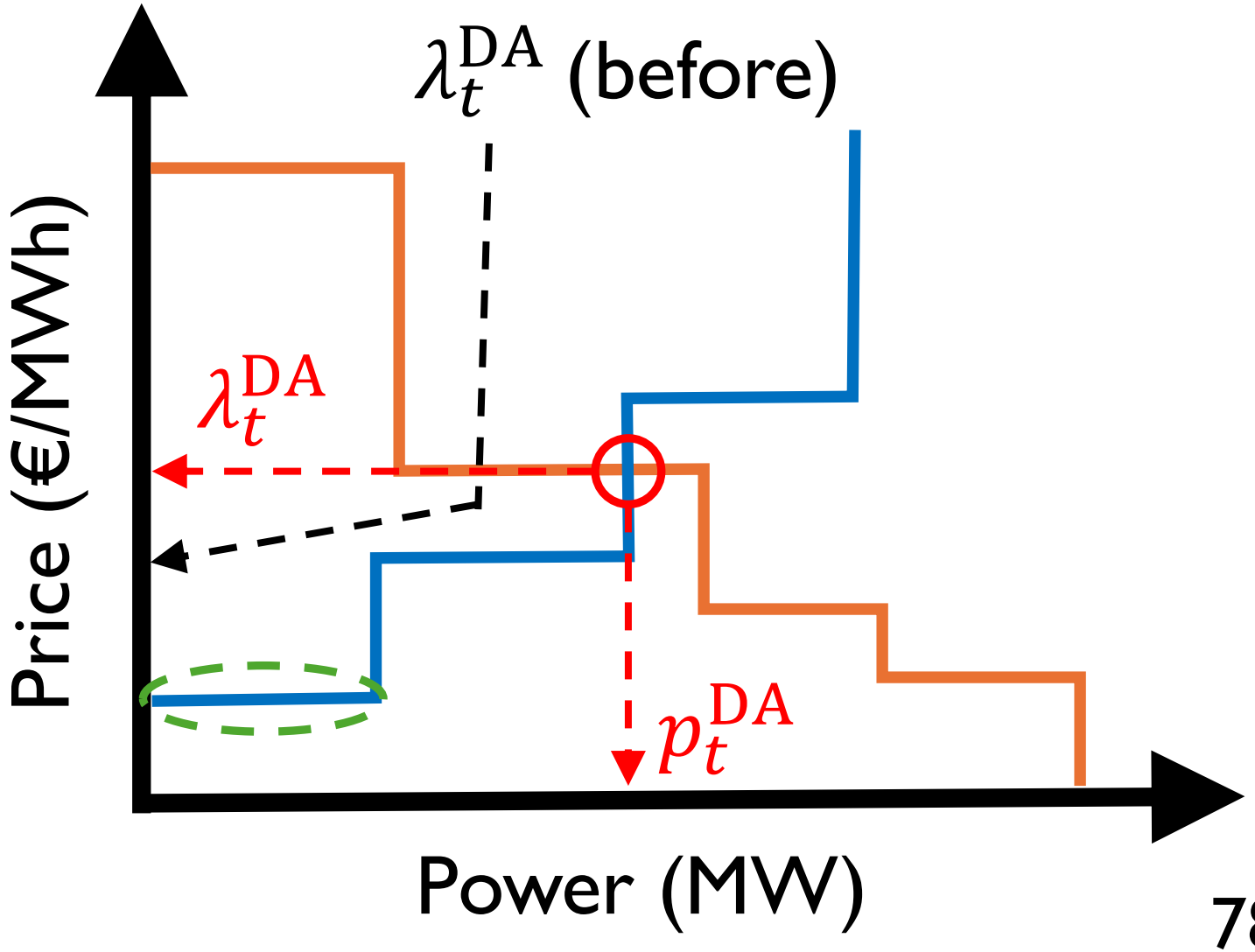
Price-maker: the participant can influence λ_t^{DA}.



Price-taker vs price-maker

Price-taker: the participant cannot influence λ_t^{DA}.

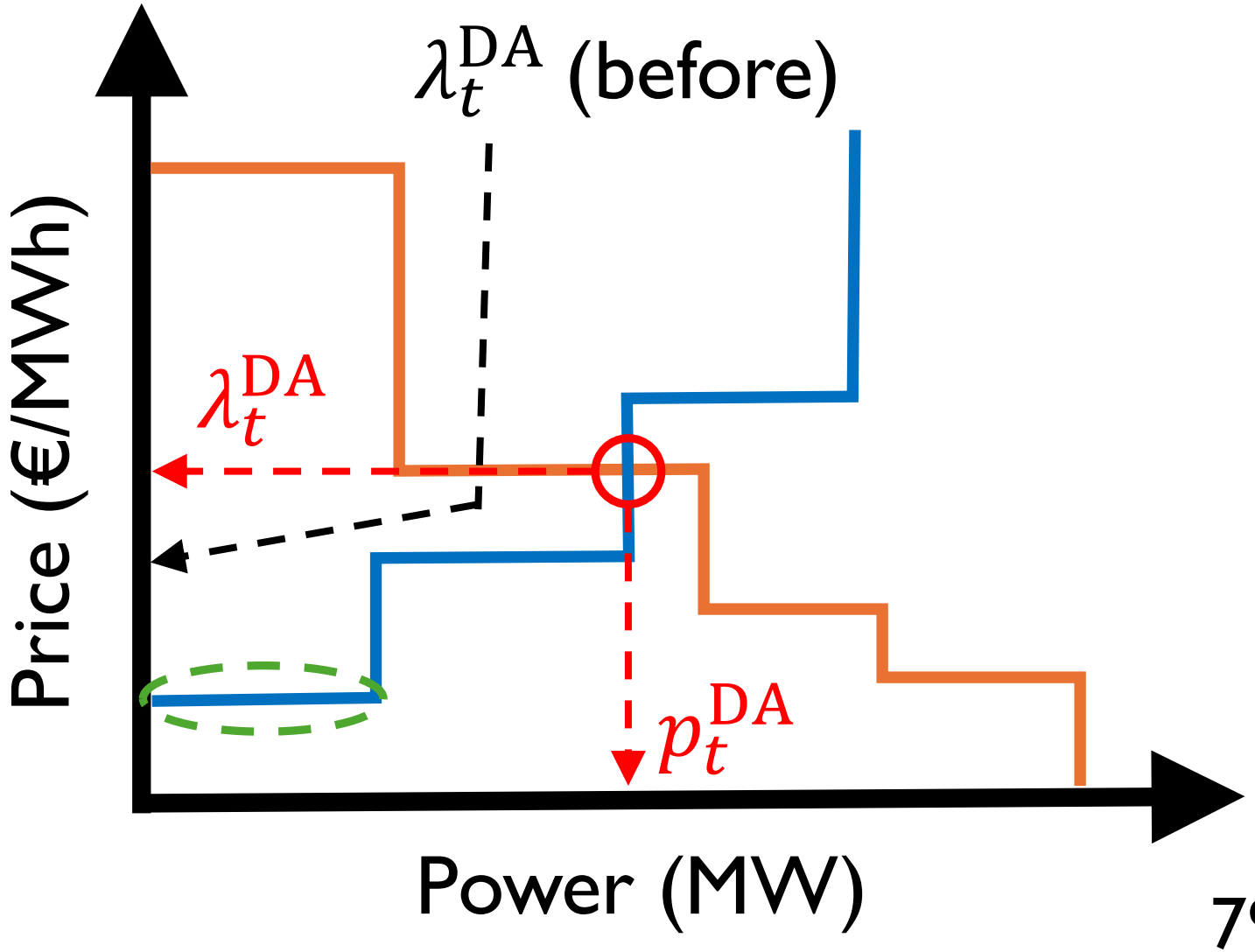
Price-maker: the participant can influence λ_t^{DA}.



Price-taker vs price-maker

Price-taker: the participant cannot influence λ_t^{DA}.

Price-maker: the participant can influence λ_t^{DA}.



Self-scheduling

Viewpoint:

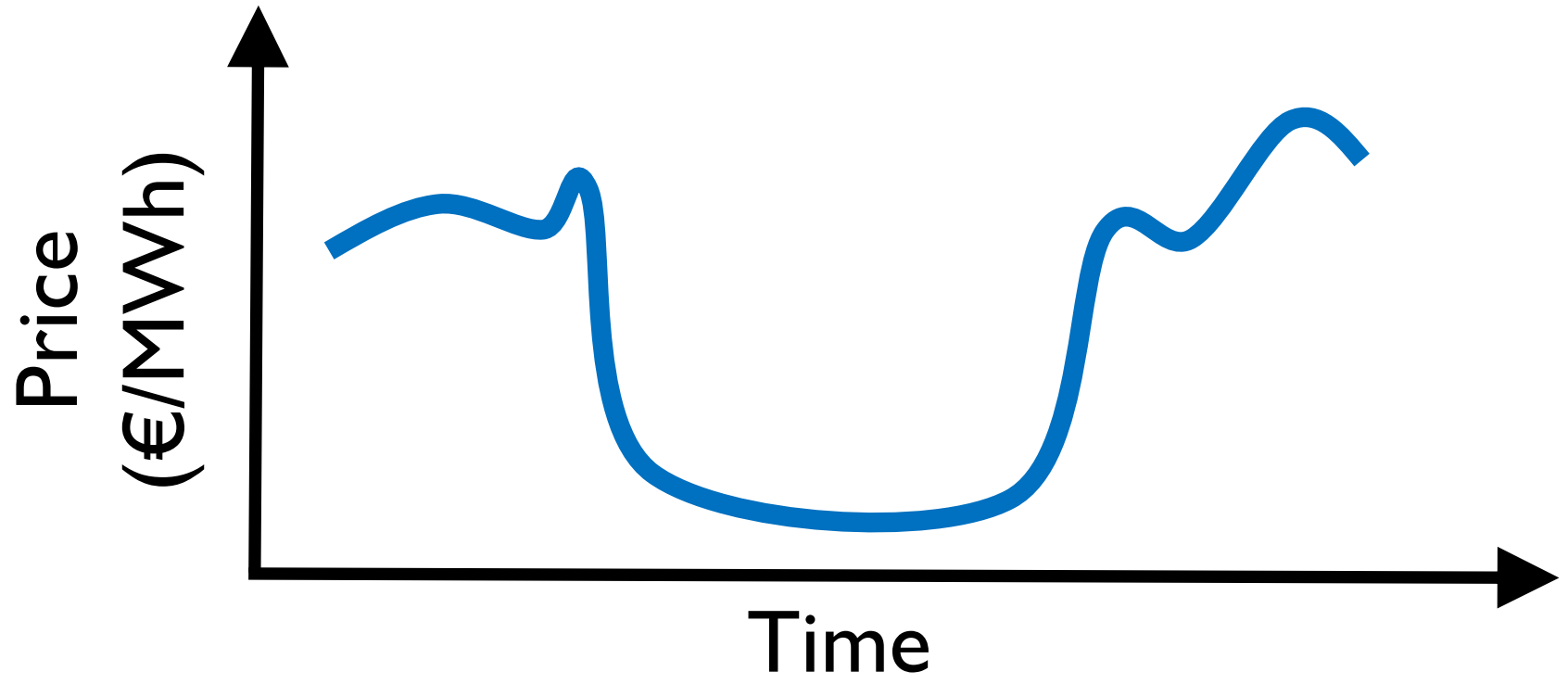
- **Conventional producer**.
- Inelastic and flexible demands.
- Renewable energy source & battery.
- Virtual power plant.

Self-scheduling – Conventional producer

$$\max_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \left(\lambda_t^{\mathrm{DA,S}} p_t^{\mathrm{DA,S}} - \left(C^{\mathrm{G,F}} u_t^{\mathrm{G}} + C^{\mathrm{G,V}} p_t^{\mathrm{G}} + C^{\mathrm{G,SU}} y_t^{\mathrm{G}} + C^{\mathrm{G,SD}} z_t^{\mathrm{G}} \right) \right)$$

Self-scheduling – Conventional producer

The self-scheduling is not trivial since the selling day-ahead electricity market price $\lambda_t^{\mathrm{DA},S}$ depends on the time period.



Self-scheduling – Conventional producer



OMIE (market operator of the wholesale electricity spot markets for Spain and Portugal): https://www.omie.es/en

Self-scheduling – Conventional producer

$$p_t^{\mathrm{DA,S}} = p_t^{\mathrm{G}}, \forall t$$

$$0 \leq p_t^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$$

Constraints of the conventional generating unit (status, ramping limits, etc.)

Self-scheduling – Conventional producer

$$\max_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \lambda_t^{\mathrm{DA,S}} p_t^{\mathrm{DA,S}} - \left(C^{\mathrm{G,F}} u_t^{\mathrm{G}} + C^{\mathrm{G,V}} p_t^{\mathrm{G}} + C^{\mathrm{G,SU}} y_t^{\mathrm{G}} + C^{\mathrm{G,SD}} z_t^{\mathrm{G}} \right)$$

s.t.:

$$p_t^{\mathrm{DA,S}} = p_t^{\mathrm{G}}, \forall t$$

$$0 \leq p_t^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$$

Constraints of the conventional generating unit (status, ramping limits, etc.)

$$\Phi = \left\{ p_t^{\mathrm{DA,S}}, p_t^{\mathrm{G}}, u_t^{\mathrm{G}}, y_t^{\mathrm{G}}, z_t^{\mathrm{G}}, \forall t \right\}$$

Self-scheduling – Conventional producer

$$\max_{\Phi} \sum_{t \in \Omega^{T}} \lambda_t^{DA,S} \boldsymbol{p_t^G} - \left(C^{G,F} u_t^G + C^{G,V} p_t^G + C^{G,SU} y_t^G + C^{G,SD} z_t^G\right)$$

s.t.:

$$0 \leq \boldsymbol{p_t^G} \leq \overline{P}^{DA,S}, \forall t$$

Constraints of the conventional generating unit (status, ramping limits, etc.)

$$\Phi = \{p_t^G, u_t^G, y_t^G, z_t^G, \forall t\}$$

Self-scheduling

Viewpoint:

- Conventional producer.
- **Inelastic and flexible demands**.
- Renewable energy source & battery.
- Virtual power plant.

Self-scheduling – Inelastic and flexible demands

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \lambda_t^{\mathrm{DA,P}} p_t^{\mathrm{DA,P}}$$

$$p_t^{\mathrm{DA,P}} = p_t^{\mathrm{D}}, \forall t$$

$$0 \leq p_t^{\mathrm{DA,P}} \leq \overline{P}^{\mathrm{DA,P}}, \forall t$$

Self-scheduling – Inelastic and flexible demands

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \lambda_t^{\mathrm{DA,P}} p_t^{\mathrm{D}}$$

$$0 \leq p_t^{\mathrm{D}} \leq \overline{P}^{\mathrm{DA,P}}, \forall t$$

Self-scheduling – Inelastic and flexible demands

$$p_t^{\mathrm{D}} = P_t^{\mathrm{D,I}} + p_t^{\mathrm{D,F}}, \forall t$$

$$\underline{P}_t^{\mathrm{D,F}} \leq p_t^{\mathrm{D,F}} \leq \overline{P}_t^{\mathrm{D,F}}, \forall t$$

Self-scheduling – Inelastic and flexible demands

$$p_t^{\mathrm{D,F}} - P^{\mathrm{D,F,0}} \leq R^{\mathrm{D,U}}, \forall t = 1$$

$$p_t^{\mathrm{D,F}} - p_{t-1}^{\mathrm{D,F}} \leq R^{\mathrm{D,U}}, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

$$P^{\mathrm{D,F,0}} - p_t^{\mathrm{D,F}} \leq R^{\mathrm{D,D}}, \forall t = 1$$

$$p_{t-1}^{\mathrm{D,F}} - p_t^{\mathrm{D,F}} \leq R^{\mathrm{D,D}}, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

Self-scheduling – Inelastic and flexible demands

Assume that $P^{\mathrm{D,F},0}$ is equal to $\underline{P}_{t=1}^{\mathrm{D,F}}$.

What is the optimal solution of this problem?

Self-scheduling – Inelastic and flexible demands

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \lambda_t^{\mathrm{DA,P}} p_t^{\mathrm{D}}$$

Self-scheduling – Inelastic and flexible demands

Assume that $P^{\mathrm{D,F},0}$ is equal to $\underline{P}_{t=1}^{\mathrm{D,F}}$.

What is the optimal solution of this problem?

$p_t^{\mathrm{D,F}^*} = \underline{P}_t^{\mathrm{D,F}}\ \forall t$ because we are minimizing costs.

Self-scheduling – Inelastic and flexible demands

Flexible demands typically have the following constraint:

$$\sum_{t \in \Omega^{\mathrm{T}}} p_t^{\mathrm{D,F}} \geq \underline{E}^{\mathrm{D,F}}$$

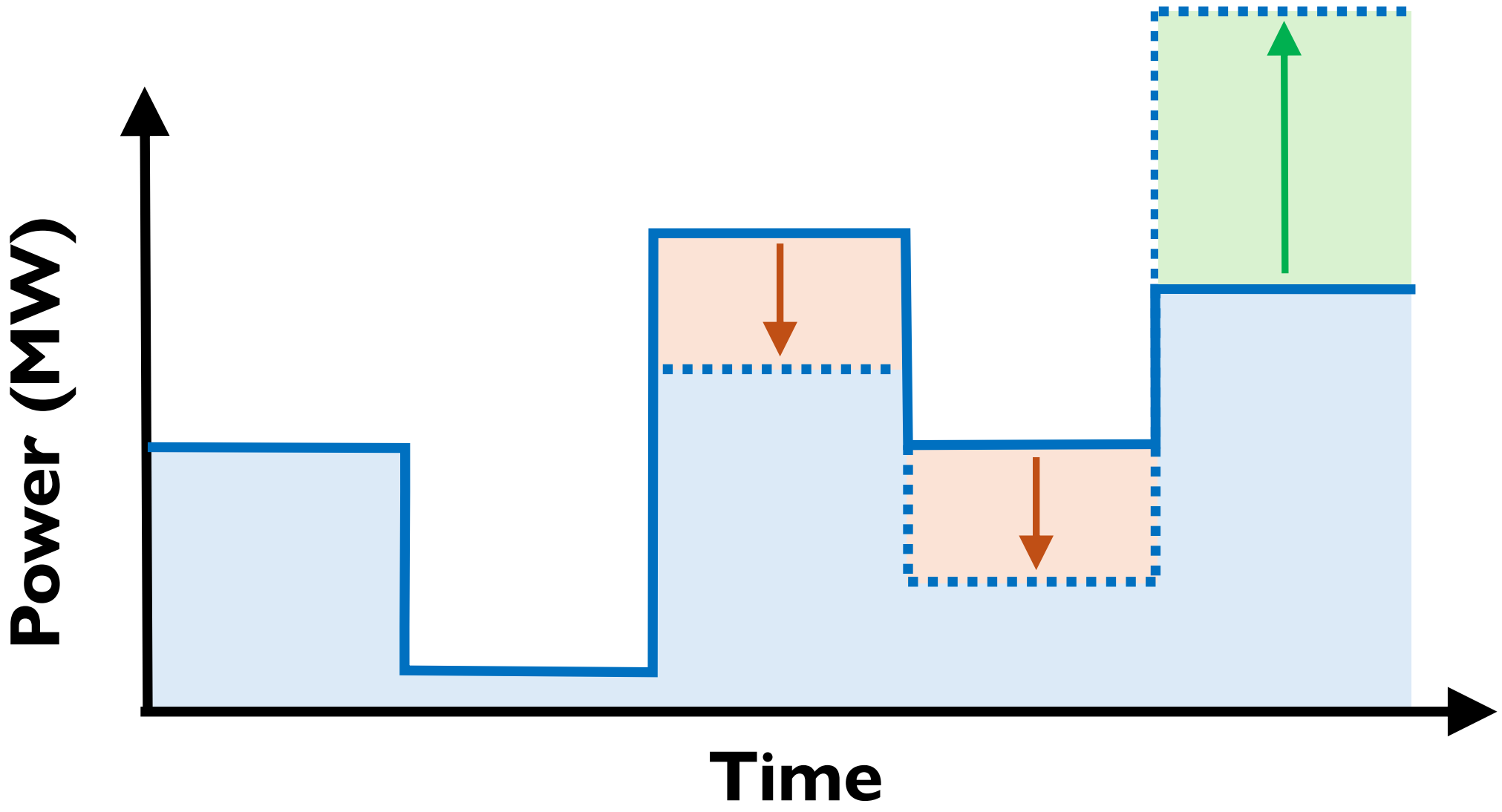
Self-scheduling – Inelastic and flexible demands

Flexible demands typically have the following constraint:

$$\sum_{t \in \Omega^{\mathrm{T}}} p_t^{\mathrm{D,F}} \geq \underline{E}^{\mathrm{D,F}}$$

"I don't care when the dishwasher is turned on, but I need my favorite cup clean for tomorrow's breakfast."

Self-scheduling – Inelastic and flexible demands



Self-scheduling – Inelastic and flexible demands

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \lambda_t^{\mathrm{DA,P}} p_t^{\mathrm{D}}$$

s.t.:

$$0 \leq p_t^{\mathrm{D}} \leq \overline{P}^{\mathrm{DA,P}}, \forall t$$

$$p_t^{\mathrm{D}} = P_t^{\mathrm{D,I}} + p_t^{\mathrm{D,F}}, \forall t$$

$$\underline{P}_t^{\mathrm{D,F}} \leq p_t^{\mathrm{D,F}} \leq \overline{P}_t^{\mathrm{D,F}}, \forall t$$

$$p_t^{\mathrm{D,F}} - p_{t-1}^{\mathrm{D,F}} \leq R^{\mathrm{D,U}}, \forall t$$

$$p_{t-1}^{\mathrm{D,F}} - p_t^{\mathrm{D,F}} \leq R^{\mathrm{D,D}}, \forall t$$

$$\sum_{t \in \Omega^{\mathrm{T}}} p_t^{\mathrm{D,F}} \geq \underline{E}^{\mathrm{D,F}}$$

$$\Phi = \left\{ p_t^{\mathrm{D}}, \forall t \right\}$$

Exercise 2 – Self-scheduling – Flexible demand

Solve the self-scheduling problem of a flexible demand using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_02.gms

Solution:

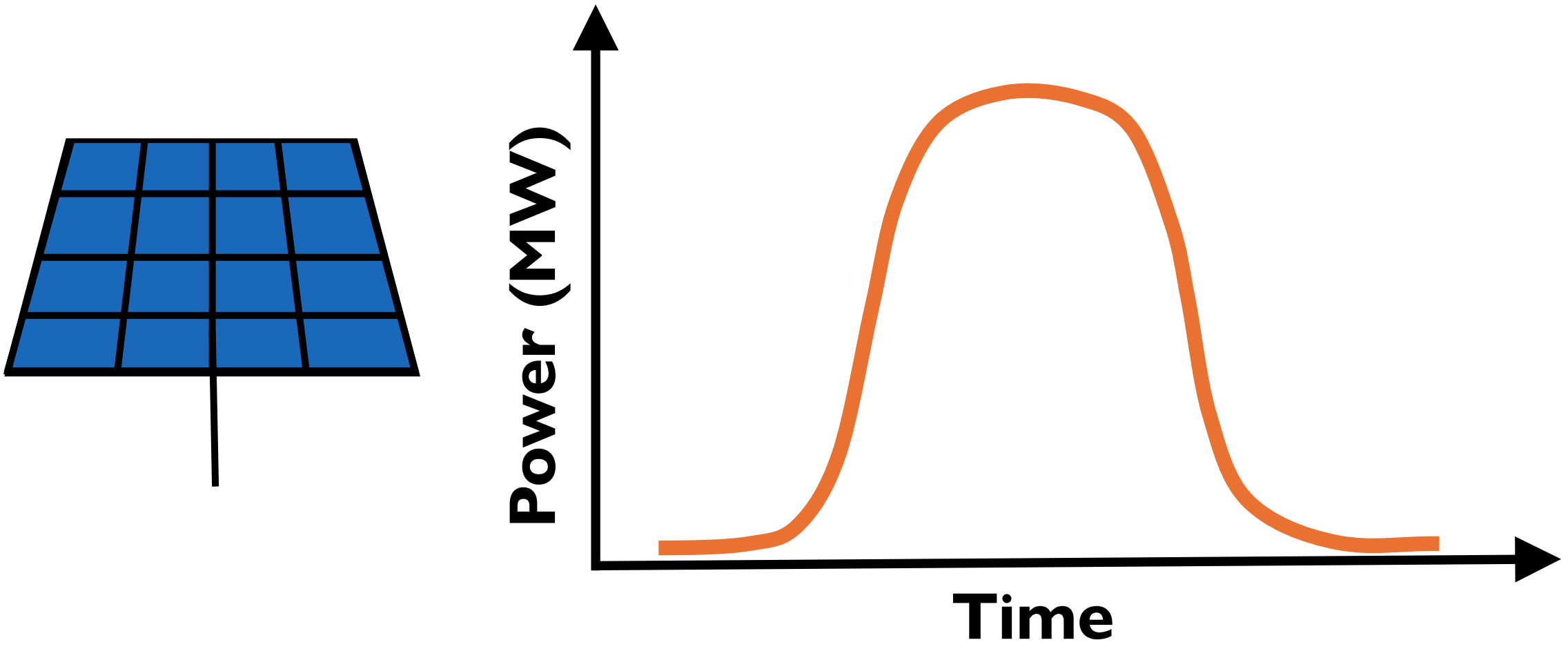
Objective function value = €8580.5.

Self-scheduling

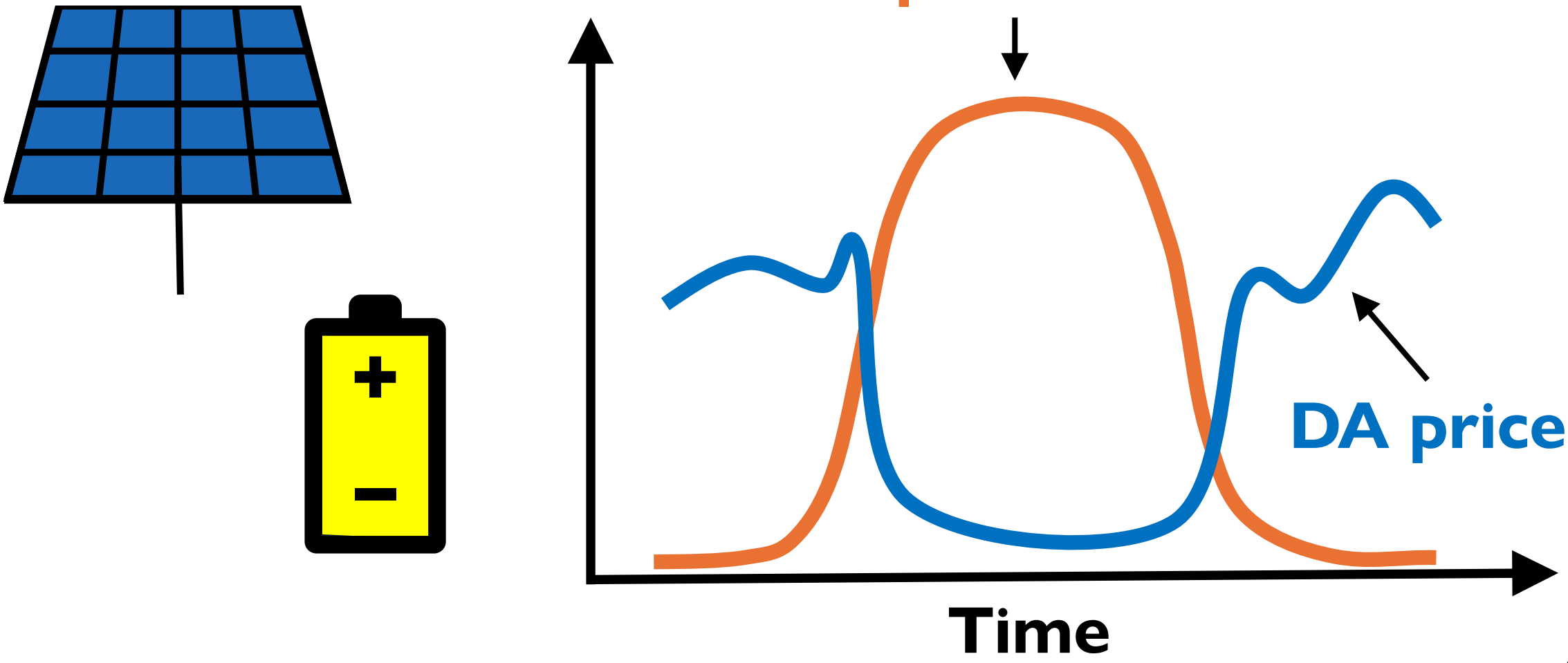
Viewpoint:

- Conventional producer.
- Inelastic and flexible demands.
- **Renewable energy source (RES) & battery**.
- Virtual power plant.

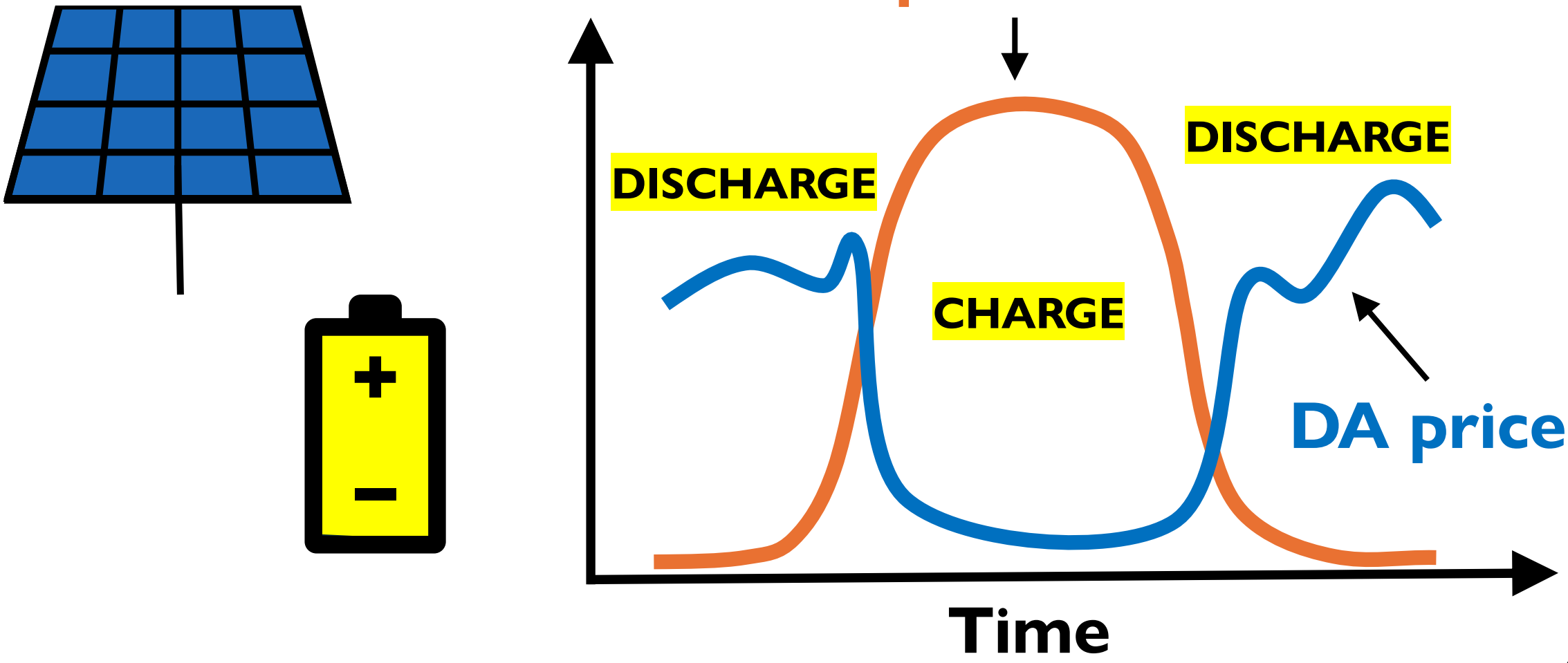
Self-scheduling – RES & Battery



Self-scheduling – RES & Battery



Self-scheduling – RES & Battery



Self-scheduling – RES & Battery

$$\max_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \left(\lambda_t^{\mathrm{DA,S}} p_t^{\mathrm{DA,S}} - C^{\mathrm{PV,V}} p_t^{\mathrm{PV}} \right)$$

Note that $C^{\mathrm{PV,V}}$ is considered to be equal to 0 in many works from the literature.

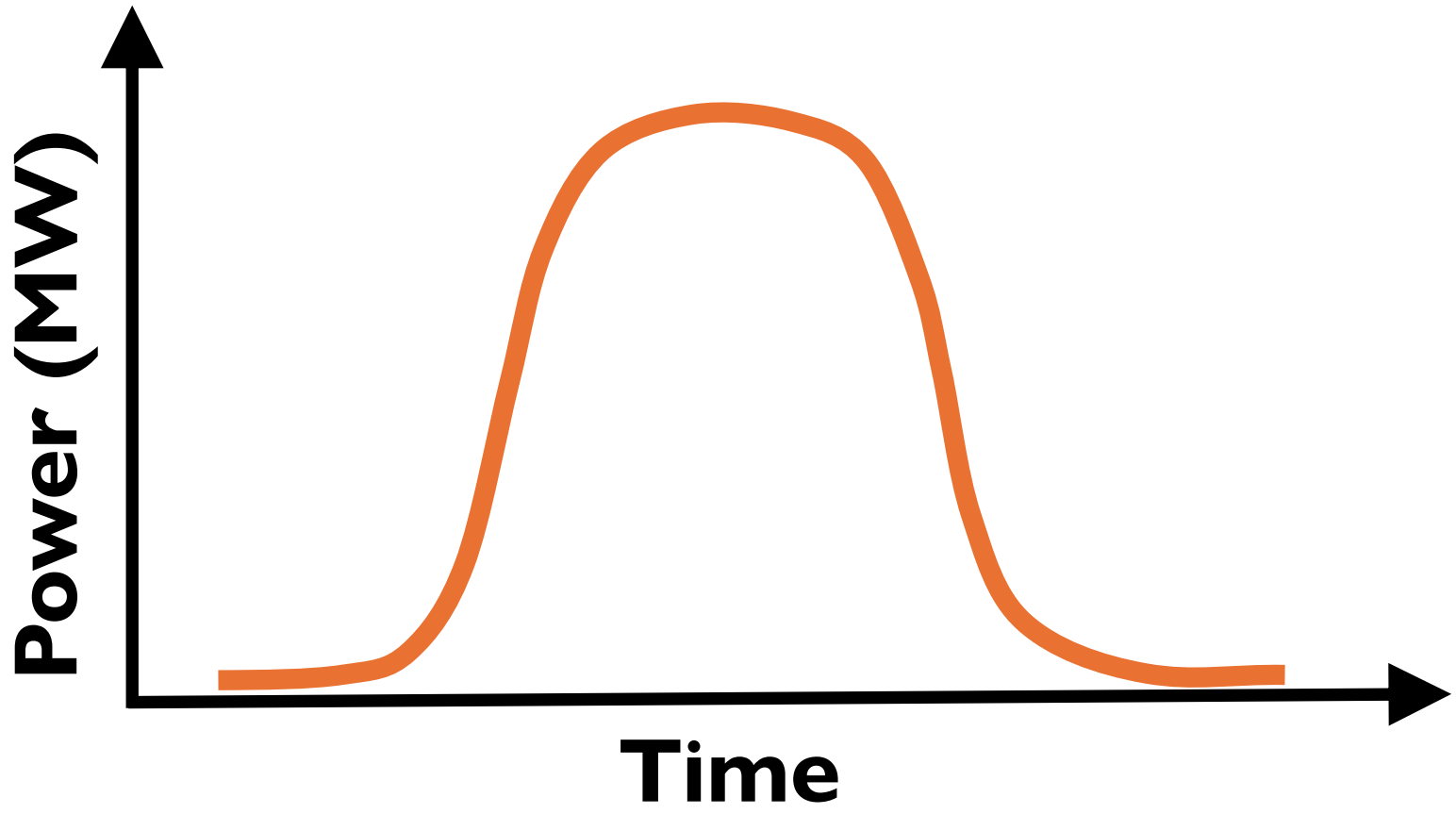
Self-scheduling – RES & Battery

$$p_t^{\mathrm{DA,S}} = p_t^{\mathrm{PV}} + p_t^{\mathrm{B,D}} - p_t^{\mathrm{B,C}}, \forall t$$

$$0 \leq p_t^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$$

Self-scheduling – RES & Battery

$$0 \leq p_t^{\mathrm{PV}} \leq \overline{P}^{\mathrm{PV}} A_t^{\mathrm{PV}}, \forall t$$



Self-scheduling – RES & Battery

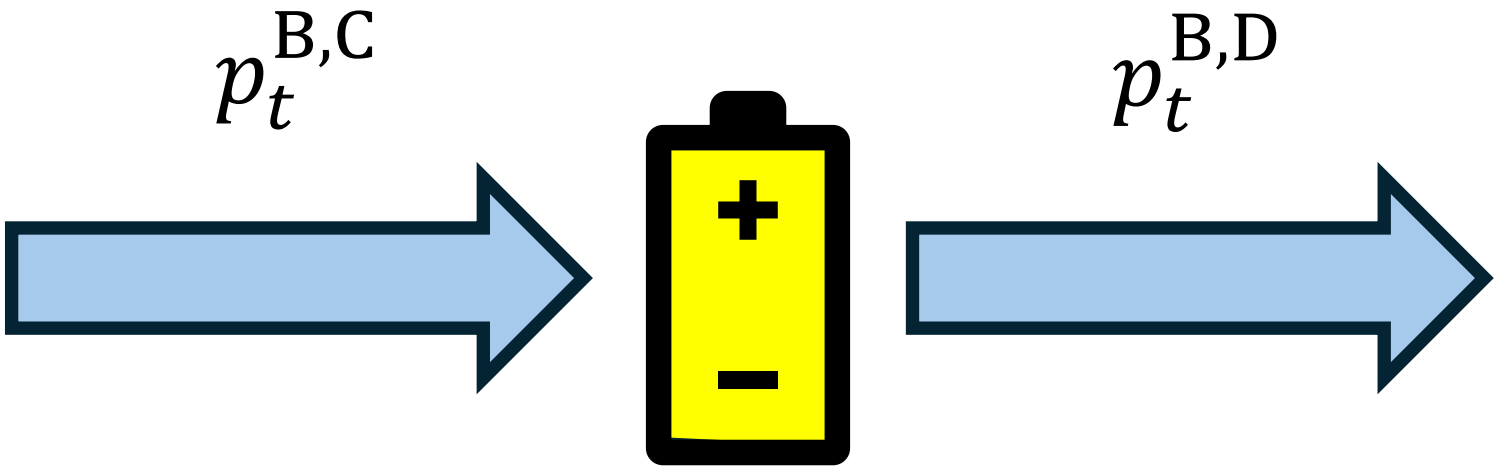
$$e_t^{\mathrm{B}} = \text{previous state + charging – discharging}, \forall t$$

Self-scheduling – RES & Battery

$$e_t^{\mathrm{B}} = E^{\mathrm{B},0} + \text{charging} - \text{discharging}\,, t = 1$$

$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + \text{charging} - \text{discharging}\,, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

Self-scheduling – RES & Battery

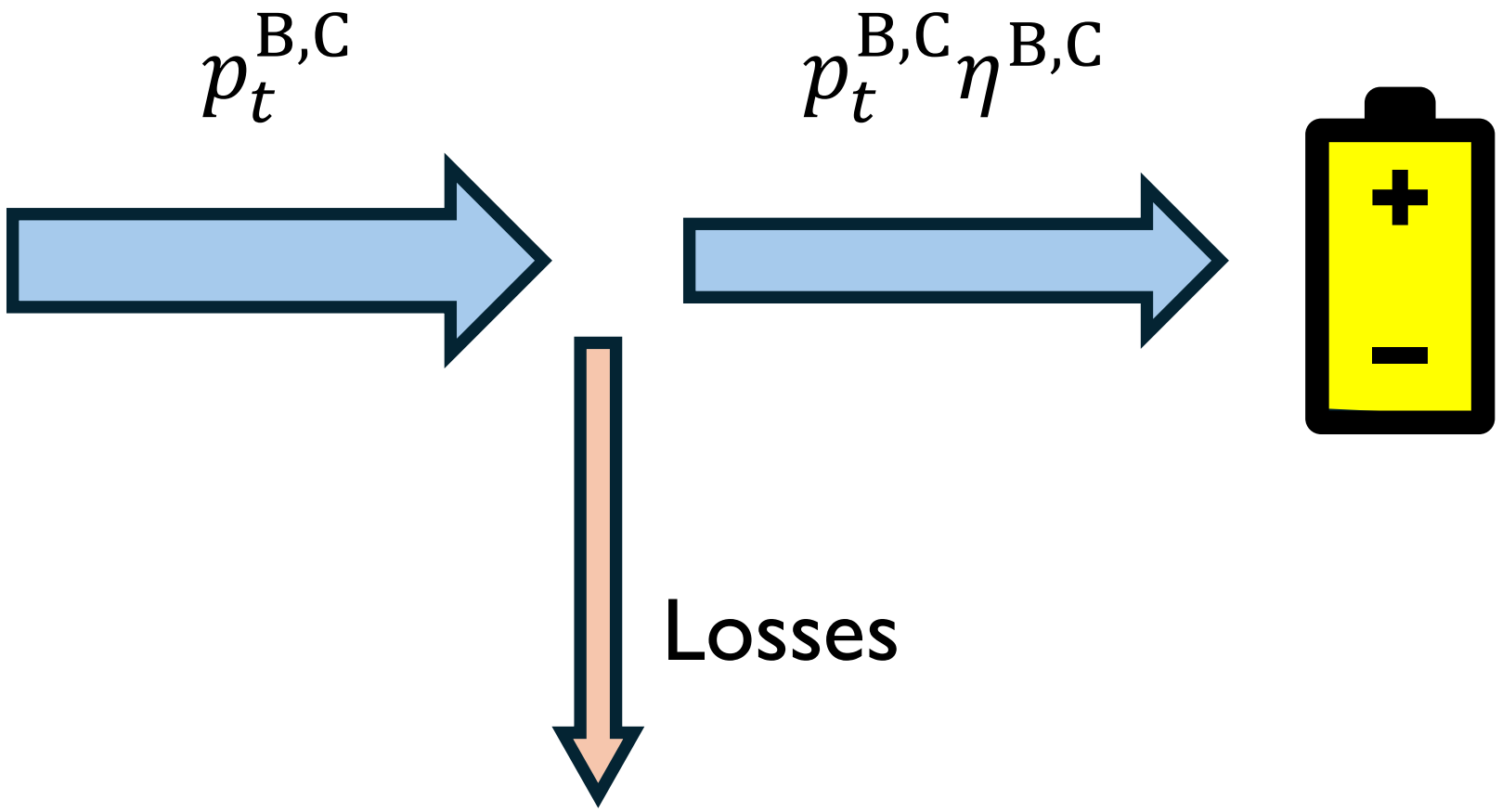


Self-scheduling – RES & Battery

$$e_t^{\mathrm{B}} = E^{\mathrm{B},0} + p_t^{\mathrm{B,C}} - p_t^{\mathrm{B,D}}\,, t = 1$$

$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + p_t^{\mathrm{B,C}} - p_t^{\mathrm{B,D}}\,, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

Self-scheduling – RES & Battery



Self-scheduling – RES & Battery

$$e_t^{\mathrm{B}} = E^{\mathrm{B},0} + p_t^{\mathrm{B,C}}\eta^{\mathrm{B,C}} - p_t^{\mathrm{B,D}}\eta^{\mathrm{B,D}}, t = 1$$

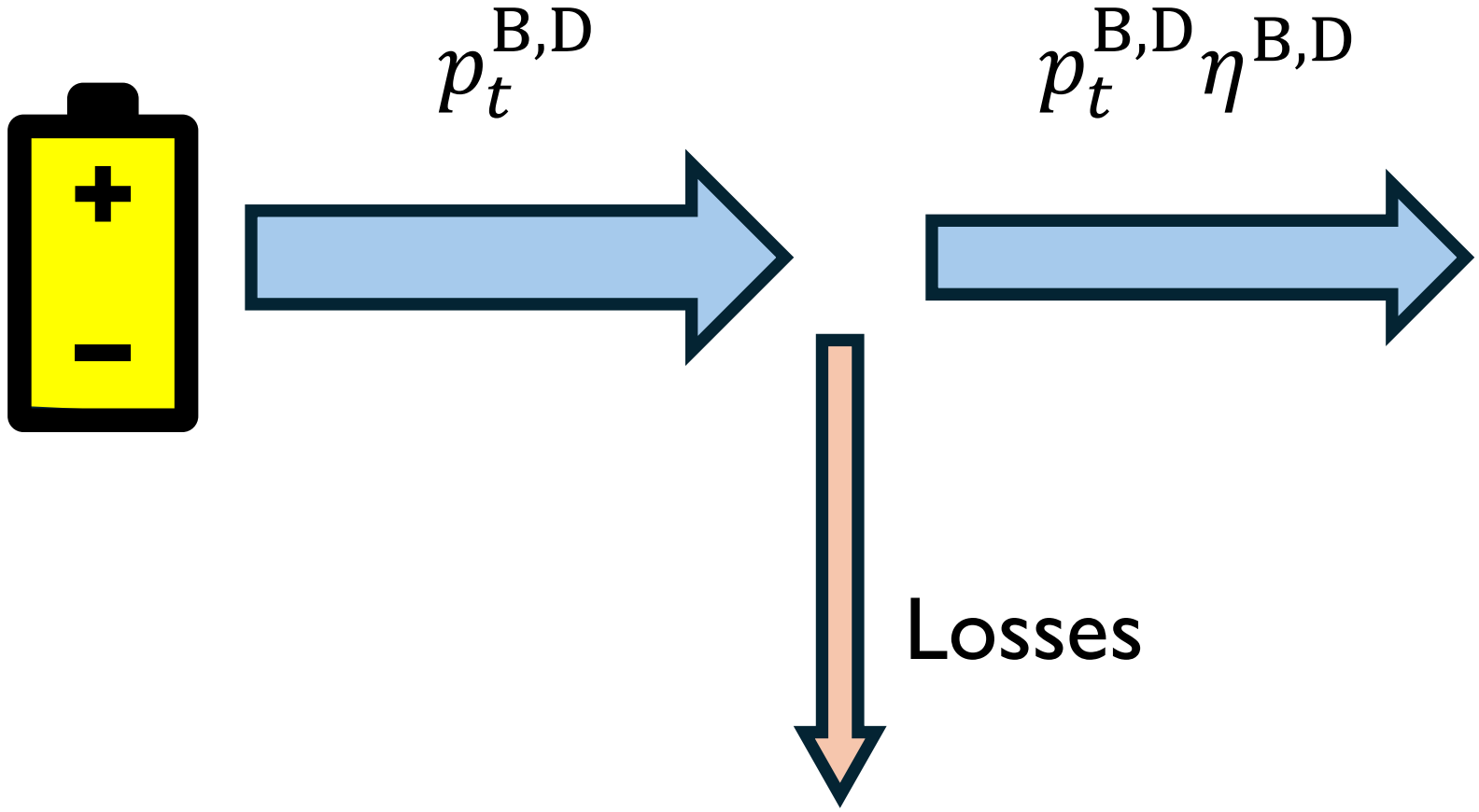
$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + p_t^{\mathrm{B,C}}\eta^{\mathrm{B,C}} - p_t^{\mathrm{B,D}}\eta^{\mathrm{B,D}}, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

Self-scheduling – RES & Battery

$$e_t^{\mathrm{B}} = E^{\mathrm{B},0} + p_t^{\mathrm{B,C}}\eta^{\mathrm{B,C}} - \boldsymbol{p_t^{\mathrm{B,D}}\eta^{\mathrm{B,D}}}, t = 1$$

$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + p_t^{\mathrm{B,C}}\eta^{\mathrm{B,C}} - \boldsymbol{p_t^{\mathrm{B,D}}\eta^{\mathrm{B,D}}}, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

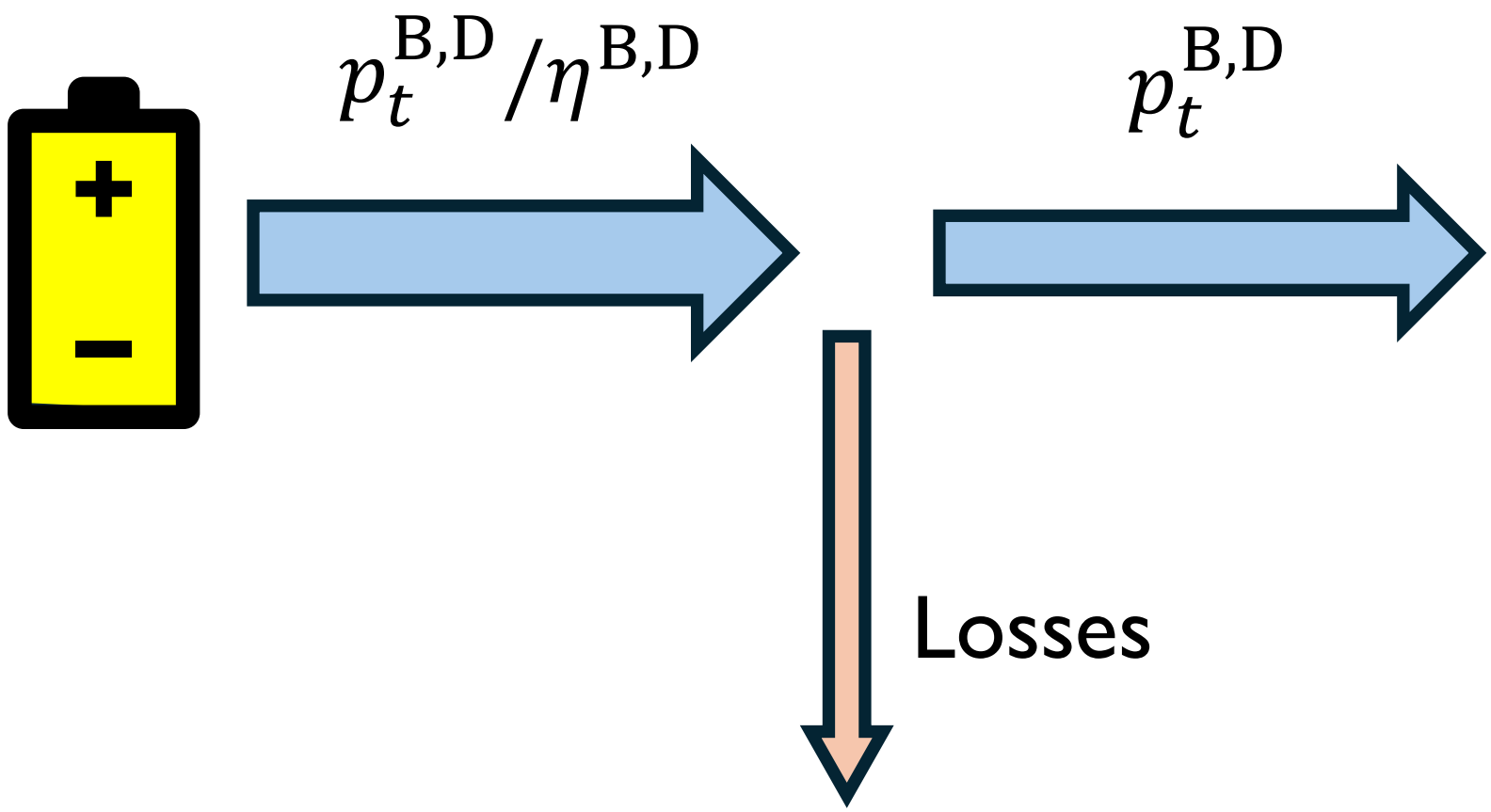
Self-scheduling – RES & Battery



Self-scheduling – RES & Battery

$$p_t^{\mathrm{DA,S}} - p_t^{\mathrm{DA,P}} = p_t^{\mathrm{PV}} + \boldsymbol{p_t^{\mathrm{B,D}}} - p_t^{\mathrm{B,C}}, \forall t$$

Self-scheduling – RES & Battery



Self-scheduling – RES & Battery

$$e_t^{\mathrm{B}} = E^{\mathrm{B},0} + p_t^{\mathrm{B,C}} \eta^{\mathrm{B,C}} - \boldsymbol{p_t^{\mathrm{B,D}} / \eta^{\mathrm{B,D}}}, t = 1$$

$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + p_t^{\mathrm{B,C}} \eta^{\mathrm{B,C}} - \boldsymbol{p_t^{\mathrm{B,D}} / \eta^{\mathrm{B,D}}}, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

Self-scheduling – RES & Battery

Remember that we are using hourly time periods. If the duration Δ_t is not one hour, then:

$$e_t^{\mathrm{B}} = E^{\mathrm{B},0} + p_t^{\mathrm{B,C}} \eta^{\mathrm{B,C}} \boldsymbol{\Delta_t} - \frac{p_t^{\mathrm{B,D}}}{\eta^{\mathrm{B,D}}} \boldsymbol{\Delta_t}, t = 1$$

$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + p_t^{\mathrm{B,C}} \eta^{\mathrm{B,C}} \boldsymbol{\Delta_t} - \frac{p_t^{\mathrm{B,D}}}{\eta^{\mathrm{B,D}}} \boldsymbol{\Delta_t}, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

Formulation

Changes related to the **ramping limits** of the generating units and flexible demands would be needed if the duration Δ_t is not one hour.

Self-scheduling – RES & Battery

$$e_t^{\mathrm{B}} = E^{\mathrm{B},0} + p_t^{\mathrm{B,C}} \eta^{\mathrm{B,C}} - \boldsymbol{p_t^{\mathrm{B,D}} / \eta^{\mathrm{B,D}}}, t = 1$$

$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + p_t^{\mathrm{B,C}} \eta^{\mathrm{B,C}} - \boldsymbol{p_t^{\mathrm{B,D}} / \eta^{\mathrm{B,D}}}, \forall t \in \Omega^{\mathrm{T}} \setminus \{1\}$$

Self-scheduling – RES & Battery

$$\underline{E}^{\mathrm{B}} \leq e_t^{\mathrm{B}} \leq \overline{E}^{\mathrm{B}}, \forall t$$

Self-scheduling – RES & Battery

"Anti carpe diem / anti apocalypsis constraint"

Self-scheduling – RES & Battery

"Anti carpe diem / anti apocalypsis constraint"

$$e_t^{\mathrm{B}} \geq E^{\mathrm{B,F}}, t = \left|\Omega^{\mathrm{T}}\right|$$

Self-scheduling – RES & Battery

What does it happen if we do not include this constraint?

Self-scheduling – RES & Battery

What does it happen if we do not include this constraint?

In the context of this problem, considering a time horizon of 24 hourly time periods ($|\Omega^T| = 24$), $e^B_{t=|\Omega^T|}$ will be equal to $\underline{E}^B$ if it is possible due to the other constraints **since there is no tomorrow**.

Self-scheduling – RES & Battery

$$u_t^{\mathrm{B}} \in \{0,1\}, \forall t$$

$$0 \leq p_t^{\mathrm{B,C}} \leq \overline{P}^{\mathrm{B,C}} u_t^{\mathrm{B}}, \forall t$$

$$0 \leq p_t^{\mathrm{B,D}} \leq \overline{P}^{\mathrm{B,D}} \left(1 - u_t^{\mathrm{B}}\right), \forall t$$

Self-scheduling – RES & Battery

Importance of using binary variables to model the charging and discharging statuses:

J. M. Arroyo, L. Baringo, A. Baringo, R. Bolaños, N. Alguacil, and N. G. Cobos, "On the use of a convex model for bulk storage in MIP-based power system operation and planning", *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4964–4967, Nov. 2020, DOI: 10.1109/TPWRS.2020.3020730.

Self-scheduling – RES & Battery

The model of the battery described in this course is very simple.

We are not modeling the degradation of the battery, the relationship between the state-of-charge level and the efficiencies, etc.

Self-scheduling – RES & Battery

$$\max_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \left(\lambda_t^{\mathrm{DA,S}} p_t^{\mathrm{DA,S}} - C^{\mathrm{PV,V}} p_t^{\mathrm{PV}} \right)$$

s.t.:

$$p_t^{\mathrm{DA,S}} = p_t^{\mathrm{PV}} + p_t^{\mathrm{B,D}} - p_t^{\mathrm{B,C}}, \forall t$$

$$0 \leq p_t^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$$

$$0 \leq p_t^{\mathrm{PV}} \leq \overline{P}^{\mathrm{PV}} A_t^{\mathrm{PV}}, \forall t$$

Self-scheduling – RES & Battery

$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + p_t^{\mathrm{B,C}}\eta^{\mathrm{B,C}} - p_t^{\mathrm{B,D}}/\eta^{\mathrm{B,D}}, \forall t$$

$$\underline{E}^{\mathrm{B}} \leq e_t^{\mathrm{B}} \leq \overline{E}^{\mathrm{B}}, \forall t$$

$$e_t^{\mathrm{B}} \geq E^{\mathrm{B,F}}, t = \left|\Omega^{\mathrm{T}}\right|$$

$$u_t^{\mathrm{B}} \in \{0,1\}, \forall t$$

$$0 \leq p_t^{\mathrm{B,C}} \leq \overline{P}^{\mathrm{B,C}} u_t^{\mathrm{B}}, \forall t$$

$$0 \leq p_t^{\mathrm{B,D}} \leq \overline{P}^{\mathrm{B,D}} \left(1 - u_t^{\mathrm{B}}\right), \forall t$$

$$\Phi = \left\{e_t^{\mathrm{B}}, p_t^{\mathrm{B,C}}, p_t^{\mathrm{B,D}}, p_t^{\mathrm{DA,S}}, p_t^{\mathrm{PV}}, u_t^{\mathrm{B}}, \forall t\right\}$$

Self-scheduling – RES & Battery

Let's think now in a case in which we can buy and sell:

- Low DA price → Buy → Charge.
- High DA price → Discharge → Sell.

Self-scheduling – RES & Battery

$$\max_{\Phi} \sum_{t \in \Omega^{T}} \left(\lambda_t^{DA,S} p_t^{DA,S} - \boldsymbol{\lambda_t^{DA,P} p_t^{DA,P}} - C^{PV,V} p_t^{PV} \right)$$

Note that:

- $\lambda_t^{DA,P}$ (purchasing price) $>$ $\lambda_t^{DA,S}$ (selling price).
- In many models $C^{PV,V}$ is considered to be equal to 0.

Self-scheduling – RES & Battery

$$p_t^{\mathrm{DA,S}} - \boldsymbol{p_t^{\mathrm{DA,P}}} = p_t^{\mathrm{PV}} + p_t^{\mathrm{B,D}} - p_t^{\mathrm{B,C}}, \forall t$$

$$\boldsymbol{0 \leq p_t^{\mathrm{DA,P}} \leq \overline{P}^{\mathrm{DA,P}}, \forall t}$$

$$0 \leq p_t^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$$

Self-scheduling – RES & Battery

$$\max_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \left(\lambda_t^{\mathrm{DA,S}} p_t^{\mathrm{DA,S}} - \lambda_t^{\mathrm{DA,P}} p_t^{\mathrm{DA,P}} - C^{\mathrm{PV,V}} p_t^{\mathrm{PV}} \right)$$

s.t.:

$$p_t^{\mathrm{DA,S}} - p_t^{\mathrm{DA,P}} = p_t^{\mathrm{PV}} + p_t^{\mathrm{B,D}} - p_t^{\mathrm{B,C}}, \forall t$$

$$0 \leq p_t^{\mathrm{DA,P}} \leq \overline{P}^{\mathrm{DA,P}}, \forall t$$

$$0 \leq p_t^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$$

$$0 \leq p_t^{\mathrm{PV}} \leq \overline{P}^{\mathrm{PV}} A_t^{\mathrm{PV}}, \forall t$$

Self-scheduling – RES & Battery

$$e_t^{\mathrm{B}} = e_{t-1}^{\mathrm{B}} + p_t^{\mathrm{B,C}}\eta^{\mathrm{B,C}} - p_t^{\mathrm{B,D}}/\eta^{\mathrm{B,D}}, \forall t$$

$$\underline{E}^{\mathrm{B}} \leq e_t^{\mathrm{B}} \leq \overline{E}^{\mathrm{B}}, \forall t$$

$$e_t^{\mathrm{B}} \geq E^{\mathrm{B,F}}, t = \left|\Omega^{\mathrm{T}}\right|$$

$$u_t^{\mathrm{B}} \in \{0,1\}, \forall t$$

$$0 \leq p_t^{\mathrm{B,C}} \leq \overline{P}^{\mathrm{B,C}} u_t^{\mathrm{B}}, \forall t$$

$$0 \leq p_t^{\mathrm{B,D}} \leq \overline{P}^{\mathrm{B,D}} \left(1 - u_t^{\mathrm{B}}\right), \forall t$$

$$\Phi = \left\{e_t^{\mathrm{B}}, p_t^{\mathrm{B,C}}, p_t^{\mathrm{B,D}}, p_t^{\mathrm{DA,P}}, p_t^{\mathrm{DA,S}}, p_t^{\mathrm{PV}}, u_t^{\mathrm{B}}, \forall t\right\}$$

Exercise 3 – Self-scheduling – RES & Battery

Solve the self-scheduling problem of a RES and a battery using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_03.gms

Solution:

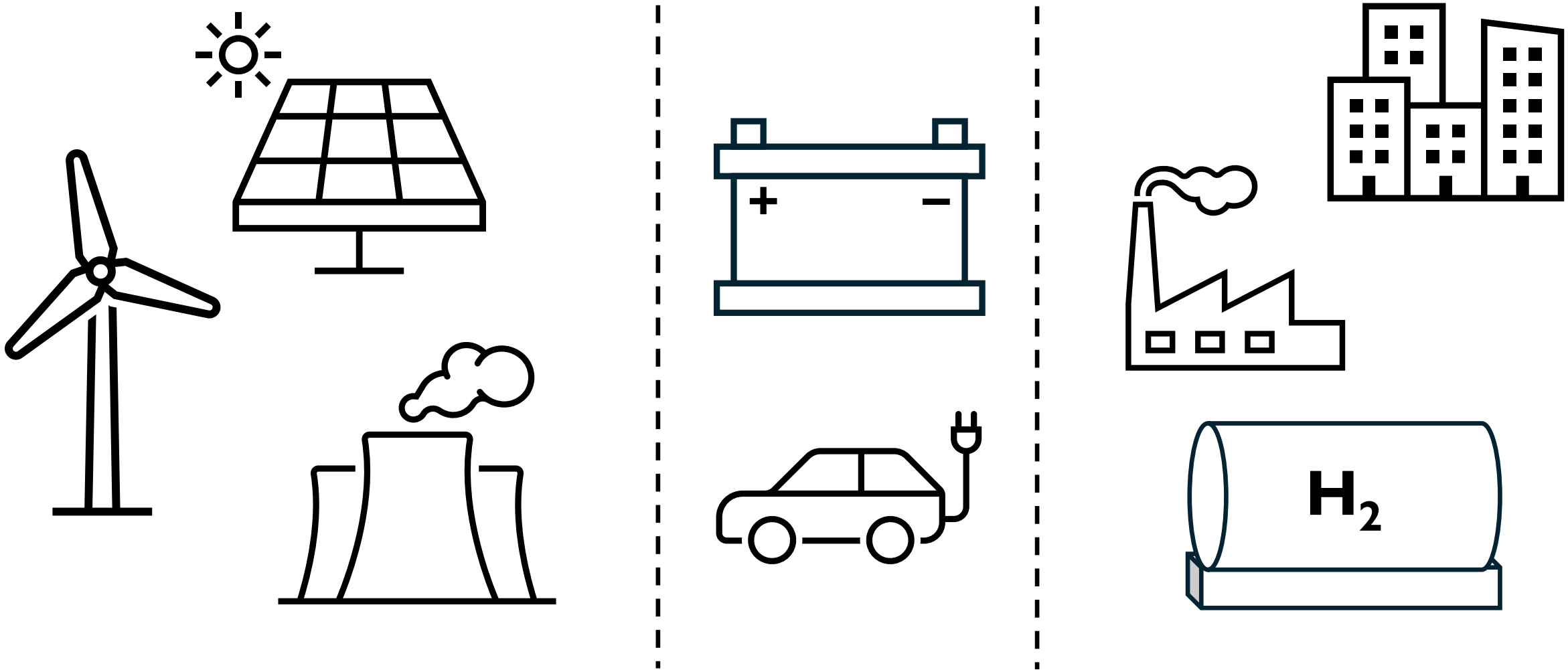
Objective function value = €6046.10.

Self-scheduling

Viewpoint:

- Conventional producer.
- Inelastic and flexible demands.
- Renewable energy source & battery.
- **Virtual power plant**.

Self-scheduling – Virtual power plant



Self-scheduling – Virtual power plant

$$\max_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \left(\lambda_t^{\mathrm{DA,S}} p_t^{\mathrm{DA,S}} - \lambda_t^{\mathrm{DA,P}} p_t^{\mathrm{DA,P}} - \text{costs of assets} \right)$$

s.t.:

$p_t^{\mathrm{DA,S}} - p_t^{\mathrm{DA,P}} =$ power of generators, demands, and storages, $\forall t$

$0 \leq p_t^{\mathrm{DA,P}} \leq \overline{P}^{\mathrm{DA,P}}, \forall t$

$0 \leq p_t^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$

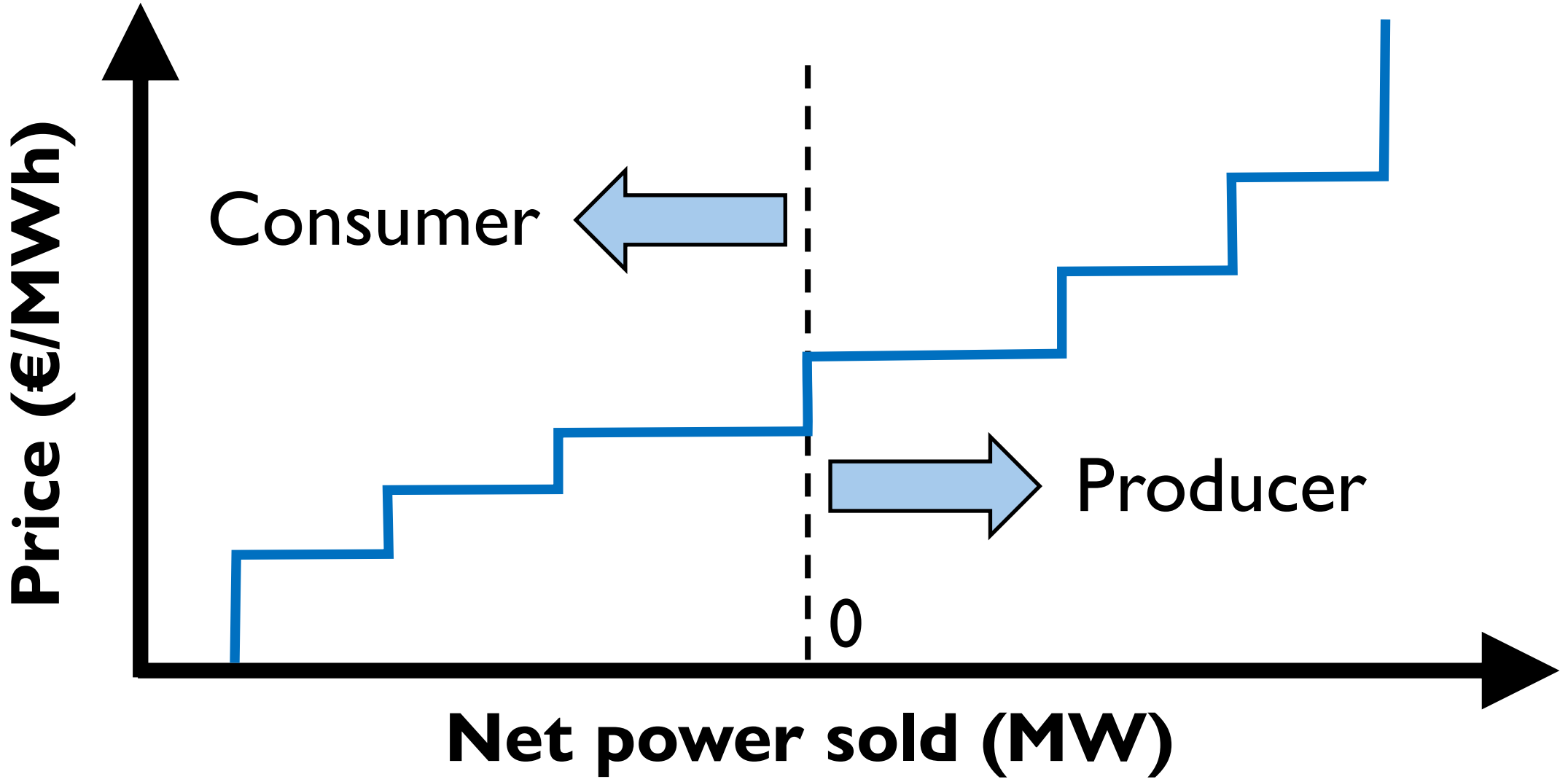
Constraints of generators, demands, and storages.

Self-scheduling – Bidding and offering curves

We are obtaining one value of the power purchased / sold for the given day-ahead electricity price.

How can we build the bidding and offering curves?

Self-scheduling – Bidding and offering curves



Self-scheduling – Bidding and offering curves

We are obtaining one value of the power purchased / sold for the given day-ahead electricity price.

How can we build the bidding and offering curves?

We will see this in lecture 4 (spoiler: we will use scenarios).

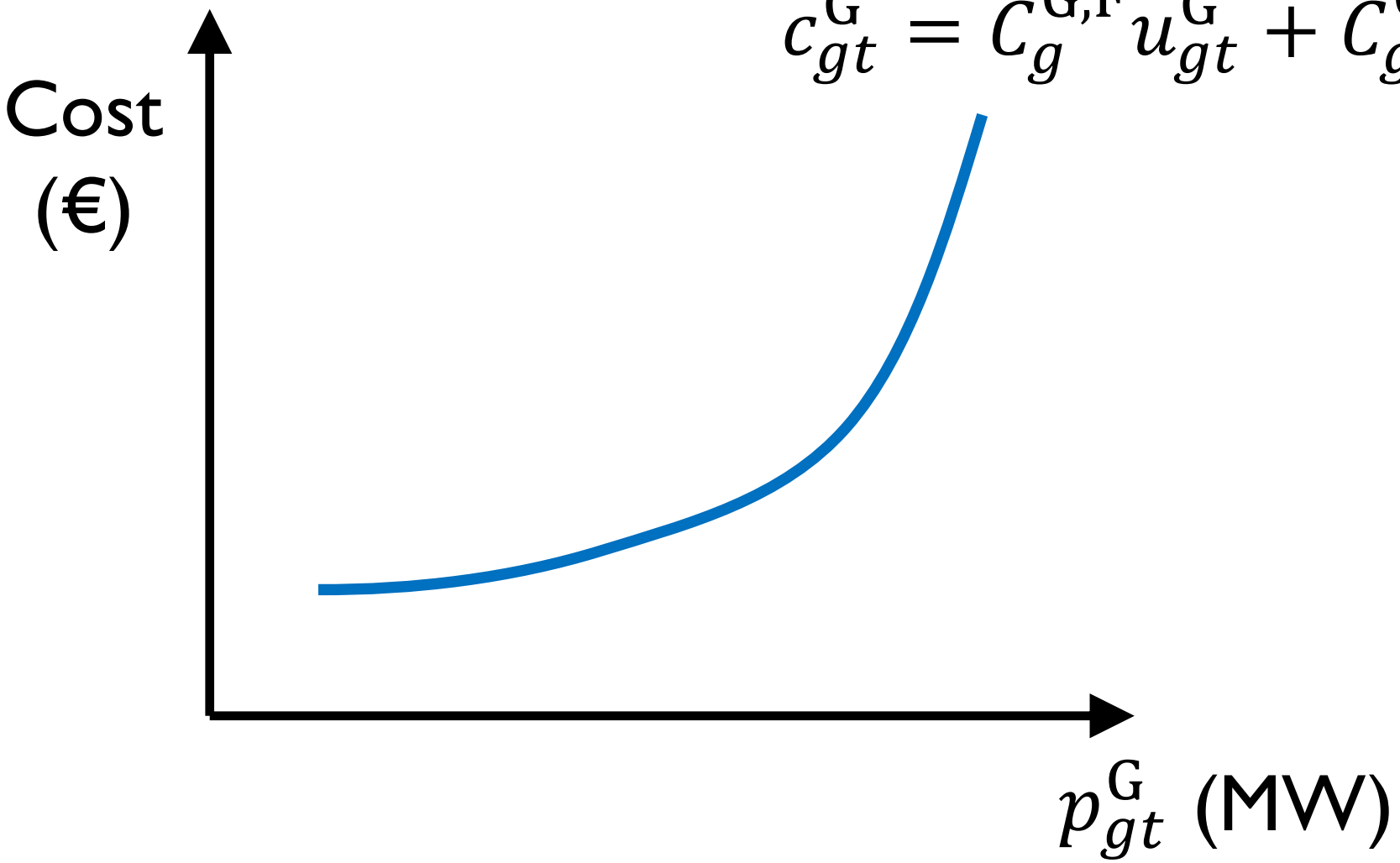
Operation

Operation optimization problems solved by:

- A centralized entity → Unit commitment.
- Private entities → Self-scheduling.
- **Cost function with intervals**.

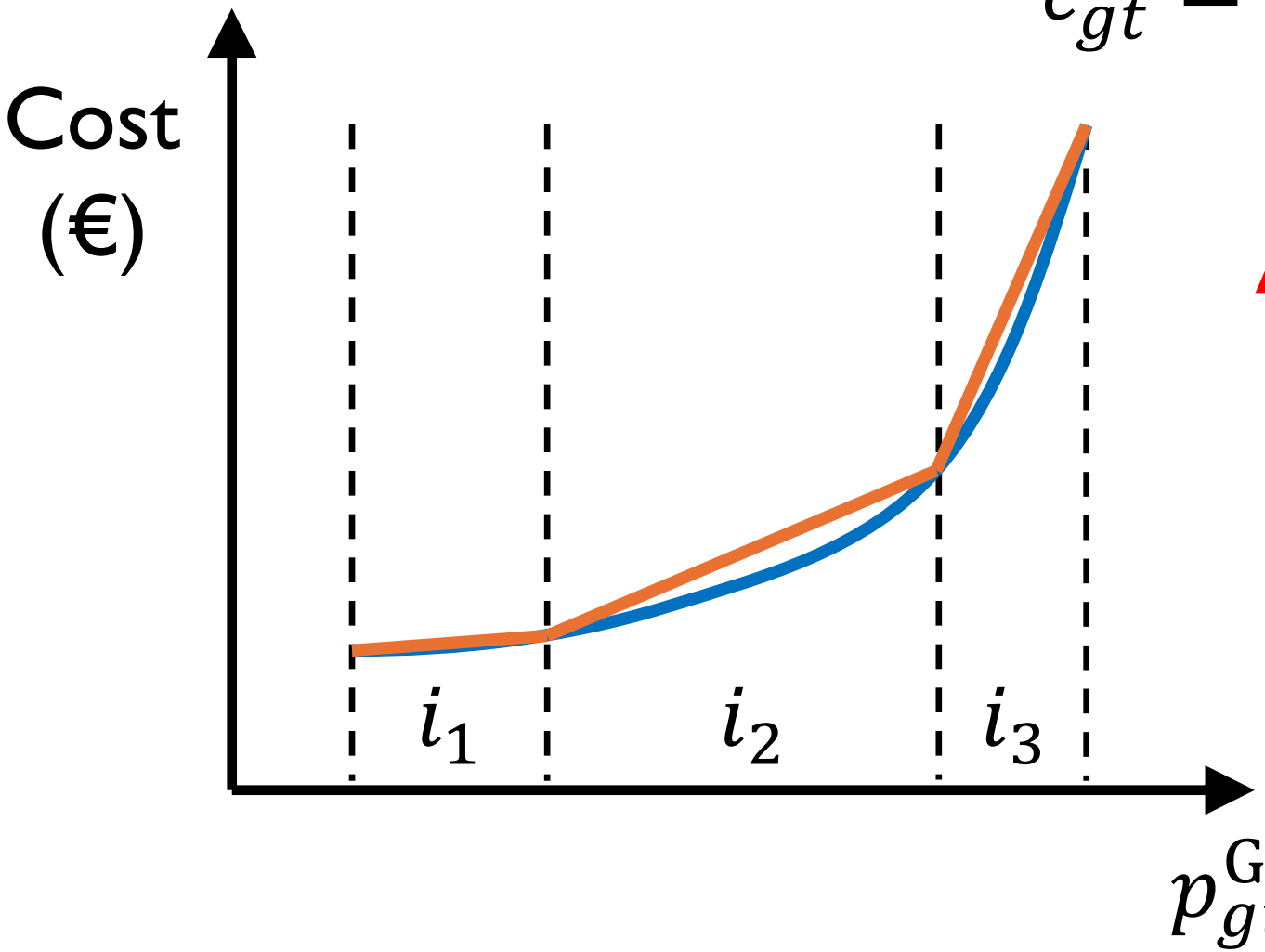
Cost function with intervals

$$c_{gt}^{\mathrm{G}} = C_g^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,L}} p_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,Q}} {p_{gt}^{\mathrm{G}}}^2$$



Cost function with intervals

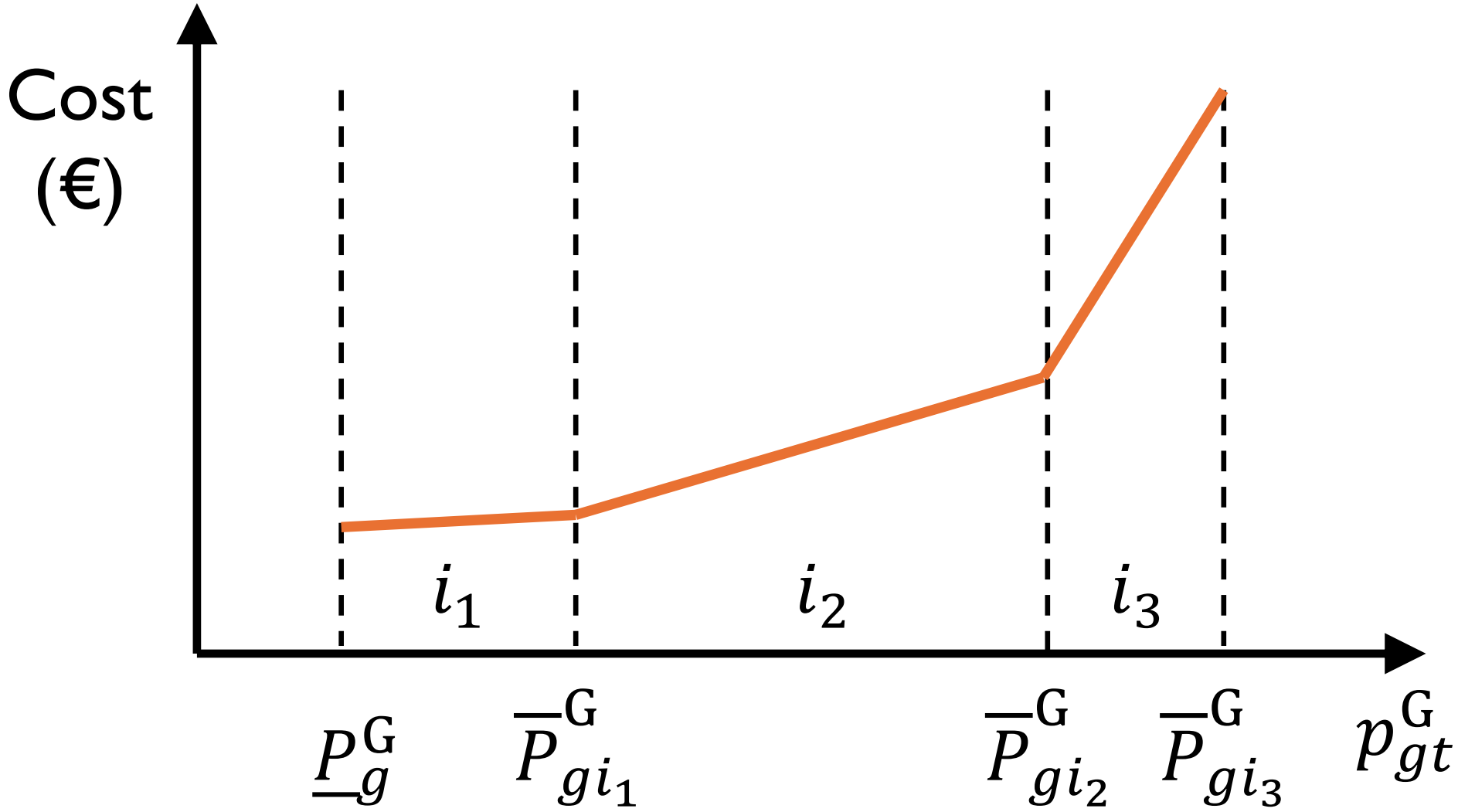
$$c_{gt}^{\mathrm{G}} = C_{g}^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + \sum_{i \in \Omega_{g}^{\mathrm{I}}} C_{gi}^{\mathrm{G,V}} p_{git}^{\mathrm{G}}$$



APPROXIMATION

Intervals → i

Cost function with intervals



Cost function with intervals

$$c_{gt}^{\mathrm{G}} = C_{g}^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + \sum_{i \in \Omega_{g}^{\mathrm{I}}} C_{gi}^{\mathrm{G,V}} p_{git}^{\mathrm{G}}, \forall g, \forall t$$

$$p_{gt}^{\mathrm{G}} = \underline{P}_{g}^{\mathrm{G}} u_{gt}^{\mathrm{G}} + \sum_{i \in \Omega_{g}^{\mathrm{I}}} p_{git}^{\mathrm{G}}, \forall g, \forall t$$

$$0 \leq p_{git}^{\mathrm{G}} \leq \overline{P}_{gi}^{\mathrm{G}} - \underline{P}_{g}^{\mathrm{G}}, \forall g, i = 1, \forall t$$

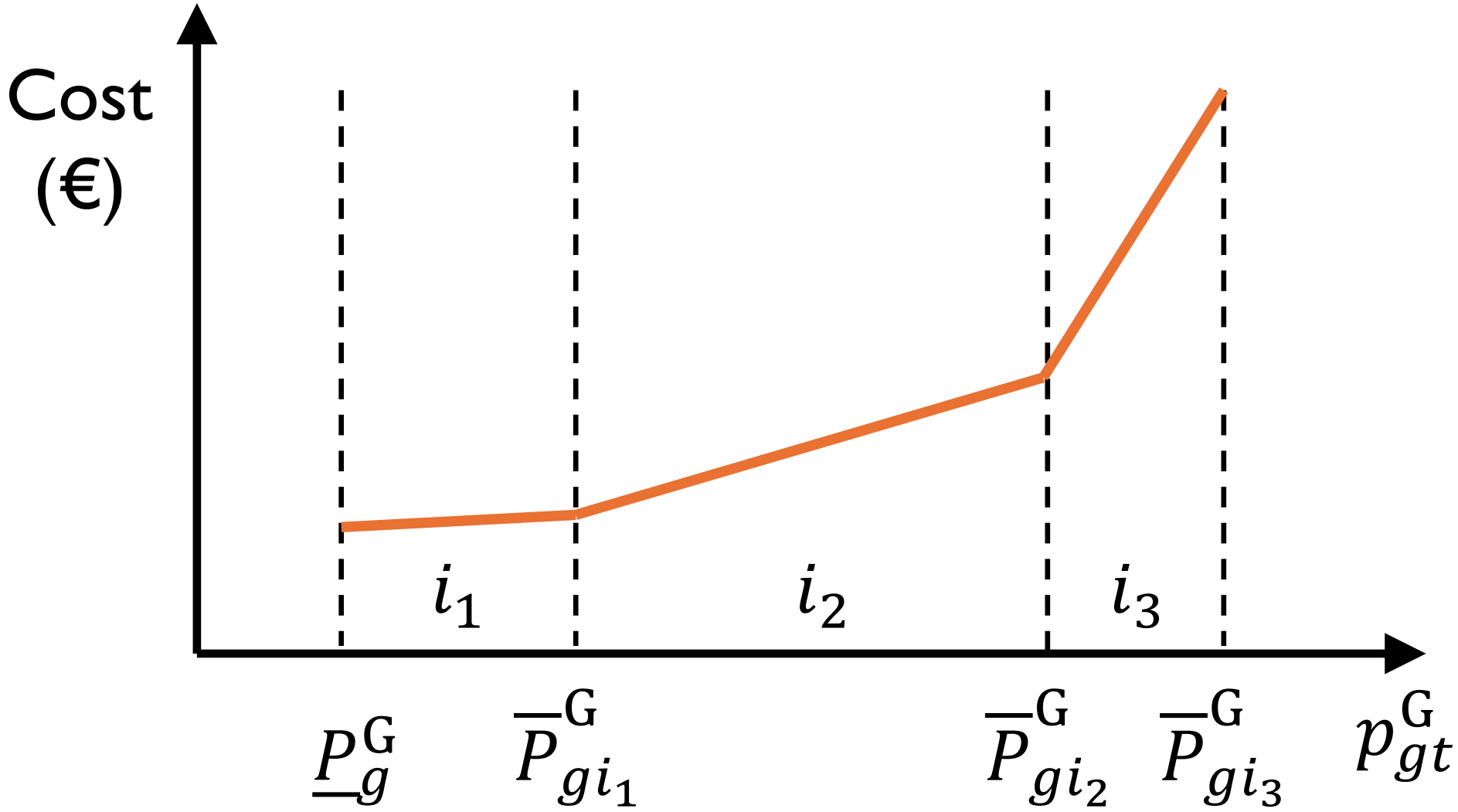
$$0 \leq p_{git}^{\mathrm{G}} \leq \overline{P}_{gi}^{\mathrm{G}} - \overline{P}_{g,i-1}^{\mathrm{G}}, \forall g, \forall i \in \Omega_{g}^{\mathrm{I}} \backslash \{1\}, \forall t$$

$$\underline{P}_{g}^{\mathrm{G}} u_{gt}^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_{g}^{\mathrm{G}} u_{gt}^{\mathrm{G}}, \forall g, \forall t$$

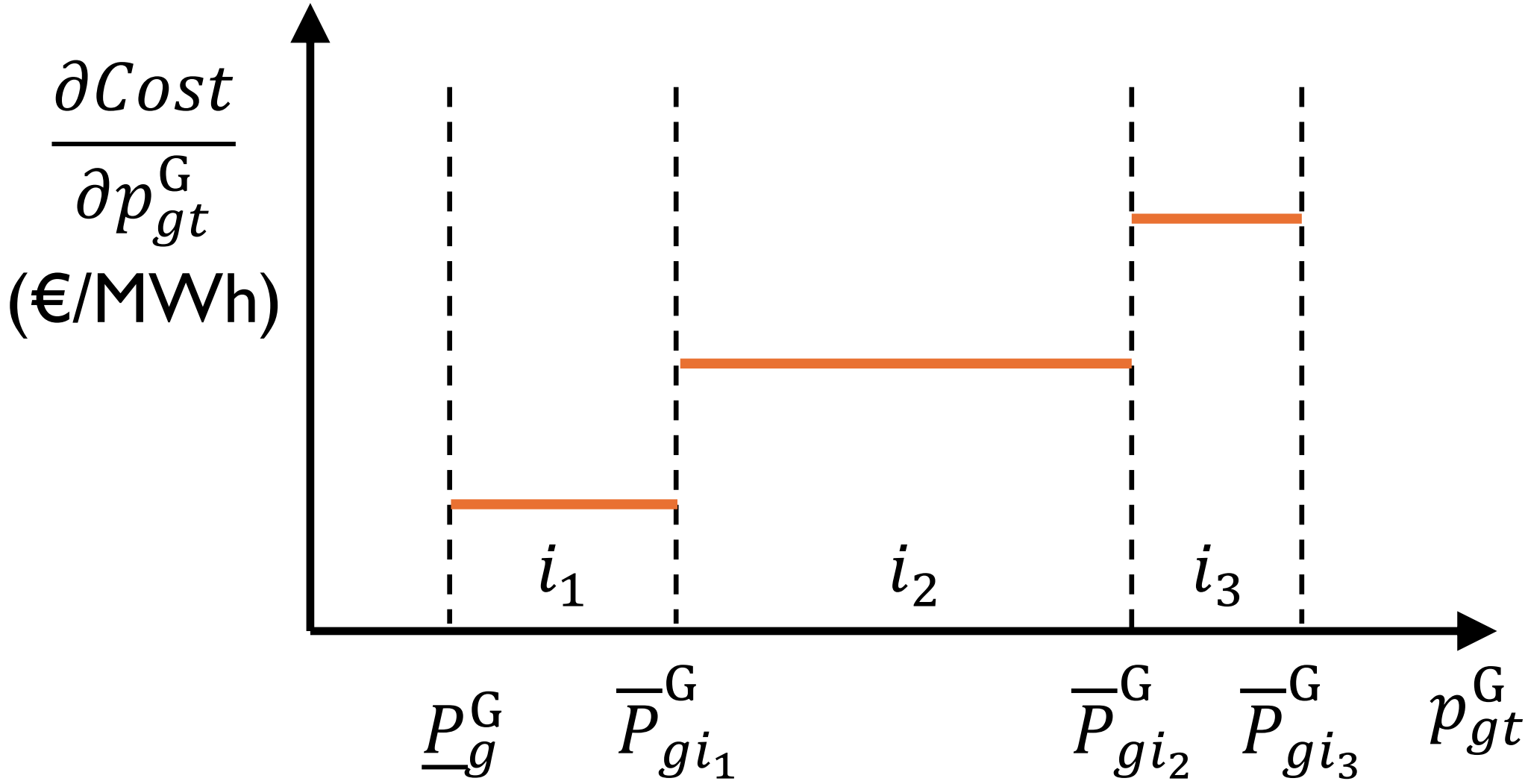
Cost function with intervals

In what order are the intervals filled?

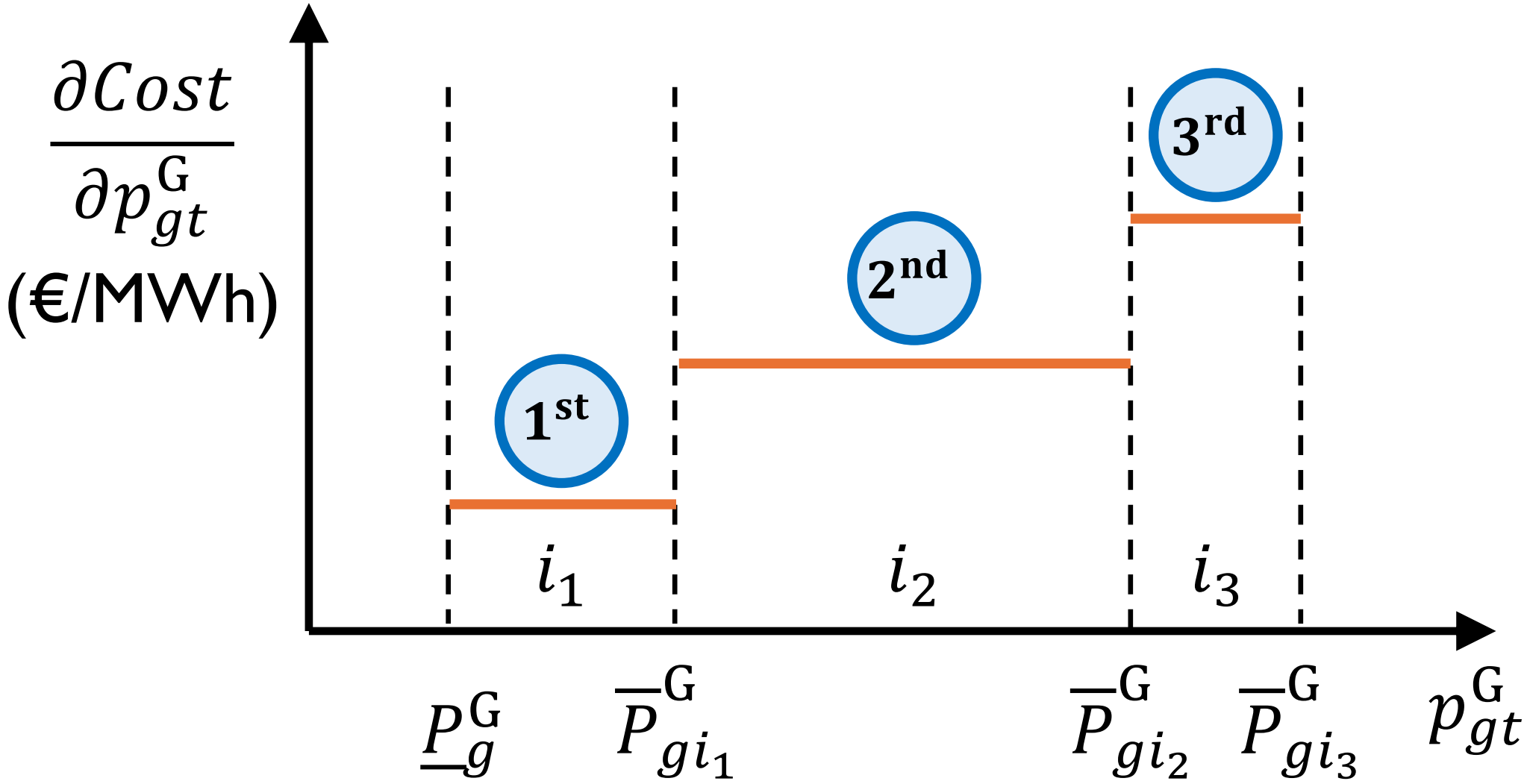
Cost function with intervals



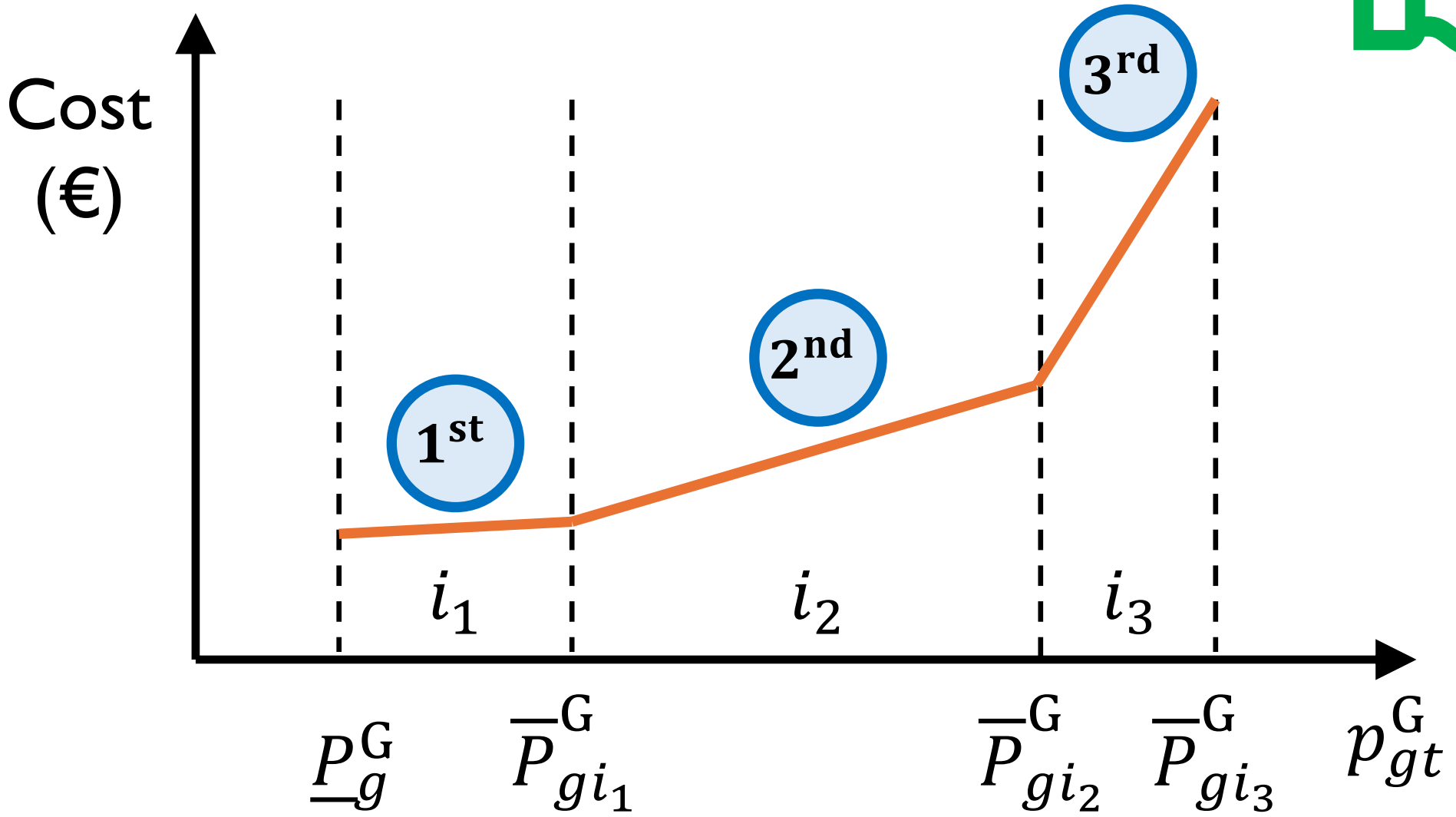
Cost function with intervals



Cost function with intervals



Cost function with intervals



Exercise 4 – Cost function with intervals

Solve the unit commitment problem considering a cost function with intervals using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_04.gms

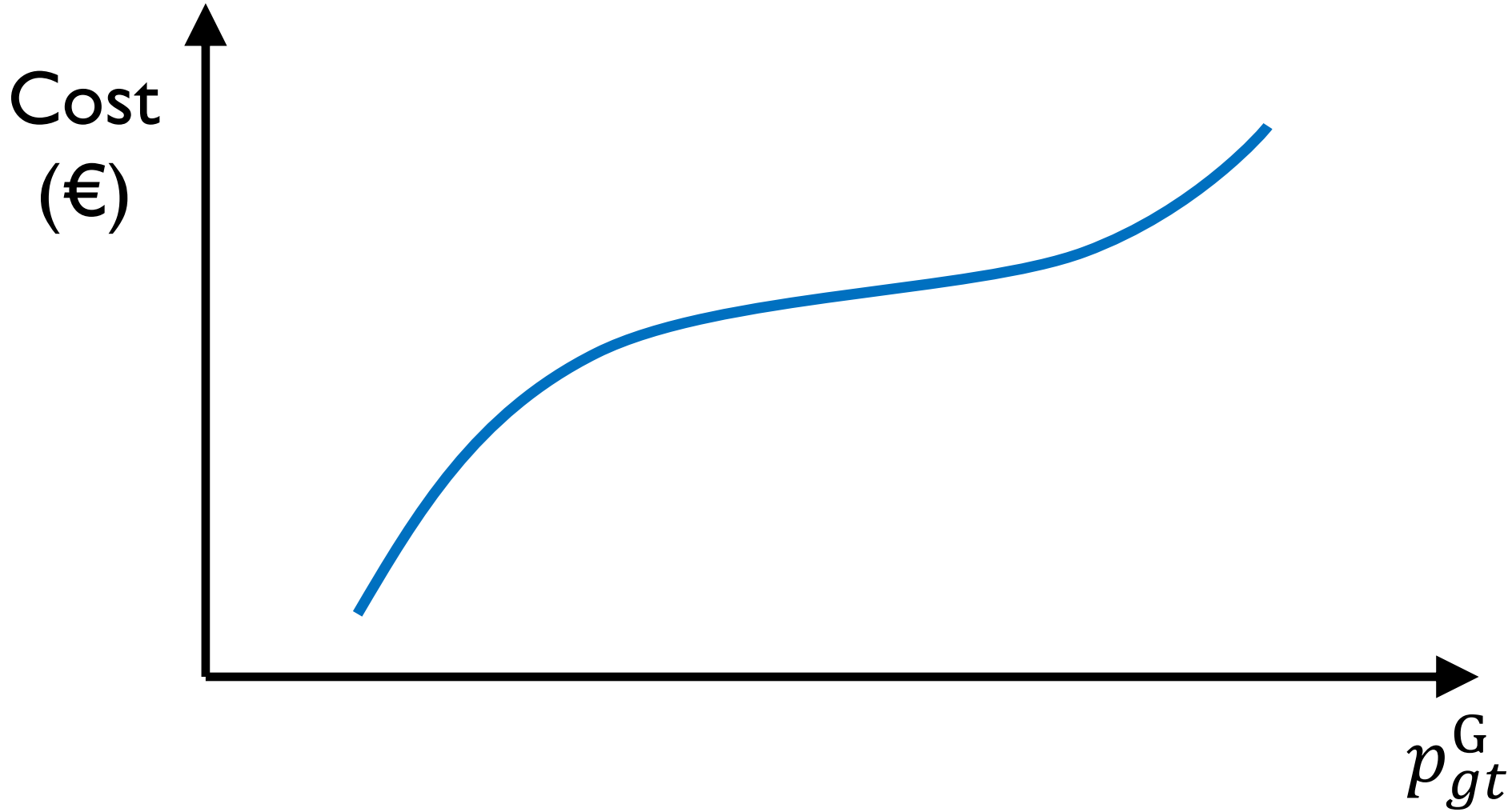
Solution:

a) $P^{\mathrm{D}} = 80$ MW $\rightarrow$ Objective function value = €15.

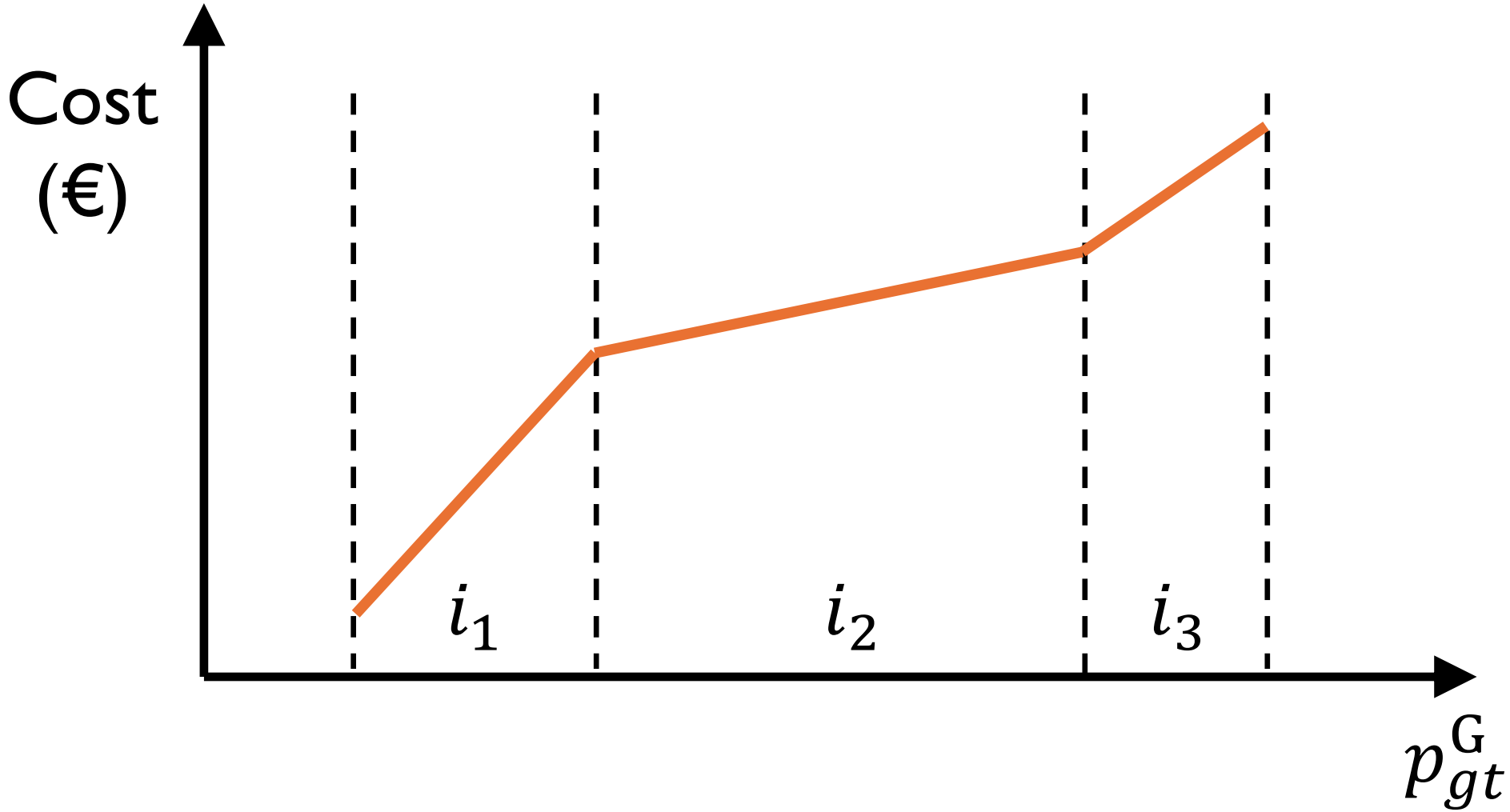
b) $P^{\mathrm{D}} = 140$ MW $\rightarrow$ Objective function value = €25.

c) $P^{\mathrm{D}} = 260$ MW $\rightarrow$ Objective function value = €65.5.

Cost function with intervals



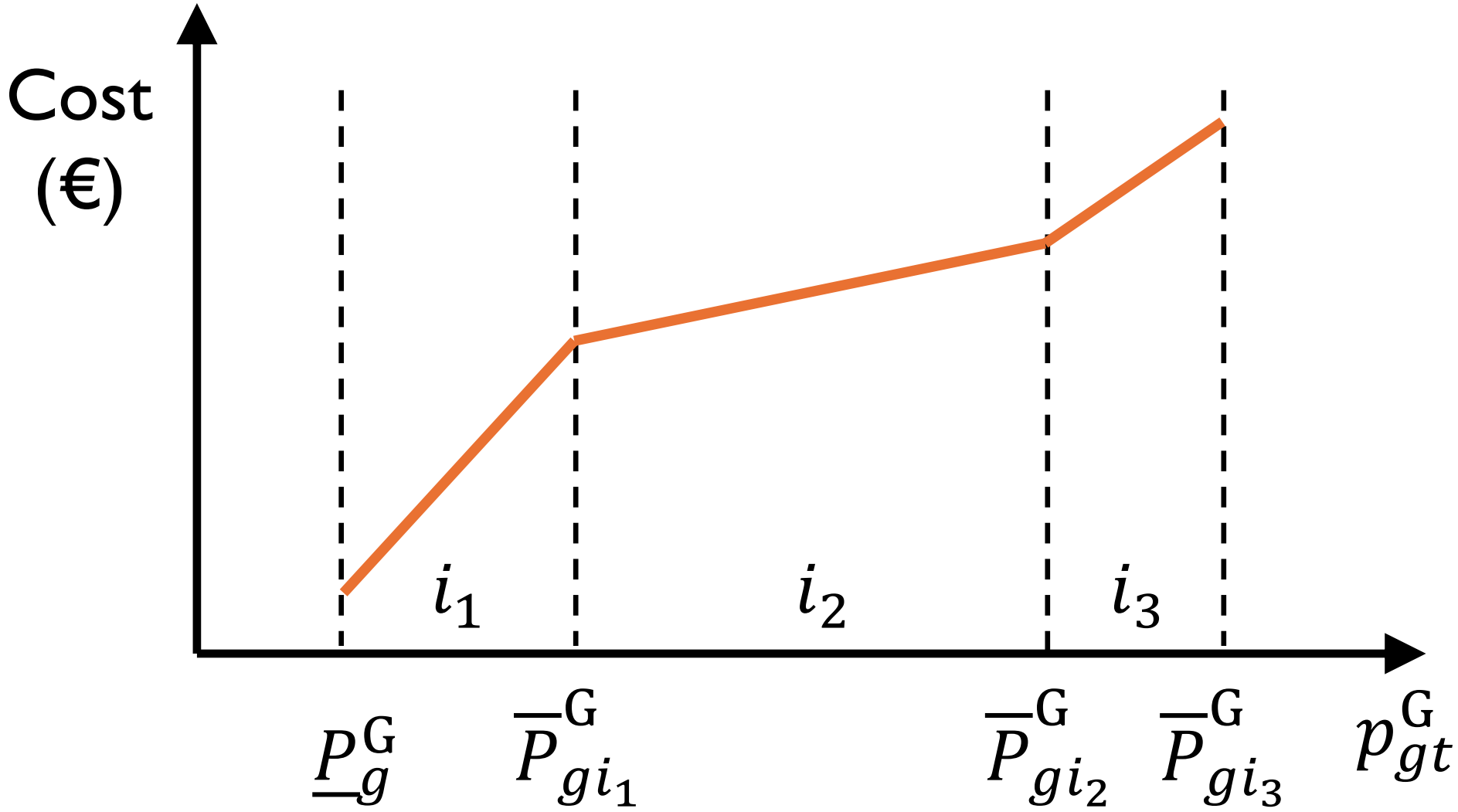
Cost function with intervals



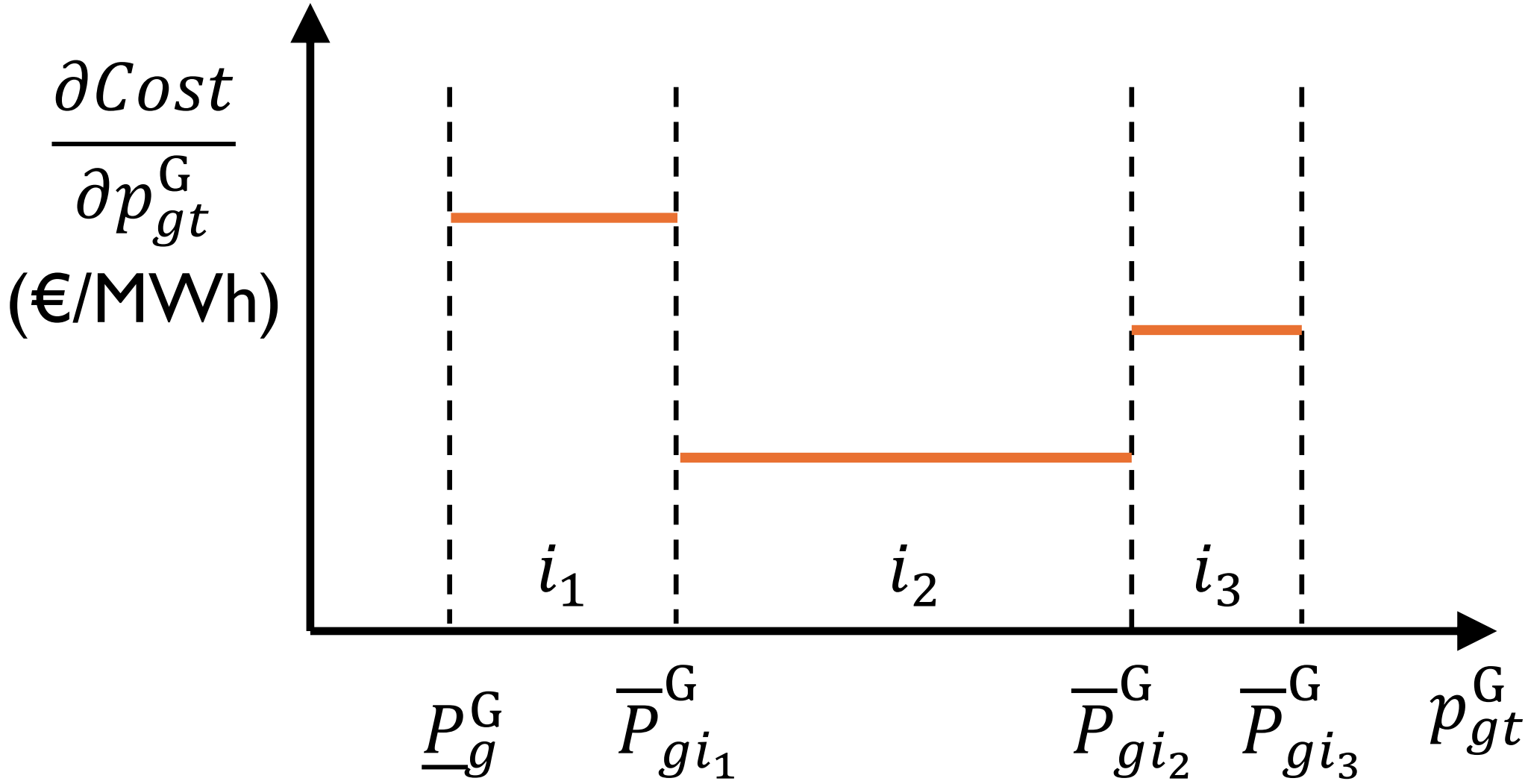
Cost function with intervals

In what order are the intervals filled?

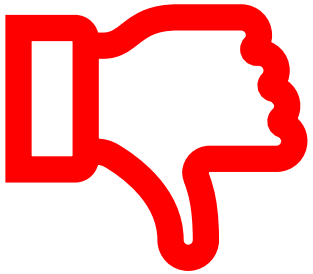
Cost function with intervals

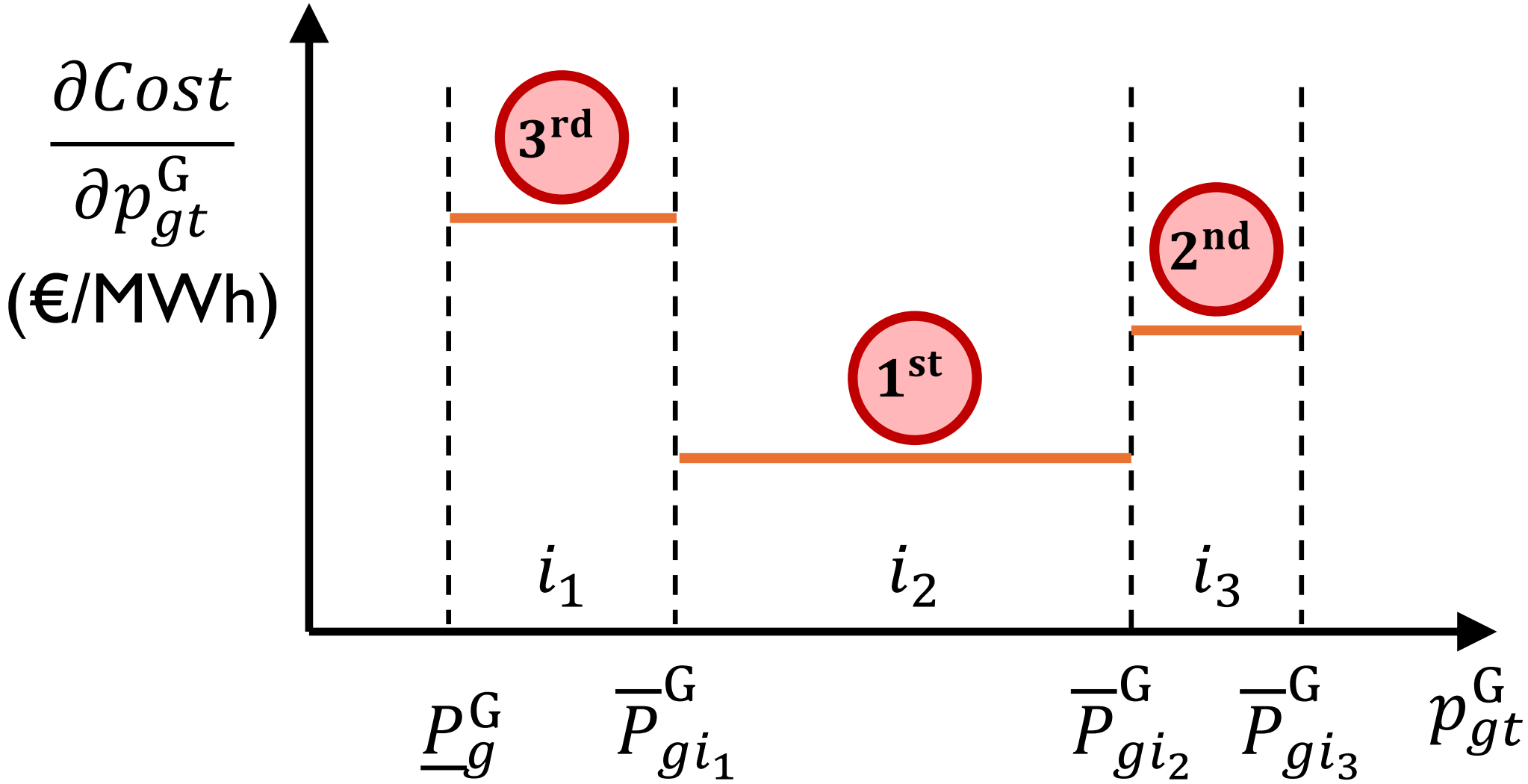


Cost function with intervals

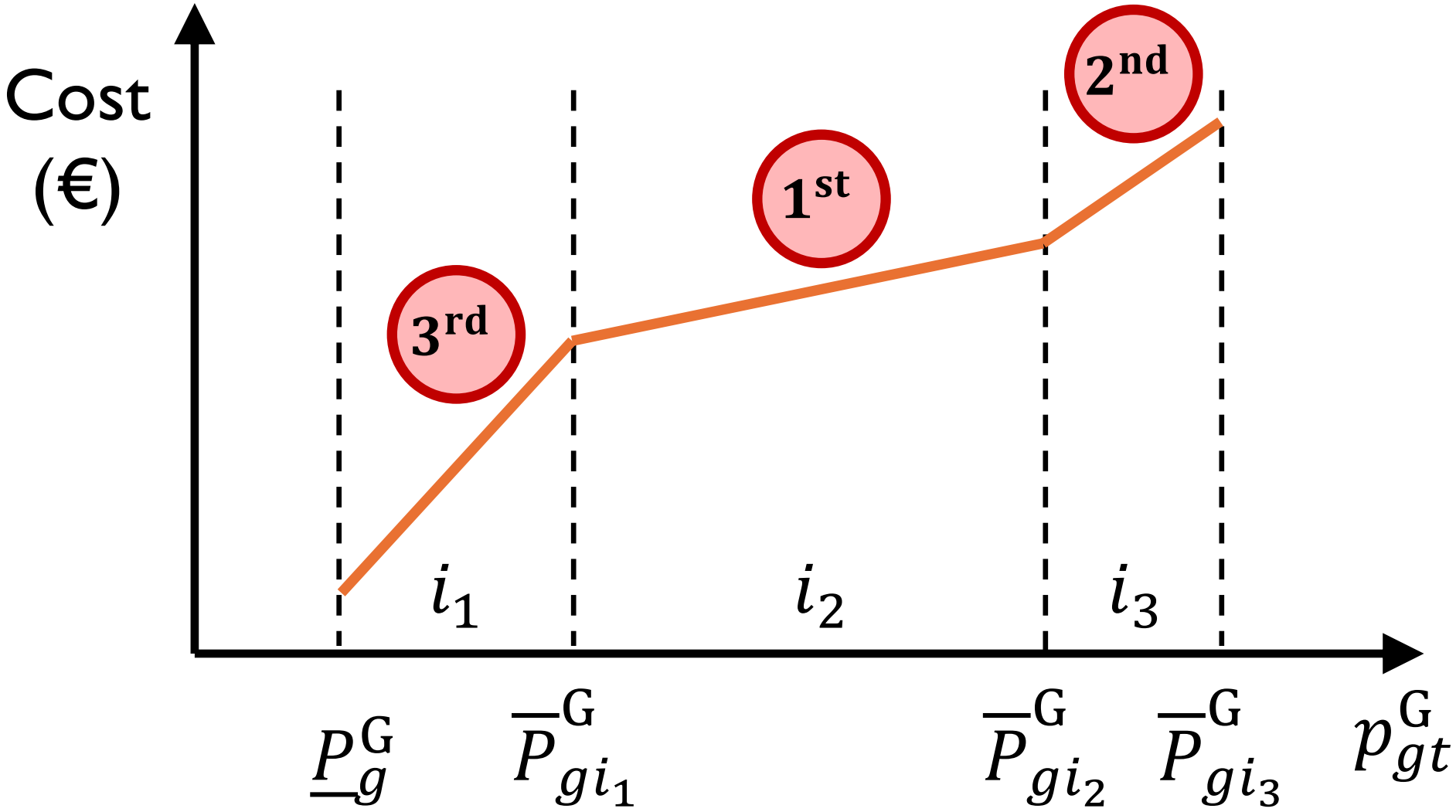


Cost function with intervals

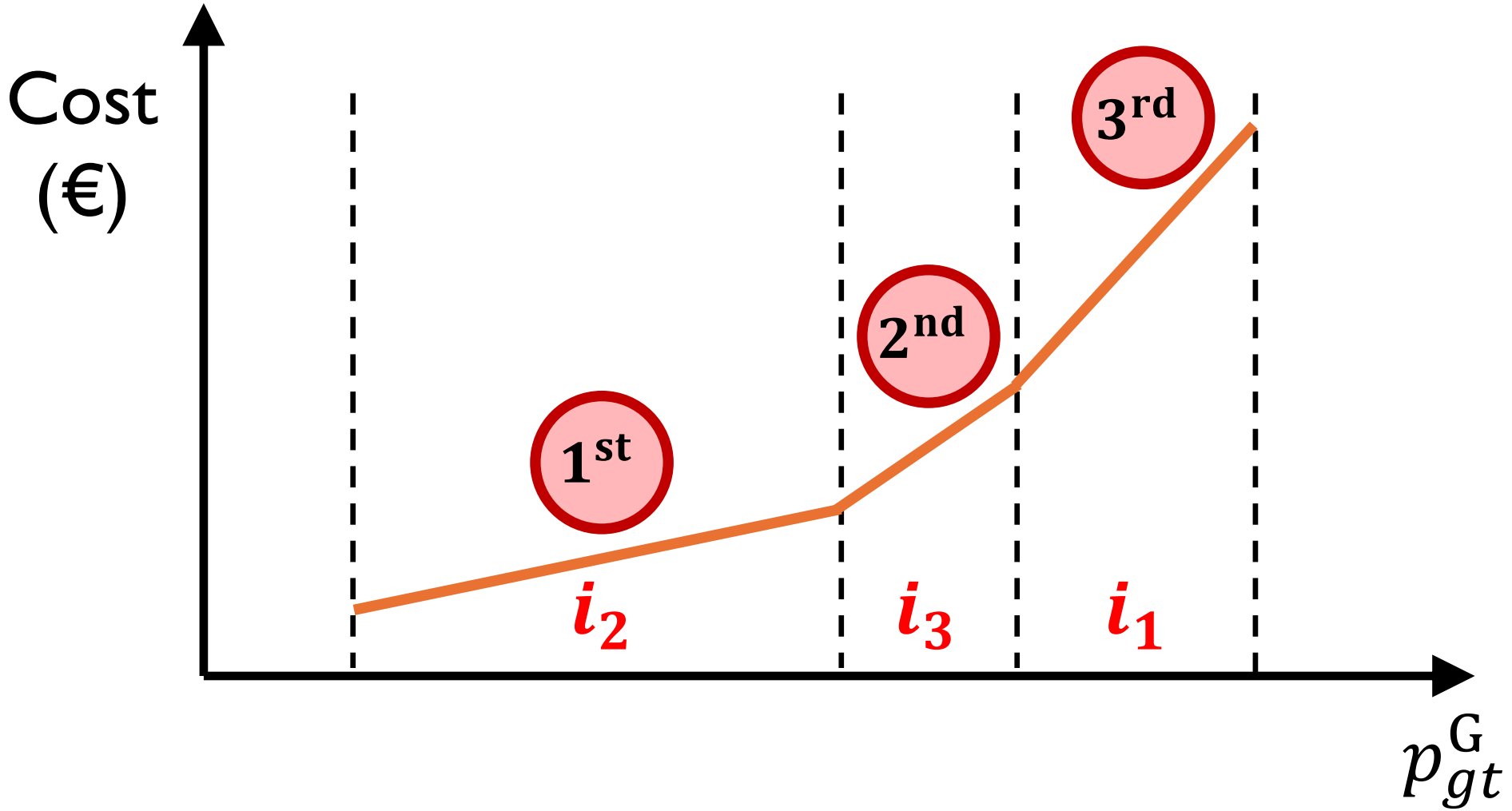




Cost function with intervals



Cost function with intervals



Exercise 5 – Cost function with intervals

Solve the unit commitment problem considering a cost function with intervals using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_05.gms

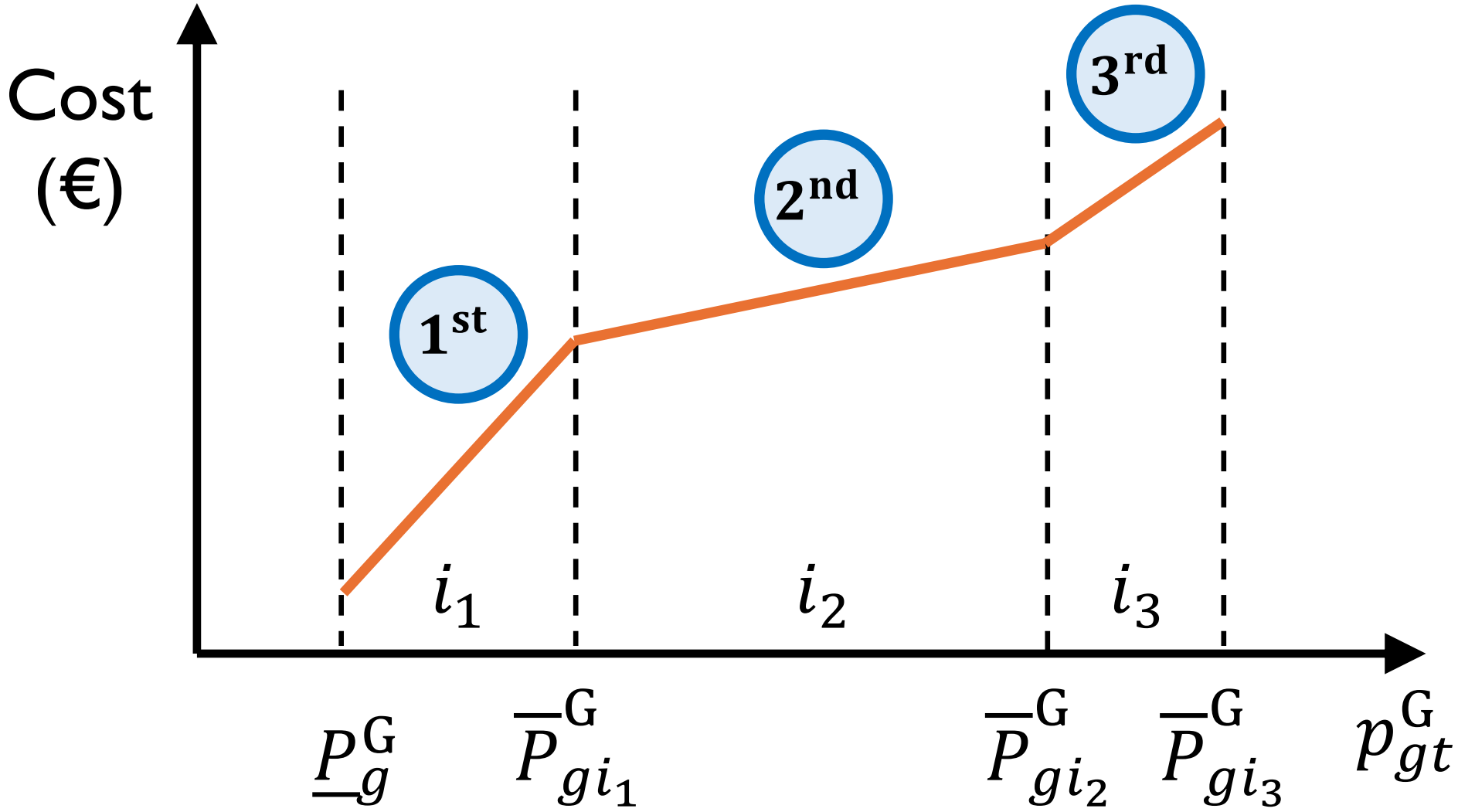
Solution:

a) $P^{\mathrm{D}} = 80$ MW $\rightarrow$ Objective function value = €15.

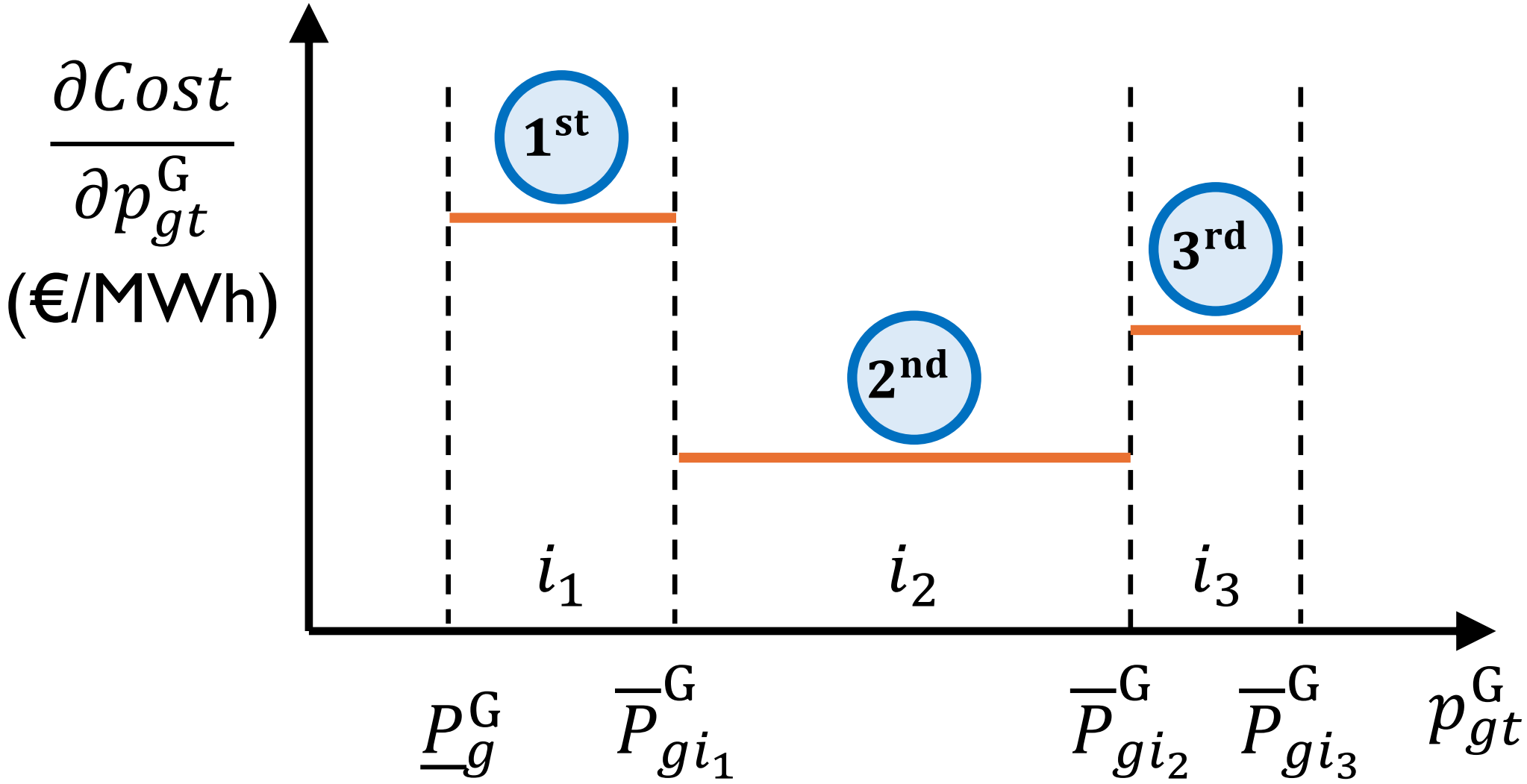
b) $P^{\mathrm{D}} = 140$ MW $\rightarrow$ Objective function value = €27.

c) $P^{\mathrm{D}} = 260$ MW $\rightarrow$ Objective function value = €55.5.

Cost function with intervals



Cost function with intervals



Cost function with intervals

We must include binary variables $\rightarrow v_{git}^{\mathrm{G}}$.

$$v_{git}^{\mathrm{G}} \in \{0,1\}, \forall g, \forall i, \forall t$$

$$v_{git}^{\mathrm{G}} = 1 \rightarrow \text{ Interval filled}$$

$$v_{git}^{\mathrm{G}} = 0 \rightarrow \text{Interval not filled}$$

Cost function with intervals

$$c_{gt}^{\mathrm{G}} = C_{g}^{\mathrm{G,F}} u_{gt}^{\mathrm{G}} + \sum_{i \in \Omega_{g}^{\mathrm{I}}} C_{gi}^{\mathrm{G,V}} p_{git}^{\mathrm{G}}, \forall g, \forall t$$

$$p_{gt}^{\mathrm{G}} = \underline{P}_{g}^{\mathrm{G}} u_{gt}^{\mathrm{G}} + \sum_{i \in \Omega_{g}^{\mathrm{I}}} p_{git}^{\mathrm{G}}, \forall g, \forall t$$

$$\left(\overline{P}_{gi}^{\mathrm{G}} - \underline{P}_{g}^{\mathrm{G}}\right) \boldsymbol{v}_{\boldsymbol{git}}^{\mathbf{G}} \leq p_{git}^{\mathrm{G}} \leq \overline{P}_{gi}^{\mathrm{G}} - \underline{P}_{g}^{\mathrm{G}}, \forall g, i = 1, \forall t$$

$$\left(\overline{P}_{gi}^{\mathrm{G}} - \overline{P}_{g,i-1}^{\mathrm{G}}\right) \boldsymbol{v}_{\boldsymbol{git}}^{\mathbf{G}} \leq p_{git}^{\mathrm{G}} \leq \left(\overline{P}_{gi}^{\mathrm{G}} - \overline{P}_{g,i-1}^{\mathrm{G}}\right) \boldsymbol{v}_{\boldsymbol{g,i-1,t}}^{\mathbf{G}}, \forall g, \forall i \in \Omega_{g}^{\mathrm{I}} \backslash \{1\}, \forall t$$

$$\underline{P}_{g}^{\mathrm{G}} u_{gt}^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_{g}^{\mathrm{G}} u_{gt}^{\mathrm{G}}, \forall g, \forall t$$

Cost function with intervals

$$\left(\overline{P}^{\mathrm{G}}_{gi_1} - \underline{P}^{\mathrm{G}}_{g}\right) \boldsymbol{v}^{\mathbf{G}}_{\boldsymbol{gi_1t}} \leq p^{\mathrm{G}}_{gi_1t} \leq \overline{P}^{\mathrm{G}}_{gi_1} - \underline{P}^{\mathrm{G}}_{g}, \forall g, \forall t$$

$$\left(\overline{P}^{\mathrm{G}}_{gi_2} - \overline{P}^{\mathrm{G}}_{gi_1}\right) \boldsymbol{v}^{\mathbf{G}}_{\boldsymbol{gi_2t}} \leq p^{\mathrm{G}}_{gi_2t} \leq \left(\overline{P}^{\mathrm{G}}_{gi_2} - \overline{P}^{\mathrm{G}}_{gi_1}\right) \boldsymbol{v}^{\mathbf{G}}_{\boldsymbol{gi_1t}}, \forall g, \forall t$$

$$\left(\overline{P}^{\mathrm{G}}_{gi_3} - \overline{P}^{\mathrm{G}}_{gi_2}\right) \boldsymbol{v}^{\mathbf{G}}_{\boldsymbol{gi_3t}} \leq p^{\mathrm{G}}_{gi_3t} \leq \left(\overline{P}^{\mathrm{G}}_{gi_3} - \overline{P}^{\mathrm{G}}_{gi_2}\right) \boldsymbol{v}^{\mathbf{G}}_{\boldsymbol{gi_2t}}, \forall g, \forall t$$

Cost function with intervals

If $v^{\mathrm{G}}_{gi_1t} = 1$

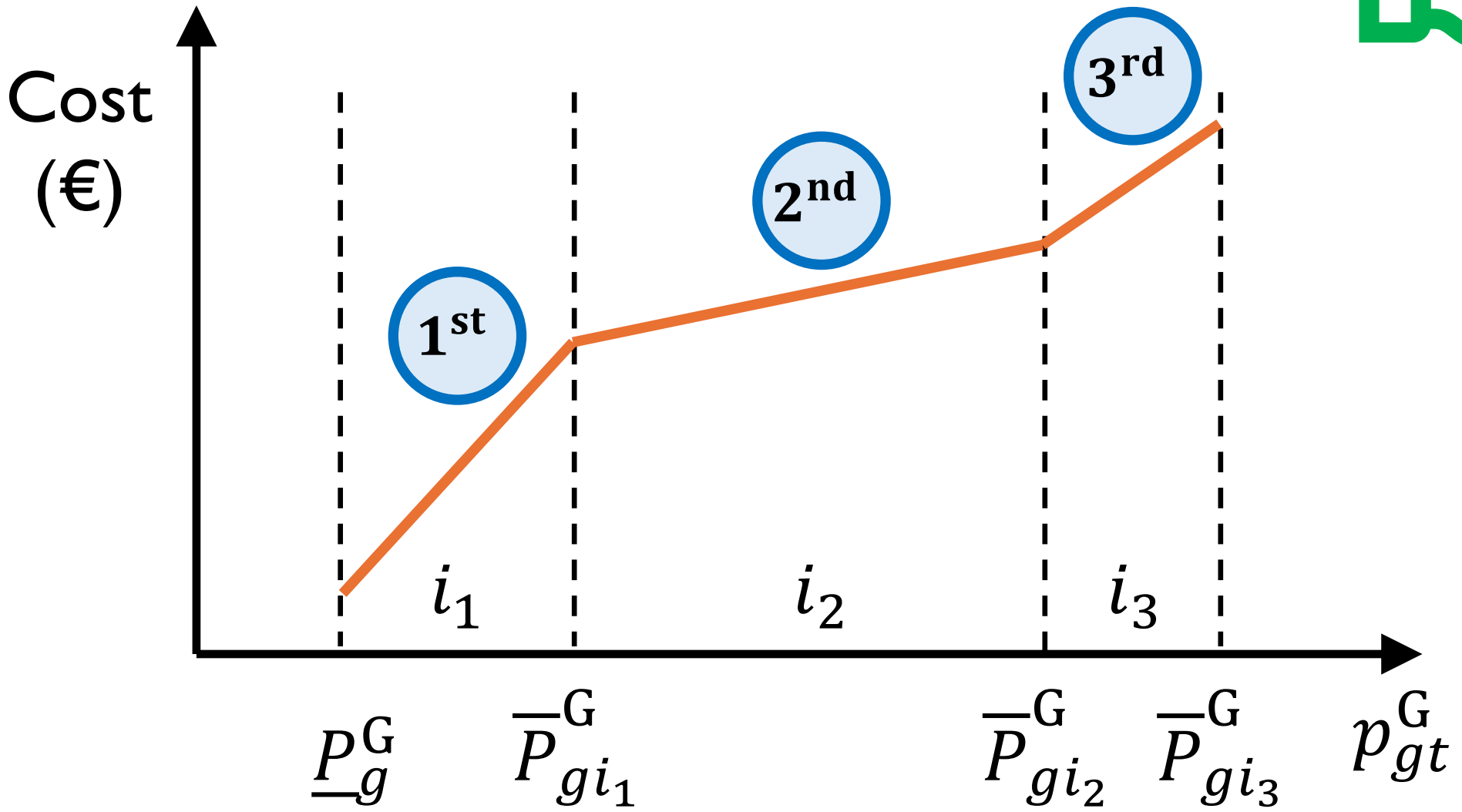
$p^{\mathrm{G}}_{gi_1t} = \overline{P}^{\mathrm{G}}_{gi_1} - \underline{P}^{\mathrm{G}}_{g} \rightarrow$ Interval 1 filled.

$$\left(\overline{P}^{\mathrm{G}}_{gi_2} - \overline{P}^{\mathrm{G}}_{gi_1}\right) \boldsymbol{v^{\mathrm{G}}_{gi_2t}} \leq p^{\mathrm{G}}_{gi_2t} \leq \overline{P}^{\mathrm{G}}_{gi_2} - \overline{P}^{\mathrm{G}}_{gi_1}$$

Note that $v^{\mathrm{G}}_{gi_1t} = 0$ and $v^{\mathrm{G}}_{gi_2t} = 1$ is infeasible:

$$\overline{P}^{\mathrm{G}}_{gi_2} - \overline{P}^{\mathrm{G}}_{gi_1} \leq p^{\mathrm{G}}_{gi_2t} \leq \mathbf{0}, \text{ where } \overline{P}^{\mathrm{G}}_{gi_2} > \overline{P}^{\mathrm{G}}_{gi_1}.$$

Cost function with intervals



Exercise 6 – Cost function with intervals

Solve the unit commitment problem considering a cost function with intervals using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_06.gms

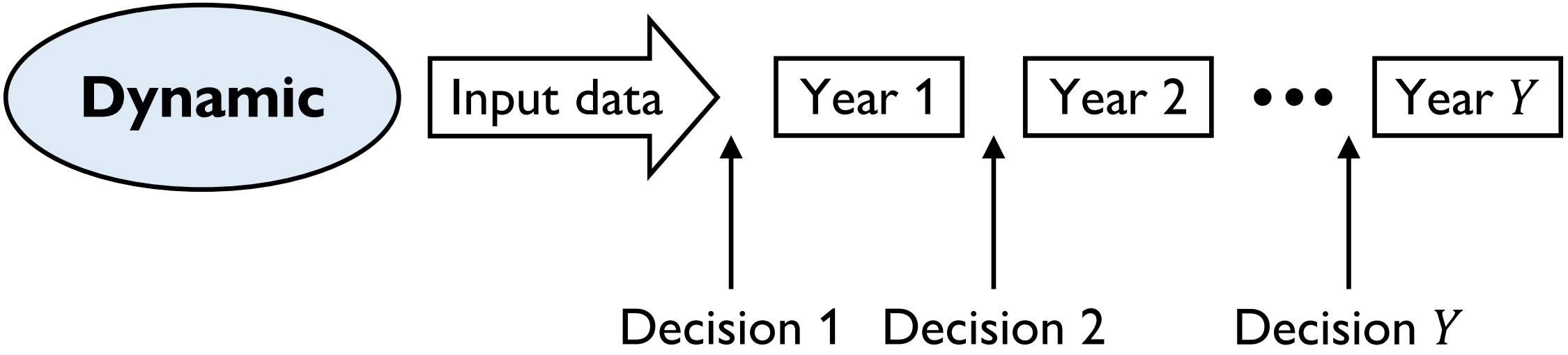
Solution:

		Exercise 5
a)	$P^{\mathrm{D}} = 80$ MW → Objective function value = €27.5.	€15
b)	$P^{\mathrm{D}} = 140$ MW → Objective function value = €38.5.	€27
c)	$P^{\mathrm{D}} = 260$ MW → Objective function value = €61.5.	€55.5

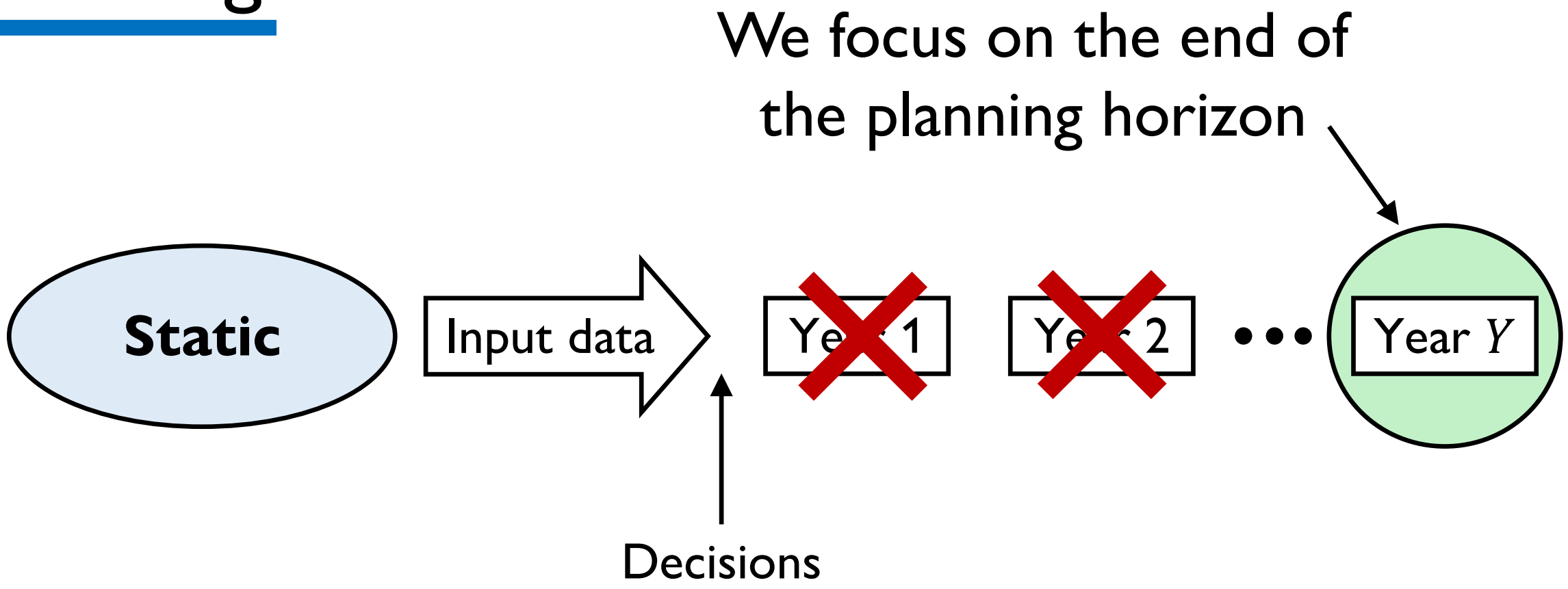
3. Planning

Operation vs Planning

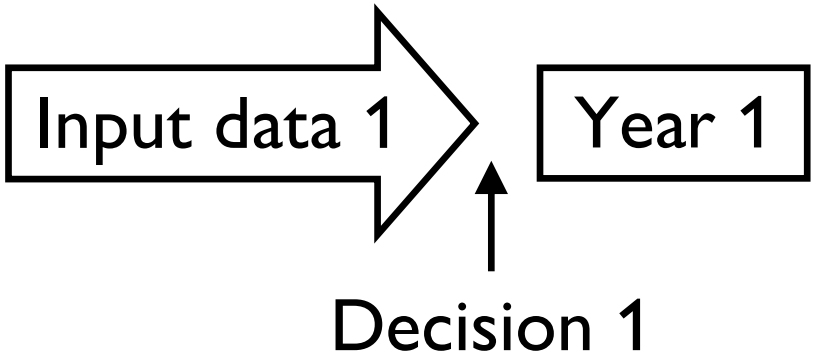
Planning


Dynamic
Input data
Year 1
Year 2
• • •
Year Y
Decision 1
Decision 2
Decision Y

Planning

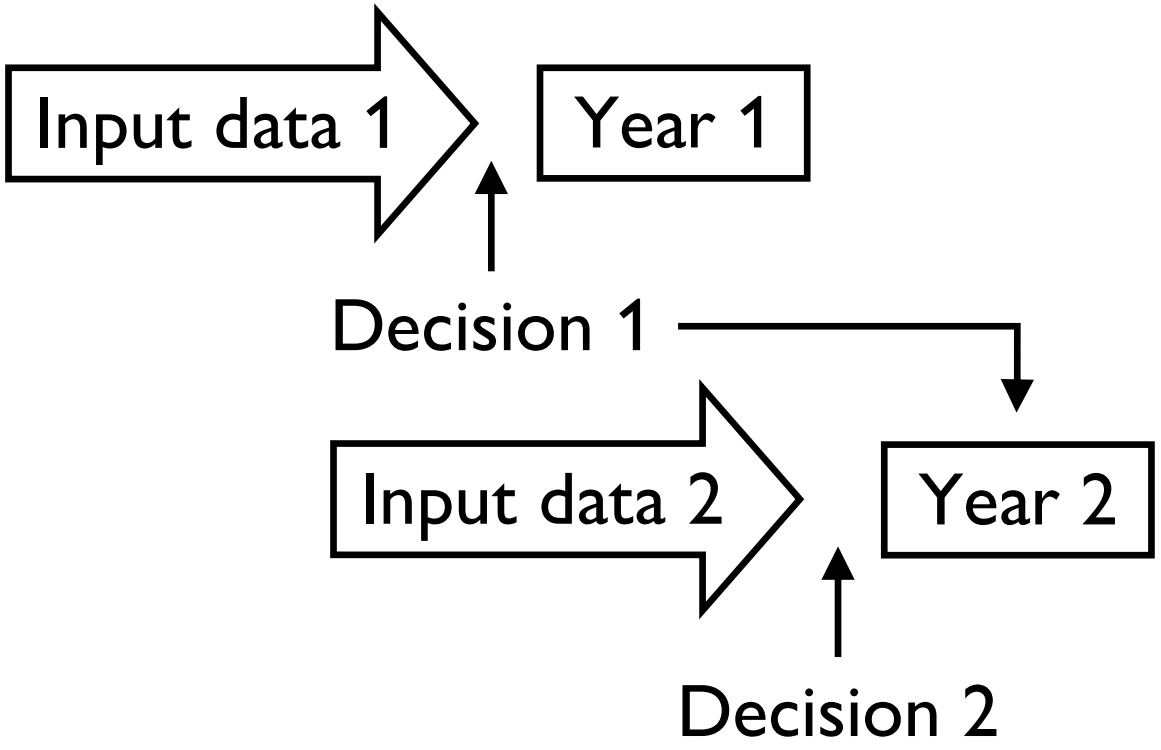

We focus on the end of the planning horizon
Static
Input data
Year 1
Year 2
Year Y
Decisions

Planning



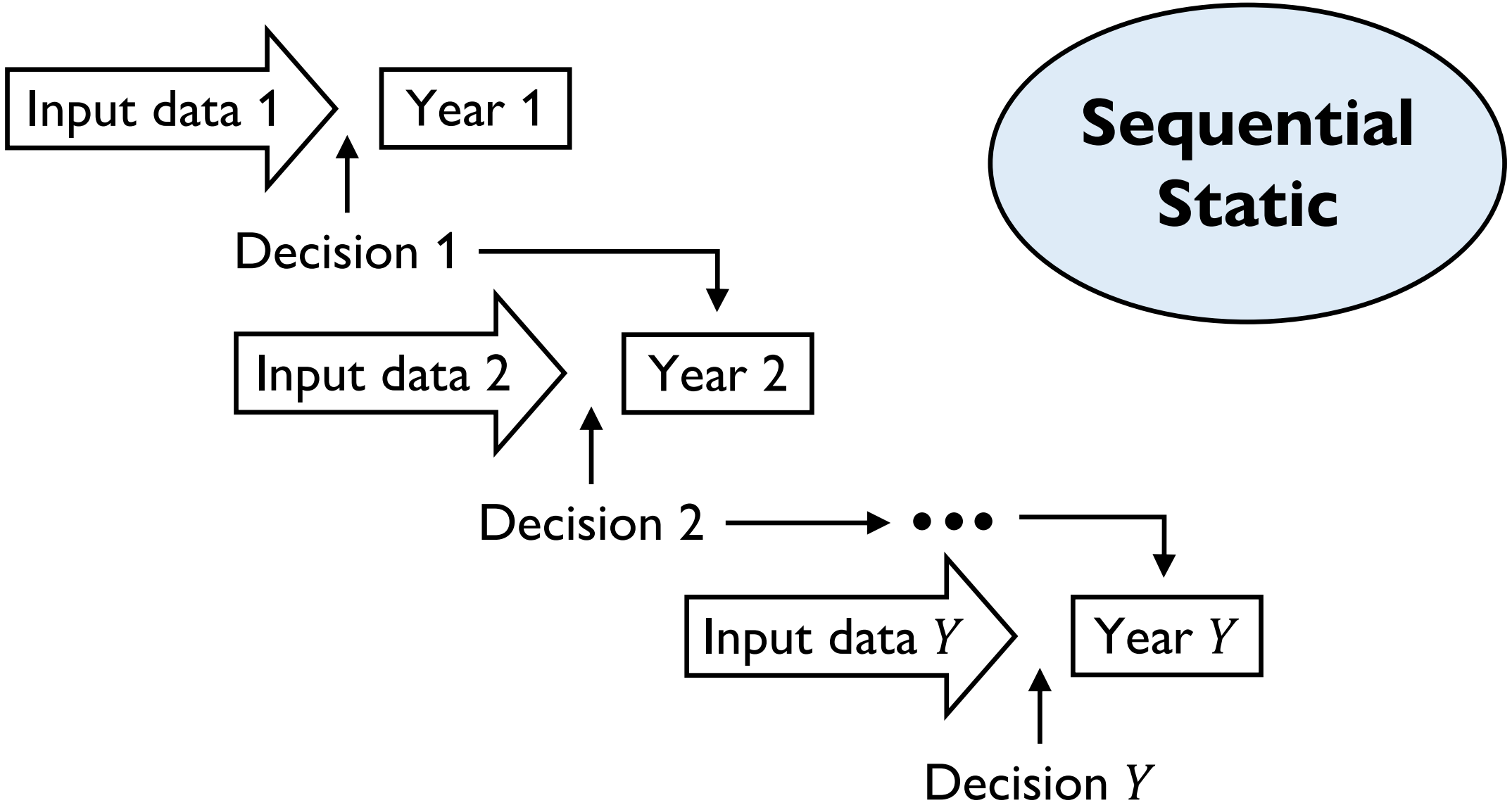
Sequential
Static

Planning

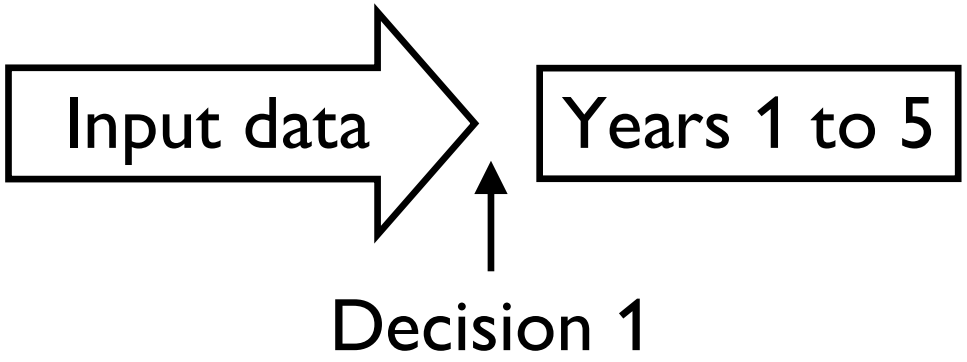


Sequential
Static

Planning


Input data 1
Year 1
Decision 1
Input data 2
Year 2
Decision 2
Input data Y
Year Y
Decision Y
Sequential
Static

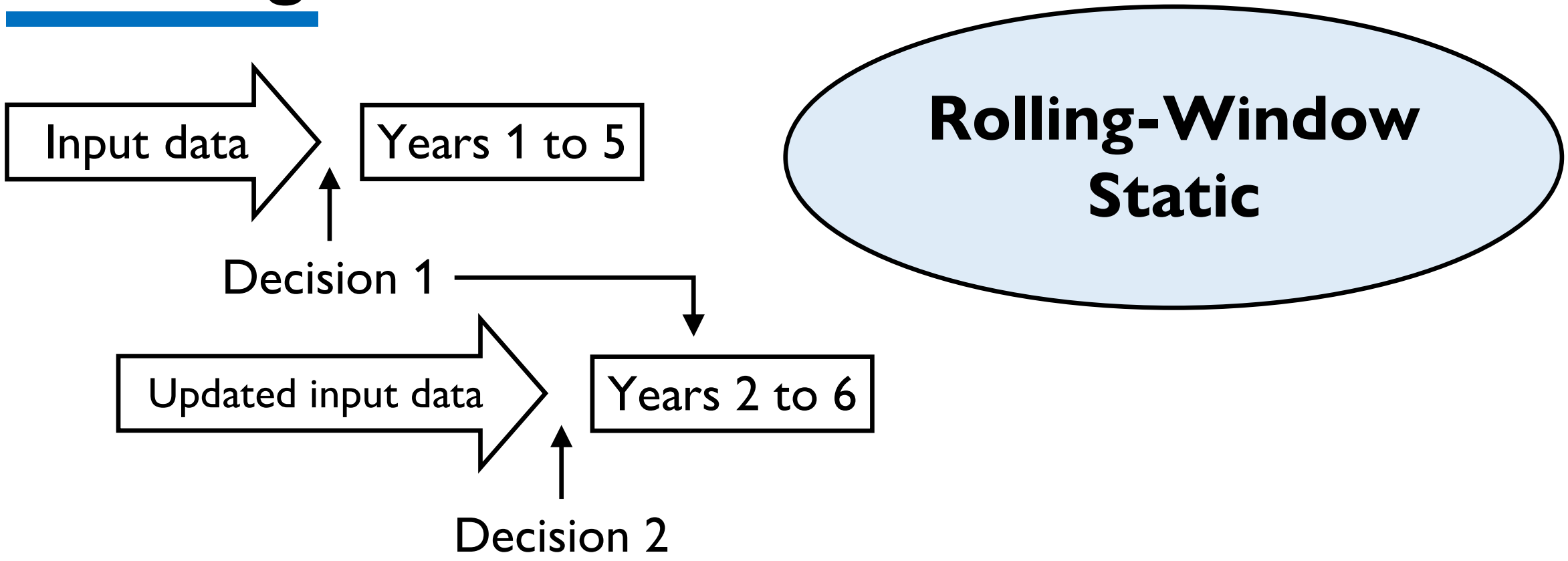
Planning



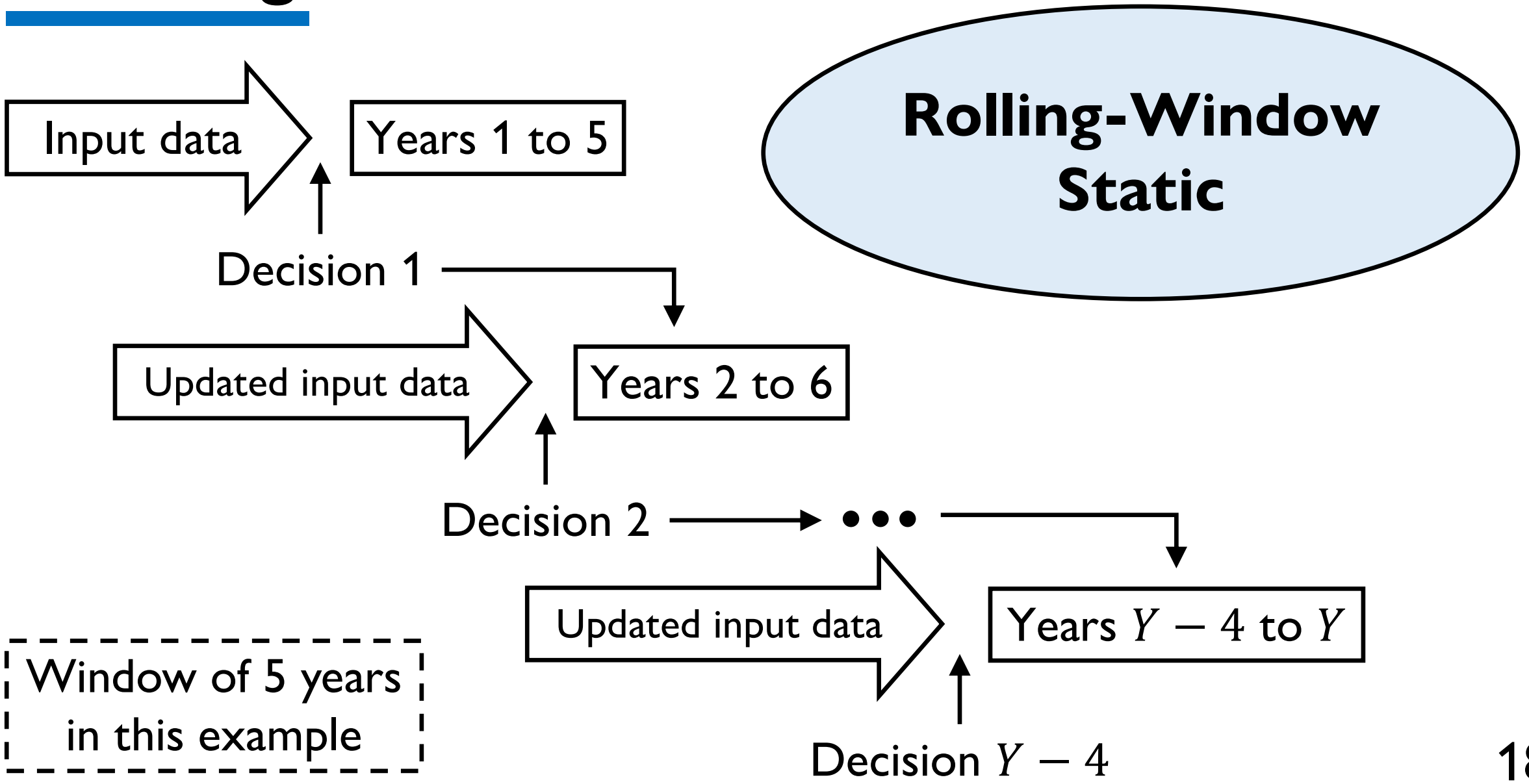
Rolling-Window
Static

Window of 5 years
in this example

Planning

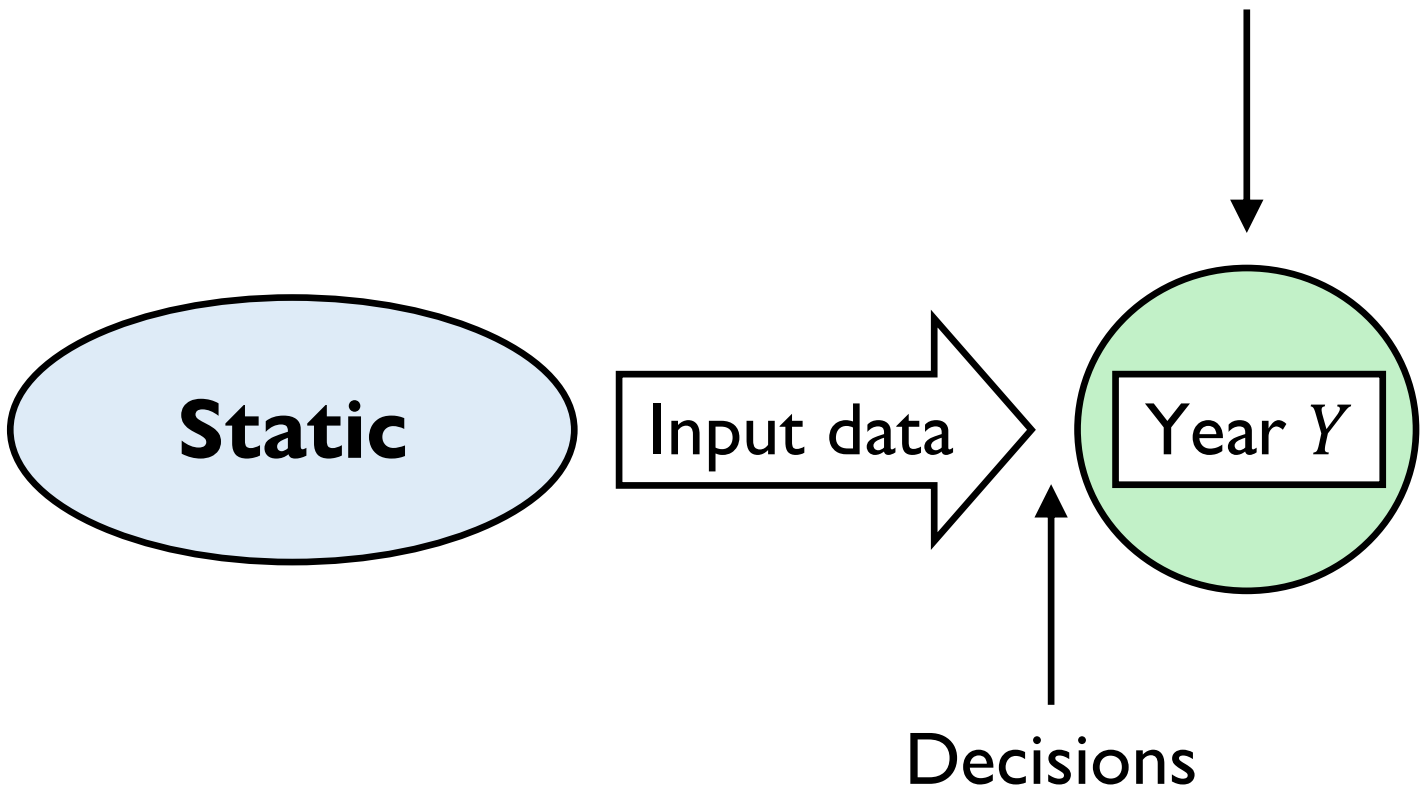

Input data
Years 1 to 5
Decision 1
Updated input data
Years 2 to 6
Decision 2
Rolling-Window
Static
Window of 5 years
in this example

Planning

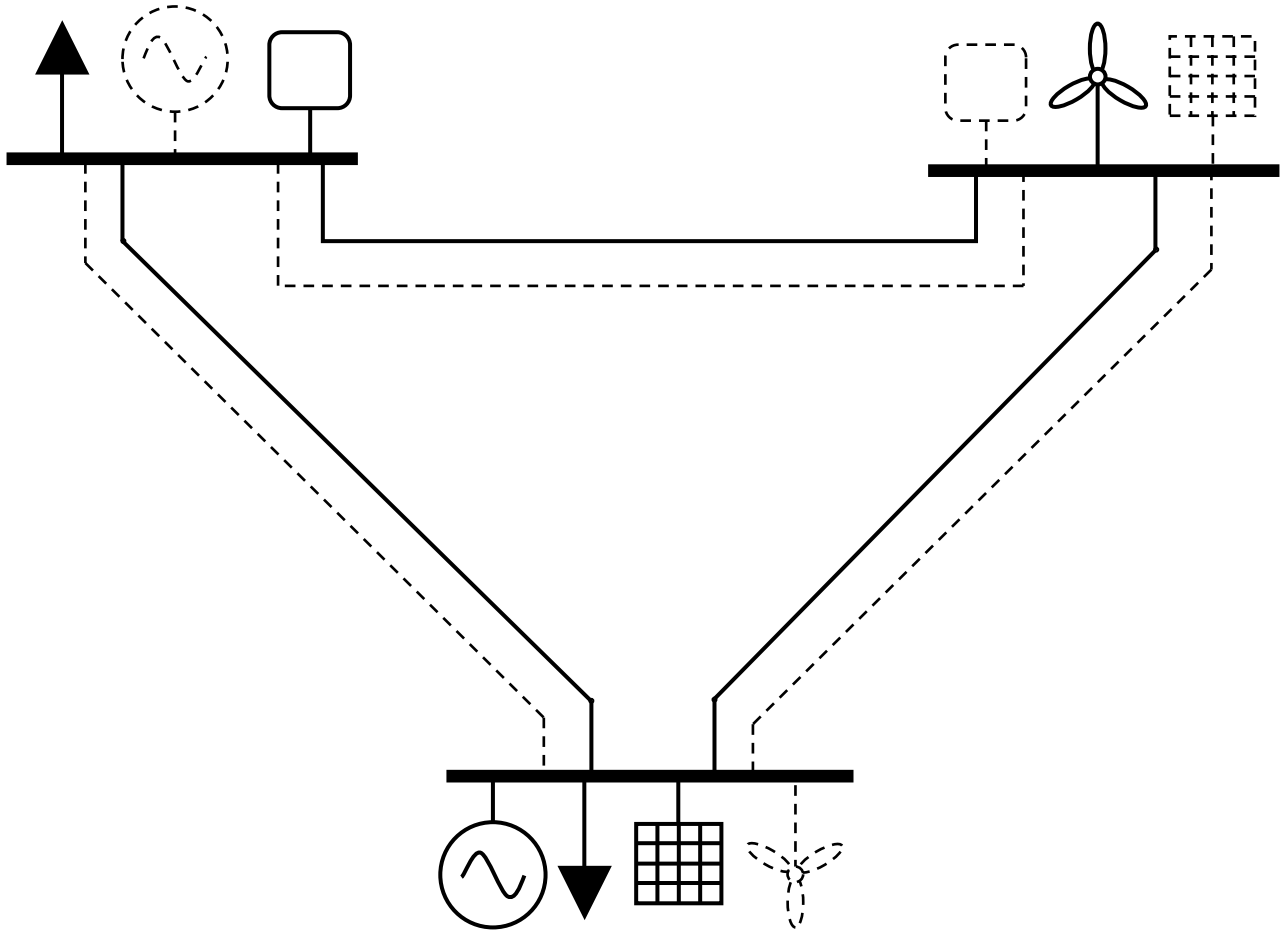

Rolling-Window
Static
Input data
Years 1 to 5
Decision 1
Updated input data
Years 2 to 6
Decision 2
Updated input data
Years $Y-4$ to Y
Decision $Y-4$
Window of 5 years
in this example

Planning

We focus on the end
of the planning horizon



Planning



Planning

min Total Costs = Operating Costs + Investment Costs

or

max Total Profit =
Operating Revenues – Operating Costs – Investment Costs

Planning

min Total Costs = Operating Costs + Investment Costs

or

max Total Profit =
Operating Revenues – Operating Costs – Investment Costs

Planning

The operation of one year is considered in the static framework.

Hence, the operating costs are associated with one year.

Planning

The operation of one year is considered in the static framework.

Hence, the operating costs are associated with one year.

Therefore, it is **unfair** to sum the operating costs of one year and the complete investment costs of the assets.

Planning

To prevent this, we must **annualize** the investment costs using the **capital recovery factor CRF**:

$$CRF = \frac{i(1+i)^Y}{(1+i)^Y - 1}$$

i: interest rate.

Y: lifetime of the facility in years.

Planning

I will consider the same capital recovery factor for all assets (generating units, transmission lines, etc.) in the exercises of this course for the sake of simplicity.

Planning

To prevent this , we must **annualize** the investment costs using the **capital recovery factor CRF**:

$$\text{Annualized Investment Costs} = CRF \cdot \text{Investment Costs}$$

Planning

Planning optimization problems solved by:

- A centralized entity:
  - Centralized generation expansion planning.
  - Transmission network expansion planning.
- Private entities:
  - Price-taker generation expansion planning.

Planning

Planning optimization problems solved by:

- A centralized entity:
  - **Centralized generation expansion planning (GEP)**.
  - Transmission network expansion planning.
- Private entities:
  - Price-taker generation expansion planning.

Centralized GEP – Formulation

I will consider the following modeling simplifications:

- Only conventional generating units g with LP operational formulation.
- Single-node model.
- Independent representative time periods t (detailed explanation in lecture 3).

Centralized GEP – Formulation

Investment decisions can be modeled using continuous or binary variables.

I will show both formulations.

Centralized GEP – Continuous invest. decisions

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} + \sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}}$$

Centralized GEP – Continuous invest. decisions

$$\min_{\Phi} \underbrace{\sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}}}_{\text{Annual operating costs}} + \sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}}$$

Centralized GEP – Continuous invest. decisions

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} + \sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}}$$

Weight of representative time period t (input data)

Centralized GEP – Continuous invest. decisions

8760 hourly time periods (1 year):

- $\rho_t^{\mathrm{T}} = 1\ \mathrm{h}, \forall t.$

- Computational issues may arise due to the high number of variables and equations.

Centralized GEP – Continuous invest. decisions

Use of representative time periods:

- Each representative time period t is associated with a value of the weight ρ_t^{T} and the parameters that vary with t, such as the electrical demand P_t^{D}.

- $\sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} = 8760$ h.

- Generation of ρ_t^{T}, P_t^{D}, etc. → Lecture 3.

Centralized GEP – Continuous invest. decisions

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} + \underbrace{\sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}}}$$

Annualized investment costs

Centralized GEP – Continuous invest. decisions

$$\sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}} \leq \overline{C}^{\mathrm{I}}$$

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}}, \forall t$$

Centralized GEP – Continuous invest. decisions

$$0 \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_{g}^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,Old}}, \forall t$$

$$0 \leq p_{gt}^{\mathrm{G}} \leq \overline{p}_{g}^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,New}}, \forall t$$

$$0 \leq \overline{p}_{g}^{\mathrm{G}} \leq \overline{P}_{g}^{\mathrm{G,New}}, \forall g \in \Omega^{\mathrm{G,New}}$$

Centralized GEP – Continuous invest. decisions

Note that if the investment budget $\overline{C}^{\mathrm{I}}$ is not large enough, it may be possible that the power balance equation is not fulfilled.

$$\sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}} \leq \overline{C}^{\mathrm{I}}$$

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}}, \forall t$$

Centralized GEP – Continuous invest. decisions

Moreover, low values of $\overline{P}_g^{\mathrm{G,New}}$ may also involve that the power balance equation is not fulfilled.

$$0 \leq \overline{p}_g^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G,New}}, \forall g \in \Omega^{\mathrm{G,New}}$$

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}}, \forall t$$

Centralized GEP – Continuous invest. decisions

This leads to an infeasible optimization problem.

Optimization software may indicate that the problem is infeasible due to a particular constraint, but the reason is not always easy to identify.

We can add slack variables. In this case, we can add load-shedding variables: p_t^{LS}.

Centralized GEP – Continuous invest. decisions

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}}, \forall t$$

Centralized GEP – Continuous invest. decisions

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}} - \boldsymbol{p_t^{\mathrm{LS}}}, \forall t$$

$$\boldsymbol{0 \leq p_t^{\mathrm{LS}} \leq P_t^{\mathrm{D}}, \forall t}$$

Centralized GEP – Continuous invest. decisions

However, the formulation is not complete.

What is the optimal solution of the problem using the formulation presented before?

Centralized GEP – Continuous invest. decisions

However, the formulation is not complete.

What is the optimal solution of the problem using the formulation presented before?

Total annualized cost = €0.

$p_{gt}^{\mathrm{G}^*} = 0, \forall g, \forall t$; $\overline{p}_{g}^{\mathrm{G}^*} = 0, \forall g \in \Omega^{\mathrm{G,New}}$; and $p_{t}^{\mathrm{LS}^*} = P_{t}^{\mathrm{D}}, \forall t$.

Centralized GEP – Continuous invest. decisions

We must include in the objective function a **penalization** associated with the slack variables:

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \left(\boldsymbol{C^{\mathrm{LS}} p_t^{\mathrm{LS}}} + \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} \right) + \sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}}$$

where C^{LS} must be large enough.

Centralized GEP – Continuous invest. decisions

This way, we can solve the problem and check the values of the load-shedding power.

If $p_t^{\mathrm{LS}^*} > 0$ for any t, then modify the input data, solve the optimization problem, and check the values of $p_t^{\mathrm{LS}^*}$.

Centralized GEP – Continuous invest. decisions

Note that the optimal solution may involve $p_t^{\mathrm{LS}^*} > 0$ due to multiplicity of solutions (different solutions that lead to the same value of the objective function).

In these cases, we can add the following constraints:

$$p_t^{\mathrm{LS}} = 0, \forall t$$

If the resulting problem is infeasible, we can conclude that $p_t^{\mathrm{LS}^*} > 0$ is not associated with multiplicity of solutions, and the input data must be modified.

Centralized GEP – Continuous invest. decisions

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \left(C^{\mathrm{LS}} p_t^{\mathrm{LS}} + \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} \right) + \sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}}$$

s.t.:

$$\sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}} \leq \overline{C}^{\mathrm{I}}$$

$$0 \leq \overline{p}_g^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G,New}}, \forall g \in \Omega^{\mathrm{G,New}}$$

Centralized GEP – Continuous invest. decisions

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}} - p_t^{\mathrm{LS}}, \forall t$$

$$0 \leq p_t^{\mathrm{LS}} \leq P_t^{\mathrm{D}}, \forall t$$

$$0 \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,Old}}, \forall t$$

$$0 \leq p_{gt}^{\mathrm{G}} \leq \overline{p}_g^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,New}}, \forall t$$

$$\Phi = \left\{ p_{gt}^{\mathrm{G}}, \forall g, \forall t; \overline{p}_g^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,New}}; \; p_t^{\mathrm{LS}}, \forall t \right\}$$

Exercise 7 – Centralized GEP

Solve the centralized GEP problem using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_07.gms

Solution:

a) Invest. budget = €200,000 → Objective function value = M€43.11.
b) Invest. budget = €180,000 →Objective function value = M€45.28.
c) Invest. budget = €145,000 → Objective function value = M€489.77.

Centralized GEP – Continuous invest. decisions

Optimal investment plan:

Build a power plant associated with a maximum capacity of 210.232 MW.

Is this realistic?

Centralized GEP – Binary investment decisions

Set of finite candidate investment options



Binary variables

Centralized GEP – Binary investment decisions

$$v_{gc}^{\mathrm{G}} \in \{0,1\}, \forall g \in \Omega^{\mathrm{G,New}}, \forall c \in \Omega_g^{\mathrm{C}}$$

$$\overline{p}_g^{\mathrm{G}} = \sum_{c \in \Omega_g^{\mathrm{C}}} \overline{P}_{gc}^{\mathrm{G,New}} v_{gc}^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,New}}$$

$$\sum_{c \in \Omega_g^{\mathrm{C}}} v_{gc}^{\mathrm{G}} \leq 1, \forall g \in \Omega^{\mathrm{G,New}}$$

Centralized GEP – Binary investment decisions

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \left(C^{\mathrm{LS}} p_t^{\mathrm{LS}} + \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} \right) + \sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}}$$

s.t.:

$$\sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}} \leq \overline{C}^{\mathrm{I}}$$

$$\overline{p}_g^{\mathrm{G}} = \sum_{c \in \Omega_g^{\mathrm{C}}} \overline{P}_{gc}^{\mathrm{G,New}} v_{gc}^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,New}}$$

Centralized GEP – Binary investment decisions

$$v_{gc}^{\mathrm{G}} \in \{0,1\}, \forall g \in \Omega^{\mathrm{G,New}}, \forall c \in \Omega_g^{\mathrm{C}}$$

$$\sum_{c \in \Omega_g^{\mathrm{C}}} v_{gc}^{\mathrm{G}} \leq 1, \forall g \in \Omega^{\mathrm{G,New}}$$

$$\sum_{g \in \Omega^{\mathrm{G}}} p_{gt}^{\mathrm{G}} = P_t^{\mathrm{D}} - p_t^{\mathrm{LS}}, \forall t$$

$$0 \leq p_t^{\mathrm{LS}} \leq P_t^{\mathrm{D}}, \forall t$$

Centralized GEP – Binary investment decisions

$$0 \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_{g}^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,Old}}, \forall t$$

$$0 \leq p_{gt}^{\mathrm{G}} \leq \overline{p}_{g}^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,New}}, \forall t$$

$$\Phi = \left\{ p_{gt}^{\mathrm{G}}, \forall g, \forall t; \overline{p}_{g}^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,New}}; p_{t}^{\mathrm{LS}}, \forall t; v_{gc}^{\mathrm{G}}, \forall g \in \Omega^{\mathrm{G,New}}, \forall c \in \Omega_{g}^{\mathrm{C}} \right\}$$

Exercise 8 – Centralized GEP

Solve the centralized GEP problem using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_08.gms

Solution:

Objective function value = M€49.83.

Exercise 7
M€45.28

Planning

Planning optimization problems solved by:

- A centralized entity:
  - Centralized generation expansion planning.
  - **Transmission network expansion planning (TNEP)**.
- Private entities:
  - Price-taker generation expansion planning.
  - Price-maker generation expansion planning.

TNEP – Formulation

I will consider the following modeling simplifications:

- Only conventional generating units g with LP formulation.
- Lossless DC power flow model.
- Independent representative time periods t (detailed explanation in lecture 3).

TNEP

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \left(\sum_{n \in \Omega^{\mathrm{N}}} C^{\mathrm{LS}} p_{nt}^{\mathrm{LS}} + \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} \right) + \sum_{l \in \Omega^{\mathrm{L,New}}} C_l^{\mathrm{L,I}} CRF_l v_l^{\mathrm{L}}$$

TNEP

$$\sum_{l \in \Omega^{\mathrm{L,New}}} C_l^{\mathrm{L,I}} CRF_l v_l^{\mathrm{L}} \leq \overline{C}^{\mathrm{I}}$$

$$v_l^{\mathrm{L}} \in \{0,1\}, \forall l \in \Omega^{\mathrm{L,New}}$$

$$\sum_{g \in \Omega_n^{\mathrm{G}}} p_{gt}^{\mathrm{G}} - \sum_{l \in \Omega^{\mathrm{L}} | FR(l)=n} p_{lt}^{\mathrm{L}} + \sum_{l \in \Omega^{\mathrm{L}} | TO(l)=n} p_{lt}^{\mathrm{L}} = P_{nt}^{\mathrm{D}} - p_{nt}^{\mathrm{LS}}, \forall n, \forall t$$

TNEP

$$0 \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_{g}^{\mathrm{G}}, \forall g, \forall t$$

$$0 \leq p_{nt}^{\mathrm{LS}} \leq P_{nt}^{\mathrm{D}}, \forall n, \forall t$$

$$-\overline{P}_{l}^{\mathrm{L}} \leq p_{lt}^{\mathrm{L}} \leq \overline{P}_{l}^{\mathrm{L}}, \forall l \in \Omega^{\mathrm{L,Old}}, \forall t$$

$$p_{lt}^{\mathrm{L}} = \frac{1}{X_{l}^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right), \forall l \in \Omega^{\mathrm{L,Old}}, \forall t$$

TNEP

$$-\overline{P}_l^{\mathrm{L}} \leq p_{lt}^{\mathrm{L}} \leq \overline{P}_l^{\mathrm{L}}, \forall l \in \Omega^{\mathrm{L,New}}, \forall t$$

$$p_{lt}^{\mathrm{L}} = \frac{v_l^{\mathrm{L}}}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right), \forall l \in \Omega^{\mathrm{L,New}}, \forall t$$

TNEP

$$-\overline{P}_l^{\mathrm{L}} \leq p_{lt}^{\mathrm{L}} \leq \overline{P}_l^{\mathrm{L}}, \forall l \in \Omega^{\mathrm{L,New}}, \forall t$$

$$p_{lt}^{\mathrm{L}} = \frac{v_l^{\mathrm{L}}}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right), \forall l \in \Omega^{\mathrm{L,New}}, \forall t$$

Product of binary and continuous variables: $v_l^{\mathrm{L}}\theta_{nt}^{\mathrm{N}}$.

TNEP

Big M method:

$$-\overline{P}_l^{\mathrm{L}} v_l^{\mathrm{L}} \leq p_{lt}^{\mathrm{L}} \leq \overline{P}_l^{\mathrm{L}} v_l^{\mathrm{L}}, \forall l \in \Omega^{\mathrm{L,New}}, \forall t$$

$$-M\left(1 - v_l^{\mathrm{L}}\right) \leq p_{lt}^{\mathrm{L}} - \frac{1}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right) \leq M\left(1 - v_l^{\mathrm{L}}\right),$$
$$\forall l \in \Omega^{\mathrm{L,New}}, \forall t$$

TNEP

If $v_l^{\mathrm{L}} = 0$:

$$-0 \leq p_{lt}^{\mathrm{L}} \leq 0 \rightarrow p_{lt}^{\mathrm{L}} = 0$$

$$-M \leq -\frac{1}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right) \leq M$$

TNEP

If $v_l^{\mathrm{L}} = 1$:

$$-\overline{P}_l^{\mathrm{L}} \leq p_{lt}^{\mathrm{L}} \leq \overline{P}_l^{\mathrm{L}}$$

$$0 \leq p_{lt}^{\mathrm{L}} - \frac{1}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right) \leq 0 \rightarrow$$

$$\rightarrow \; p_{lt}^{\mathrm{L}} = \frac{1}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right)$$

TNEP

$$\theta_{nt}^{\mathrm{N}} = 0, n{:}\, ref.., \forall t$$

TNEP – Formulation

$$\min_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \left(\sum_{n \in \Omega^{\mathrm{N}}} C^{\mathrm{LS}} p_{nt}^{\mathrm{LS}} + \sum_{g \in \Omega^{\mathrm{G}}} C_g^{\mathrm{G,V}} p_{gt}^{\mathrm{G}} \right) + \sum_{l \in \Omega^{\mathrm{L,New}}} C_l^{\mathrm{L,I}} CRF_l v_l^{\mathrm{L}}$$

s.t.:

$$\sum_{l \in \Omega^{\mathrm{L,New}}} C_l^{\mathrm{L,I}} CRF_l v_l^{\mathrm{L}} \leq \overline{C}^{\mathrm{I}}$$

$$v_l^{\mathrm{L}} \in \{0, 1\}, \forall l \in \Omega^{\mathrm{L,New}}$$

$$\sum_{g \in \Omega_n^{\mathrm{G}}} p_{gt}^{\mathrm{G}} - \sum_{l \in \Omega^{\mathrm{L}} | FR(l) = n} p_{lt}^{\mathrm{L}} + \sum_{l \in \Omega^{\mathrm{L}} | TO(l) = n} p_{lt}^{\mathrm{L}} = P_{nt}^{\mathrm{D}} - p_{nt}^{\mathrm{LS}}, \forall n, \forall t$$

TNEP – Formulation

$$0 \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_{g}^{\mathrm{G}}, \forall g, \forall t$$

$$0 \leq p_{nt}^{\mathrm{LS}} \leq P_{nt}^{\mathrm{D}}, \forall n, \forall t$$

$$-\overline{P}_{l}^{\mathrm{L}} \leq p_{lt}^{\mathrm{L}} \leq \overline{P}_{l}^{\mathrm{L}}, \forall l \in \Omega^{\mathrm{L,Old}}, \forall t$$

$$p_{lt}^{\mathrm{L}} = \frac{1}{X_{l}^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right), \forall l \in \Omega^{\mathrm{L,Old}}, \forall t$$

TNEP – Formulation

$$-\overline{P}_l^{\mathrm{L}} v_l^{\mathrm{L}} \le p_{lt}^{\mathrm{L}} \le \overline{P}_l^{\mathrm{L}} v_l^{\mathrm{L}}, \forall l \in \Omega^{\mathrm{L,New}}, \forall t$$

$$-M\left(1 - v_l^{\mathrm{L}}\right) \le p_{lt}^{\mathrm{L}} - \frac{1}{X_l^{\mathrm{L}}}\left(\theta_{FR(l),t}^{\mathrm{N}} - \theta_{TO(l),t}^{\mathrm{N}}\right) \le M\left(1 - v_l^{\mathrm{L}}\right),$$

$$\forall l \in \Omega^{\mathrm{L,New}}, \forall t$$

$$\theta_{nt}^{\mathrm{N}} = 0, n\text{: } ref., \forall t$$

$$\Phi = \left\{p_{gt}^{\mathrm{G}}, \forall g, \forall t; p_{lt}^{\mathrm{L}}, \forall l, \forall t; p_{nt}^{\mathrm{LS}}, \theta_{nt}^{\mathrm{N}}, \forall n, \forall t; v_l^{\mathrm{L}}, \forall l \in \Omega^{\mathrm{L,New}}\right\}$$

Exercise 9 – TNEP

Solve the TNEP problem considering the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_09.gms

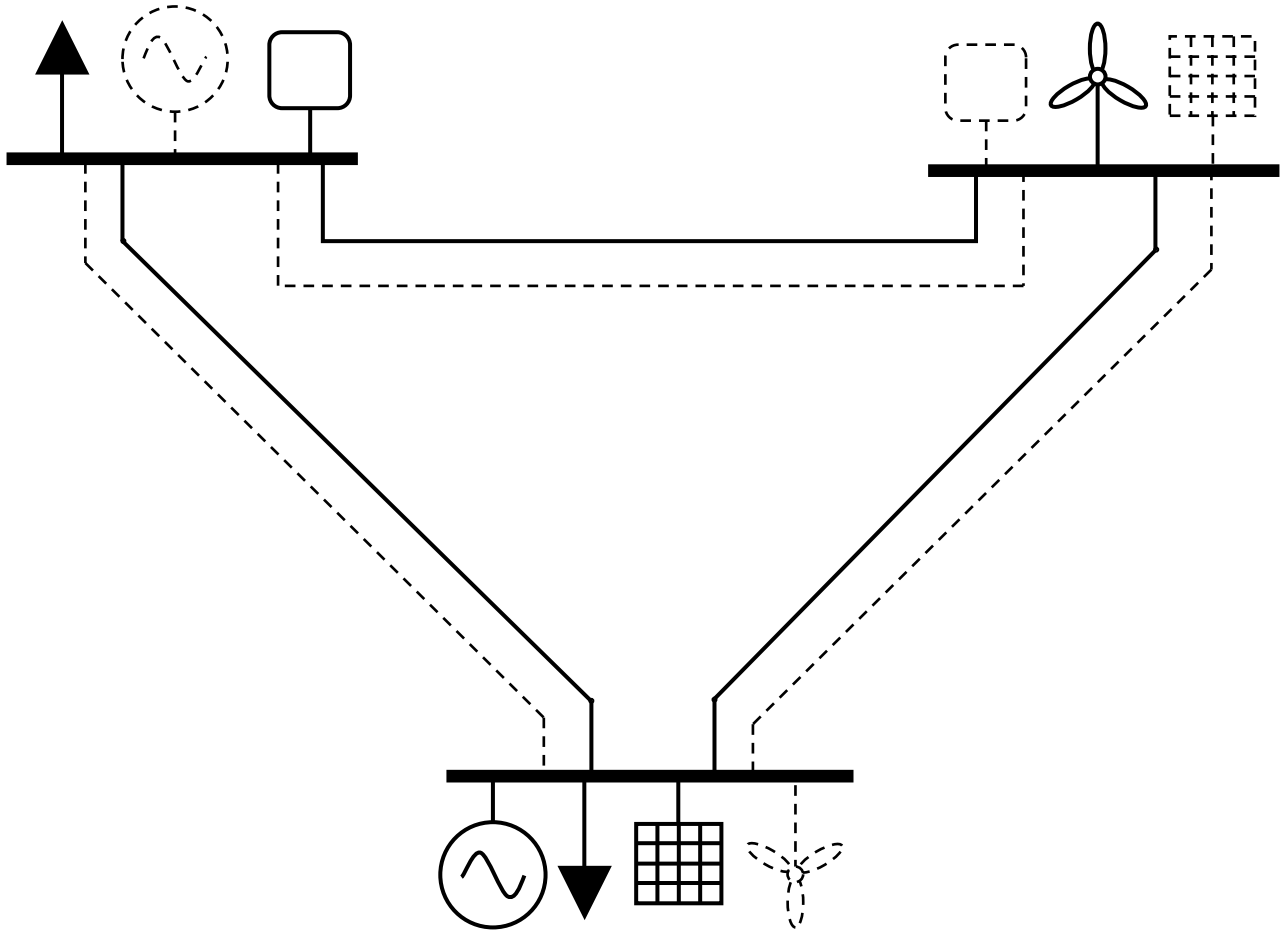
Solution:

a) $P_{n_3}^{\mathrm{D}} = 450$ MW $\rightarrow$ Transmission lines built: l_5.

b) $P_{n_3}^{\mathrm{D}} = 460$ MW $\rightarrow$ Transmission lines built: l_4 and l_5.

c) $P_{n_3}^{\mathrm{D}} = 470$ MW $\rightarrow$ Transmission lines built: l_3 and l_5.

Centralized GEP + TNEP = Centralized G&TNEP



Planning

Planning optimization problems solved by:

- A centralized entity:
  - Centralized generation expansion planning.
  - Transmission network expansion planning.
- Private entities:
  - **Price-taker generation expansion planning (GEP)**.

Price-taker GEP – Formulation

I will consider the following modeling simplifications:

- Viewpoint of a conventional producer.
- One technology without installed capacity.
- Independent representative time periods t (detailed explanation in lecture 3) → Ramping limits and start-up and shut-down costs are not modeled.

Price-taker GEP

$$\max_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \left(\lambda_t^{\mathrm{DA,S}} p_t^{\mathrm{G}} - C^{\mathrm{G,F}} u_t^{\mathrm{G}} - C^{\mathrm{G,V}} p_t^{\mathrm{G}} \right) - C^{\mathrm{G,I}} CRF \overline{p}^{\mathrm{G}}$$

$$0 \leq p_t^{\mathrm{G}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$$

Price-taker GEP

$$C^{\mathrm{G,I}} CRF \overline{p}^{\mathrm{G}} \leq \overline{C}^{\mathrm{I}}$$

$$v_c^{\mathrm{G}} \in \{0, 1\}, \forall c$$

$$\overline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \overline{P}_c^{\mathrm{G,New}} v_c^{\mathrm{G}}$$

$$\sum_{c \in \Omega^{\mathrm{C}}} v_c^{\mathrm{G}} \leq 1$$

Price-taker GEP

$$\underline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_{c}^{\mathrm{G,New}} v_{c}^{\mathrm{G}}$$

$$u_{t}^{\mathrm{G}} \in \{0, 1\}, \forall t$$

$$\underline{p}^{\mathrm{G}} u_{t}^{\mathrm{G}} \leq p_{t}^{\mathrm{G}} \leq \overline{p}^{\mathrm{G}} u_{t}^{\mathrm{G}}, \forall t$$

Price-taker GEP

$$\underline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_{c}^{\mathrm{G,New}} v_{c}^{\mathrm{G}}$$

$$u_{t}^{\mathrm{G}} \in \{0, 1\}, \forall t$$

$$\underline{p}^{\mathrm{G}} u_{t}^{\mathrm{G}} \leq p_{t}^{\mathrm{G}} \leq \overline{p}^{\mathrm{G}} u_{t}^{\mathrm{G}}, \forall t$$

Products of continuous and binary variables: $\underline{p}^{\mathrm{G}} u_{t}^{\mathrm{G}}$ and $\overline{p}^{\mathrm{G}} u_{t}^{\mathrm{G}}$.

Price-taker GEP

$$\overline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \overline{P}_{c}^{\mathrm{G,New}} v_{c}^{\mathrm{G}}$$

$$\underline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_{c}^{\mathrm{G,New}} v_{c}^{\mathrm{G}}$$

Price-taker GEP

$$\underline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_c^{\mathrm{G,New}} v_c^{\mathrm{G}}$$

$$u_t^{\mathrm{G}} \in \{0, 1\}, \forall t$$

$$\sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_c^{\mathrm{G,New}} v_c^{\mathrm{G}} u_t^{\mathrm{G}} \leq p_t^{\mathrm{G}} \leq \sum_{c \in \Omega^{\mathrm{C}}} \overline{P}_c^{\mathrm{G,New}} v_c^{\mathrm{G}} u_t^{\mathrm{G}}, \forall t$$

Price-taker GEP

$$\underline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_c^{\mathrm{G,New}} v_c^{\mathrm{G}}$$

$$u_t^{\mathrm{G}} \in \{0, 1\}, \forall t$$

$$\sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_c^{\mathrm{G,New}} v_c^{\mathrm{G}} u_t^{\mathrm{G}} \leq p_t^{\mathrm{G}} \leq \sum_{c \in \Omega^{\mathrm{C}}} \overline{P}_c^{\mathrm{G,New}} v_c^{\mathrm{G}} u_t^{\mathrm{G}}, \forall t$$

Products of binary variables: $v_c^{\mathrm{G}} u_t^{\mathrm{G}}$.

Price-taker GEP

$$p_t^{\mathrm{G}} \leq \overline{p}^{\mathrm{G}}, \forall t$$

$$\underline{p}^{\mathrm{G}} - \overline{P}_{\max}^{\mathrm{G,New}}\left(1 - u_t^{\mathrm{G}}\right) \leq p_t^{\mathrm{G}}, \forall t$$

$$p_t^{\mathrm{G}} \leq \overline{P}_{\max}^{\mathrm{G,New}} u_t^{\mathrm{G}}, \forall t$$

$$\underline{P}_{\min}^{\mathrm{G,New}} u_t^{\mathrm{G}} \leq p_t^{\mathrm{G}}, \forall t$$

where $\overline{P}_{\max}^{\mathrm{G,New}} = \max\left(\overline{P}_{\mathrm{c}}^{\mathrm{G,New}}\right)$ and $\underline{P}_{\min}^{\mathrm{G,New}} = \min\left(\underline{P}_{c}^{\mathrm{G,New}}\right)$.

Price-taker GEP

If $\sum_{c \in \Omega^C} v_c^G = 0$ and $u_t^G = 0$:

$$p_t^G \leq \overline{p}^G \rightarrow p_t^G \leq 0$$

$$\underline{p}^G - \overline{P}_{\max}^{G,New}\left(1 - u_t^G\right) \leq p_t^G \rightarrow -\overline{P}_{\max}^{G,New} \leq p_t^G$$

$$p_t^G \leq \overline{P}_{\max}^{G,New} u_t^G \rightarrow p_t^G \leq 0$$

$$\underline{P}_{\min}^{G,New} u_t^G \leq p_t^G \rightarrow 0 \leq p_t^G$$

$$\Big\} \; p_t^G = 0$$

Price-taker GEP

If $\sum_{c \in \Omega^C} v_c^G = 0$ and $u_t^G = 1$:

$$p_t^G \leq \overline{p}^G \rightarrow p_t^G \leq 0$$

$$\underline{p}^G - \overline{P}_{\max}^{G,New}\left(1 - u_t^G\right) \leq p_t^G \rightarrow 0 \leq p_t^G$$

$$p_t^G \leq \overline{P}_{\max}^{G,New} u_t^G \rightarrow p_t^G \leq \overline{P}_{\max}^{G,New}$$

$$\underline{P}_{\min}^{G,New} u_t^G \leq p_t^G \rightarrow \underline{P}_{\min}^{G,New} \leq p_t^G$$

Infeasible:

$$\underline{P}_{\min}^{G,New} \leq p_t^G \leq 0$$

Price-taker GEP

If $\sum_{c\in\Omega^{C}} v_c^{G} = 1$ and $u_t^{G} = 0$:

$$p_t^{G} \le \overline{p}^{G}$$

$$\underline{p}^{G} - \overline{P}_{\max}^{G,New}\left(1 - u_t^{G}\right) \le p_t^{G} \rightarrow \underline{p}^{G} - \overline{P}_{\max}^{G,New} \le p_t^{G}$$

$$p_t^{G} \le \overline{P}_{\max}^{G,New} u_t^{G} \rightarrow p_t^{G} \le 0$$

$$\underline{P}_{\min}^{G,New} u_t^{G} \le p_t^{G} \rightarrow 0 \le p_t^{G}$$

$$\Big\} \; p_t^{G} = 0$$

Price-taker GEP

If $\sum_{c \in \Omega^C} v_c^G = 1$ and $u_t^G = 1$:

$$\overline{p}^G \leq p_t^G$$

$$\underline{p}^G - \overline{P}_{\max}^{G,New}\left(1 - u_t^G\right) \leq p_t^G \rightarrow \underline{p}^G \leq p_t^G$$

$$p_t^G \leq \overline{P}_{\max}^{G,New} u_t^G \rightarrow p_t^G \leq \overline{P}_{\max}^{G,New}$$

$$\underline{P}_{\min}^{G,New} u_t^G \leq p_t^G \rightarrow \underline{P}_{\min}^{G,New} \leq p_t^G$$

$$\underline{p}^G \leq p_t^G \leq \overline{p}^G$$

Price-taker GEP

We can also linearize non-linear terms $v_c^{\mathrm{G}} u_t^{\mathrm{G}}$ using the procedure presented in **lecture 1** (product of two binary variables – case 2).

Price-taker GEP

$$\sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_c^{\mathrm{G,New}} a_{ct}^{\mathrm{G}} \leq p_t^{\mathrm{G}} \leq \sum_{c \in \Omega^{\mathrm{C}}} \overline{P}_c^{\mathrm{G,New}} a_{ct}^{\mathrm{G}}, \forall t$$

$$a_{ct}^{\mathrm{G}} \leq v_c^{\mathrm{G}}, \forall c, \forall t$$

$$a_{ct}^{\mathrm{G}} \leq u_t^{\mathrm{G}}, \forall c, \forall t$$

$$a_{ct}^{\mathrm{G}} \geq v_c^{\mathrm{G}} + u_t^{\mathrm{G}} - 1, \forall c, \forall t$$

$$a_{ct}^{\mathrm{G}} \geq 0, \forall c, \forall t$$

Price-taker GEP

Note that the following constraints can also be included:

$$u_t^{\mathrm{G}} \leq \sum_{c \in \Omega^{\mathrm{C}}} v_c^{\mathrm{G}}, \forall t$$

If $\sum_{c \in \Omega^{\mathrm{C}}} v_c^{\mathrm{G}} = 0$, then $u_t^{\mathrm{G}} \leq 0 \rightarrow u_t^{\mathrm{G}} = 0$.
Otherwise, $u_t^{\mathrm{G}} \leq 1$, which does not limit the value of u_t^{G}.

Price-taker GEP

Note that the following constraints can also be included:

$$u_t^{\mathrm{G}} \leq \sum_{c \in \Omega^{\mathrm{C}}} v_c^{\mathrm{G}}, \forall t$$

These constraints are not needed, but they may accelerate the solution procedure.

Price-taker GEP – Formulation

$$\max_{\Phi} \sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} \left(\lambda_t^{\mathrm{DA,S}} p_t^{\mathrm{G}} - C^{\mathrm{G,F}} u_t^{\mathrm{G}} - C^{\mathrm{G,V}} p_t^{\mathrm{G}} \right) - C^{\mathrm{G,I}} CRF \overline{p}^{\mathrm{G}}$$

s.t.:

$$0 \leq p_t^{\mathrm{G}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t$$

$$C^{\mathrm{G,I}} CRF \overline{p}^{\mathrm{G}} \leq \overline{C}^{\mathrm{I}}$$

$$v_c^{\mathrm{G}} \in \{0, 1\}, \forall c$$

Price-taker GEP – Formulation

$$\overline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \overline{P}_{c}^{\mathrm{G,New}} v_{c}^{\mathrm{G}}$$

$$\sum_{c \in \Omega^{\mathrm{C}}} v_{c}^{\mathrm{G}} \leq 1$$

$$\underline{p}^{\mathrm{G}} = \sum_{c \in \Omega^{\mathrm{C}}} \underline{P}_{c}^{\mathrm{G,New}} v_{c}^{\mathrm{G}}$$

$$u_{t}^{\mathrm{G}} \in \{0, 1\}, \forall t$$

Price-taker GEP – Formulation

$$p_t^{\mathrm{G}} \leq \overline{p}^{\mathrm{G}}, \forall t$$

$$\underline{p}^{\mathrm{G}} - \overline{P}_{\max}^{\mathrm{G,New}} \left(1 - u_t^{\mathrm{G}}\right) \leq p_t^{\mathrm{G}}, \forall t$$

$$p_t^{\mathrm{G}} \leq \overline{P}_{\max}^{\mathrm{G,New}} u_t^{\mathrm{G}}, \forall t$$

$$\underline{P}_{\min}^{\mathrm{G,New}} u_t^{\mathrm{G}} \leq p_t^{\mathrm{G}}, \forall t$$

$$u_t^{\mathrm{G}} \leq \sum_{c \in \Omega^{\mathrm{C}}} v_c^{\mathrm{G}}, \forall t$$

$$\Phi = \left\{ p_t^{\mathrm{G}}, \forall t; \overline{p}^{\mathrm{G}}; \underline{p}^{\mathrm{G}}; u_t^{\mathrm{G}}, \forall t; v_c^{\mathrm{G}}, \forall c \right\}$$

Exercise 10 – Price-taker GEP

Solve the centralized GEP problem using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%202/exercise2_10.gms

Solution:

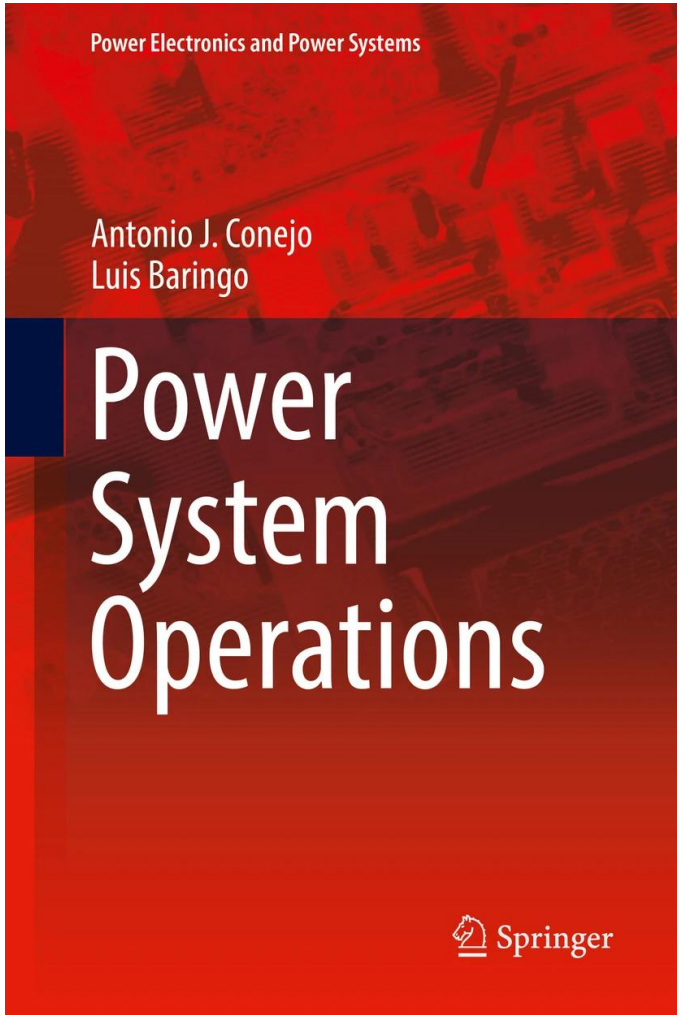
Objective function value = M€210.56.

Next lecture

- ~~Lecture 1: Introduction to operations research.~~
- ~~Lecture 2: Operation and planning problems.~~
- **Lecture 3: Operational variability.**
- Lecture 4: Uncertainty and risk.
- Lecture 5: Further applications.

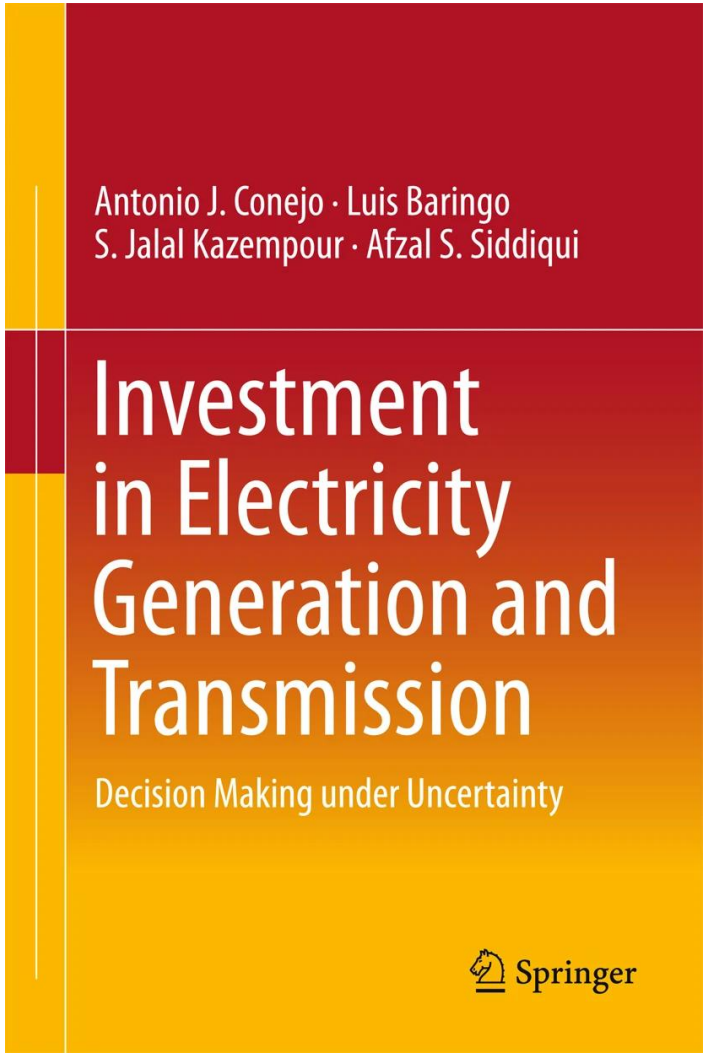
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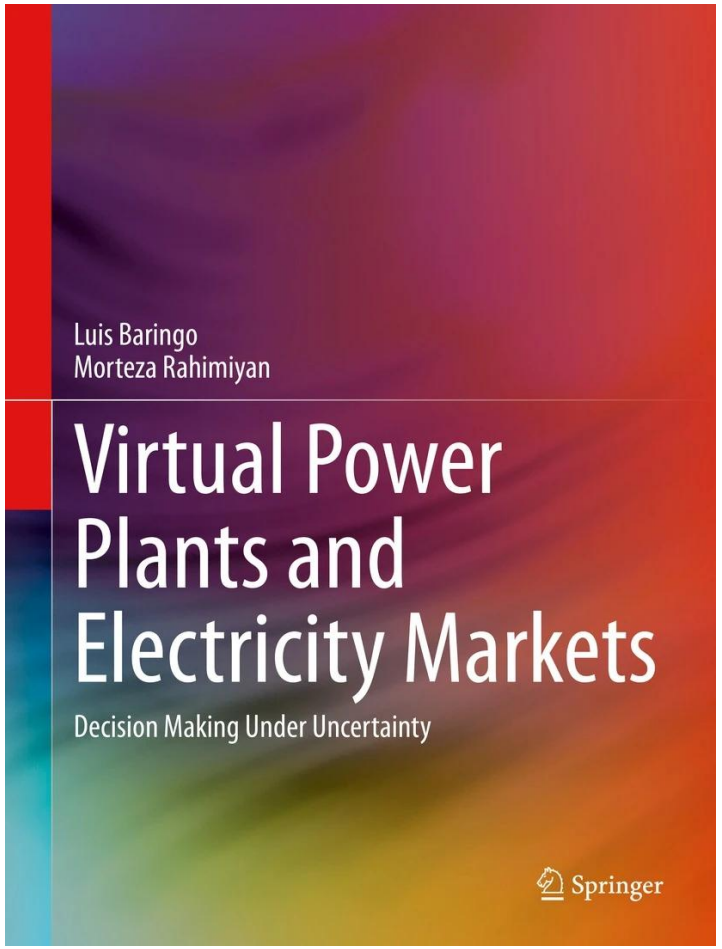
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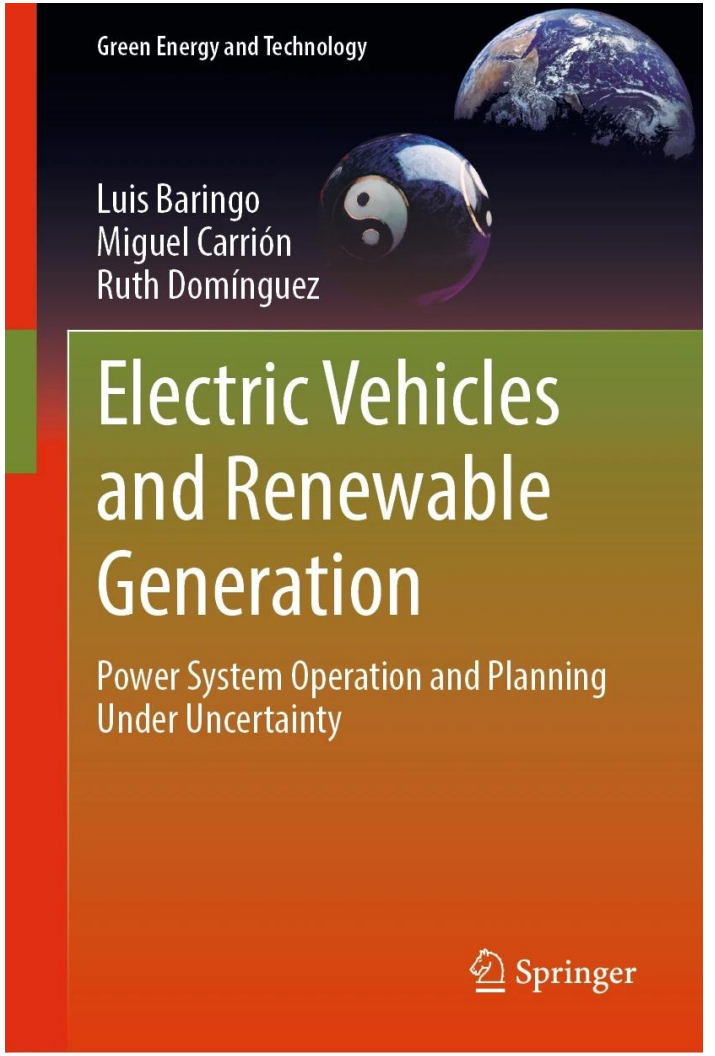
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Thank you very much for your attention!

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